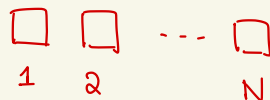


Online RL

Online learning. bandits. RL { Rmax (Tabular)
OLIVE (Bellman Rank)
LSVI-UCB (linear MDP)
↳ ...

Prediction with Expert Advice (PEA)

At round $t = 1, 2, \dots, T$:



- 1) the player first picks a weight p_t from a simplex Δ_N
- 2) and simultaneously environment pick a loss vector $\underline{l}_t \in [0, 1]^N$
- 3) the player suffers loss $\underline{IE}_{p_t}[\underline{l}_t] = \langle p_t, \underline{l}_t \rangle$,

★ observes loss $\underline{l}_t$ and update the model.

↳ p_{t+1} depends on $p_{1:t}, l_{1:t}$

< full-information
partial-information

< stochastic
adversarial Δ

Goal: minimize regret 遗憾

$$\text{Reg}_T = \sum_{t=1}^T \langle p_t, \underline{l}_t \rangle - \min_{p \in \Delta_N} \sum_{t=1}^T \langle p, \underline{l}_t \rangle$$

$$= \sum_{t=1}^T \langle p_t, \underline{l}_t \rangle - \min_{i \in [N]} \sum_{t=1}^T l_{t,i}$$

sublinear $\rightarrow o(T)$

Let $\underline{L}_{t,i} = \sum_{s=1}^t l_{s,i}$

Algorithm 1: $p_t = \underset{p \in \Delta_N}{\operatorname{argmin}} \langle p, \underline{L}_{t-1} \rangle$
(FTL,

Follow-The-Leader) = $\underset{i \in [N]}{\operatorname{argmin}} \underline{L}_{t-1,i}$

best expert in hindsight

t=1	t=2	t=3	t=4
1 [0.5]	1 [1]	0 [0]	3 [3]
2 [0.5]	0 [0]	2 [2]	0 [0]

Reg_T = o(T) ✗

Algorithm 2: $P_{t,i} \propto \exp(-\eta \cdot L_{t-1,i})$

(Hedge)

MWL (Multiplicative Weight Update)

$$P_{t,i} = \frac{\exp(-\eta \cdot L_{t-1,i})}{\sum_{j=1}^N \exp(-\eta \cdot L_{t-1,j})}$$

Greedy-update: $P_{t+1,i} \propto P_{t,i} \cdot \exp(-\eta \cdot l_{t,i})$

Softmax

Langevin Dynamics

Theorem (Regret Bound of Hedge)

Suppose $\forall t \in [T], i \in [N]: 0 \leq l_{t,i} \leq 1$

$$\underline{\text{Reg}_T} \leq \frac{\log N}{\eta} + \eta T = O(\sqrt{T \log N})$$

Proof: (potential-based)

$$\text{by } \eta = \sqrt{\frac{\log N}{T}}$$

potential: $\Phi_t = \frac{1}{\eta} \log \sum_{i=1}^N \exp(-\eta L_{t,i})$

$$\Phi_t - \Phi_{t+1} = \frac{1}{\eta} \log \frac{\sum_{i=1}^N \exp(-\eta \cdot L_{t,i})}{\sum_{j=1}^N \exp(-\eta \cdot L_{t+1,j})}$$

$$L_{t,i} = L_{t+1,i} + l_{t,i}$$

$$= \frac{1}{\eta} \log \sum_{i=1}^N \left[\frac{\exp(-\eta \cdot L_{t+1,i})}{\sum_{j=1}^N \exp(-\eta \cdot L_{t+1,j})} \right] \cdot \exp(-\eta \cdot l_{t,i})$$

$$= \frac{1}{\eta} \log \sum_{i=1}^N P_{t,i} \exp(-\eta \cdot l_{t,i})$$

$$\mathbb{E}_{P_t} [e^{-\eta l_t}]$$

$$\leq \frac{1}{\eta} \log \sum_{i=1}^N p_{t,i} \left(1 - \eta \cdot \underline{l}_{t,i} + \eta^2 \underline{l}_{t,i}^2 \right) \leq 1$$

Hoeffding's Lemma

$$\leq \frac{1}{\eta} \log \left(1 - \eta \cdot \langle p_t, l_t \rangle + \eta^2 \sum_{i=1}^N p_{t,i} \underline{l}_{t,i}^2 \right) \quad e^{-x} \leq 1 - x + \frac{x^2}{2} \text{ for } x \geq 0$$

$$= -\langle p_t, l_t \rangle + \eta \cdot \sum_{i=1}^N p_{t,i} \underline{l}_{t,i}^2$$

Telescoping:

$$\sum_{t=1}^T \langle p_t, l_t \rangle \leq \underbrace{\Phi_0 - \Phi_T}_{\downarrow} + \eta \cdot \sum_{t=1}^T \sum_{i=1}^N p_{t,i} \underline{l}_{t,i}^2$$

$$\Phi_0 = \frac{1}{\eta} \log \sum_{i=1}^N \exp(-\eta \cdot 0) = \frac{1}{\eta} \log N.$$

$$\Phi_T = \frac{1}{\eta} \log \sum_{i=1}^N \exp(-\eta \cdot \underline{l}_{T,i})$$

$$\geq \frac{1}{\eta} \log \exp(-\eta \underline{l}_{T,i^*}) = -\underline{l}_{T,i^*}$$

$$\Rightarrow \text{Reg}_T = \sum_{t=1}^T \langle p_t, l_t \rangle - \underline{l}_{T,i^*} \leq \frac{1}{\eta} \log N + \eta T.$$

$$\text{By tuning } \eta = \sqrt{\frac{\log N}{T}} \Rightarrow \underline{\text{Reg}_T = O(\sqrt{T \log N})} \quad \square$$

Lower Bound of PEA: For any algorithm A .

$$\sup_{T, N} \max_{l_1, \dots, l_T} \frac{\text{Reg}_T}{\sqrt{T \log N}} \geq \frac{1}{\sqrt{2}}$$

A Hedge is minimax optimal!

Multiplicative Weight Update. (MWU)

Weighted Majority

Hedge

FTRL

(Follow-the-Regularized Leader)

$$x_{t+1} = \operatorname{argmin}_{x \in X} \left\{ \sum_{s=1}^t f_s(x) + \frac{1}{\eta} \underline{\psi_{t+1}(x)} \right\}$$

OMD (Online Mirror Descent)

$$x_{t+1} = \operatorname{argmin}_{x \in X} \left\{ \langle x, \nabla f_t(x) \rangle + \frac{1}{\eta} \underline{D_{\psi}(x, x_t)} \right\}$$

negative entropy

KL divergence

Adversarial Multi-Armed Bandits (MAB)

I_t

At round $t = 1, 2, \dots, T$:

$\square \quad \square \quad \dots \quad \square$
1 2 $\dots$ K

1) the player first picks an arm $a_t \in [K]$ from K candidate arms

2) and simultaneously environments pick a loss vector $\ell_t \in [0, 1]^K$

3) the player suffers and only observes loss ℓ_{t, a_t} , then updates the model

partial information

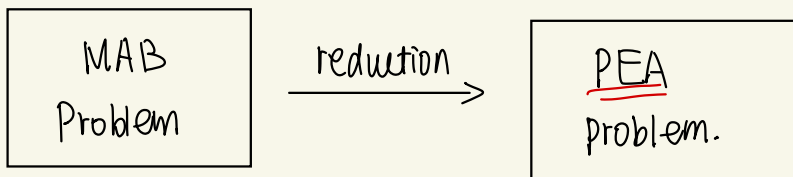
A_N

$\left\{ \begin{array}{l} \text{PEA} : I_t \\ \text{MAB} : \ell_{t, a_t} \end{array} \right.$

$\left\{ \begin{array}{l} \text{OCO} : \underline{f_t}, \nabla f_t \\ \text{BCO} : \underline{f_t(x_t)} \end{array} \right.$

domain $\{e_1, \dots, e_k\}$ is NOT convex

Idea:



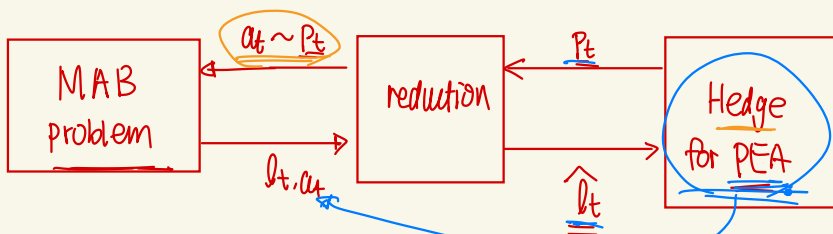
Goal: expected regret.

$$\text{Reg}_T = \sum_{t=1}^T \langle P_t, \underline{l}_t \rangle - \min_{i \in [k]} L_{t,i}$$

$$\mathbb{E}[\text{Reg}_T] = \mathbb{E} \left[\sum_{t=1}^T \underline{l}_{t,a_t} \right] - \min_{a \in [k]} \sum_{t=1}^T l_{t,a}$$

over the randomness of algorithms.

Protocol:



Importance-Weighted (IW) Loss Estimation

$$\underline{\hat{l}}_t \in \mathbb{R}^k. \quad \underline{\hat{l}}_{t,a} = \frac{\underline{l}_{t,a_t}}{P_{t,a_t}} \cdot \mathbb{1}\{a = a_t\} = \begin{cases} \frac{\underline{l}_{t,a_t}}{P_{t,a_t}}, & \text{if } a = a_t \\ 0 & \text{o.w.} \end{cases}$$

Property 1

$$\underline{l}_{t,a_t} = \langle \underline{\hat{l}}_t, P_t \rangle$$

$$\underline{\hat{l}}_t = [0, \dots, \frac{\underline{l}_{t,a_t}}{P_{t,a_t}}, \dots, 0]$$

Property 2. (unbiased estimator)

$$\mathbb{E}_{a_t \sim P_t} [\underline{\hat{l}}_{t,a}] = \underline{l}_{t,a} \quad \forall a \in [k]$$

Proof : $\mathbb{E}_{a_t \sim p_t} [\hat{L}_{t,a}] = \sum_{a_t} p_{t,a} \cdot \frac{l_{t,a}}{p_{t,a}} \cdot \mathbb{1}\{a=a_t\}$

$$= \frac{1}{p_{t,a}} \sum_{a_t} p_{t,a} \cdot l_{t,a} \cdot \mathbb{1}\{a=a_t\}$$

$$= \frac{1}{p_{t,a}} \cdot p_{t,a} \cdot l_{t,a} = l_{t,a}$$

fixed random

□

exploitation

$$\hat{L}_t = [0, \dots, \frac{l_{t,a}}{p_{t,a}}, \dots, 0]$$

exploration

Algorithm: Exp3

(Exponential-weight for
Exploration and Exploitation)

At round $t=1, 2, \dots$:

① compute $p_t \in \Delta_K$: $p_{t,a} \propto \exp(-\eta \cdot \hat{L}_{t-1,a})$

② sample $a_t \sim p_t$. observes l_{t,a_t} .

construct $\hat{L}_t$ as $\hat{L}_{t,a} = \frac{l_{t,a}}{p_{t,a}} \cdot \mathbb{1}\{a=a_t\}$

③ $\hat{L}_{t,a} = \hat{L}_{t-1,a} + \hat{L}_{t,a}$

Theorem

Exp3:

$$\mathbb{E}[\text{Reg}_T] \leq O(\sqrt{TK \log k})$$

harder than PEA

Proof: $\sum_{t=1}^T \langle P_t, \hat{l}_t \rangle - \sum_{t=1}^T \hat{l}_{t,a} \leq \frac{\log K}{\eta} + \eta \cdot \sum_{t=1}^T \sum_{k=1}^K P_{t,a} (\hat{l}_{t,a})^2$

$\forall a \in [K]$ from PEA analysis

$IE[\text{Reg}_T] = IE\left[\sum_{t=1}^T \underline{\underline{l_{t,a_t}}}\right] - \min_{a \in [K]} \sum_{t=1}^T l_{t,a}$

$\Rightarrow \sum_{t=1}^T l_{t,a_t} - \sum_{t=1}^T \hat{l}_{t,a} \leq \frac{\log K}{\eta} + \eta \cdot \sum_{t=1}^T \sum_{k=1}^K P_{t,a} (\hat{l}_{t,a})^2$

$IE_{a_t \sim p_t}[(\hat{l}_{t,a})^2] = \sum_{a_t} P_{t,a_t} \cdot \frac{l_{t,a_t}^2}{P_{t,a_t}^2} \cdot \underline{\underline{1\{a=a_t\}}}$

$= \frac{1}{P_{t,a}^2} \cdot P_{t,a} \cdot l_{t,a}^2 = \frac{l_{t,a}^2}{P_{t,a}}$

$\sum_{t=1}^T IE\left[IE_{a_t}[l_{t,a_t} - \hat{l}_{t,a}]\right]$

$= \sum_{t=1}^T IE\left[\underline{\underline{IE_{a_t}[l_{t,a_t}]}} - \underline{\underline{l_{t,a}}}\right]$ unbiasedness

$= IE\left[\sum_{t=1}^T l_{t,a_t}\right] - \sum_{t=1}^T l_{t,a} \quad \forall a$

$\leq \frac{\log K}{\eta} + \eta \cdot \sum_{t=1}^T \sum_{k=1}^K IE\left[\underline{\underline{P_{t,a}}} \cdot \underline{\underline{(\hat{l}_{t,a})^2}}\right]$

$= \frac{\log K}{\eta} + \eta \cdot \sum_{t=1}^T \sum_{k=1}^K \underline{\underline{l_{t,a}^2}} \leq 1$

$$\leq \frac{\log K}{\eta} + \eta \cdot T \cdot K \leq O(\sqrt{TK \log K})$$

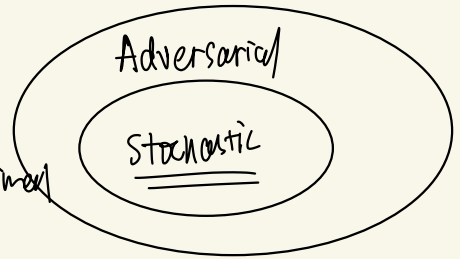
MAB lower bound: $\Omega(\sqrt{TK})$

by tuning $\eta = \sqrt{\frac{\log K}{TK}}$ $\square$

↑
stochastic



Exp3 is NOT minimax optimal



↙
Extra $\sqrt{\log K}$ dependency.

Hedge: OMD with regularizer $D_{\mathcal{L}}$ $\leftarrow$ negative entropy $\psi(p) = \sum_a p_a \log p_a$



OMD with Tsallis entropy regularizer, + IW estimator.

$$\psi(p) = \frac{1 - \sum_a p_a^\beta}{1 - \beta} \xrightarrow[\beta \rightarrow 1]{} \sum_a p_a \log p_a$$

c.f. Regret Bounds and Minimax Policies Under Partial Monitoring

Expected Regret $\rightarrow$ High-Probability Regret:

c.f. Explore no more: Improved High-Probability Regret Bounds
for Non-stochastic Bandits,

IX (Implicit exploration) : $\hat{I}_{t,a} = \frac{I_{t,a}}{p_{t,a} + \gamma} \cdot \mathbb{1}\{a = a_t\}$

Stochastic Multi-Armed Bandits

c.f. Bandits Algorithms by L & S.

At round $t=1, 2, \dots, T$:

1) the player first chooses an arm $a_t \in [K]$

2) and then environment reveals a reward $r_t(a_t) \in [0, 1]$

3) the player updates the model by the pair $(a_t, r_t(a_t))$

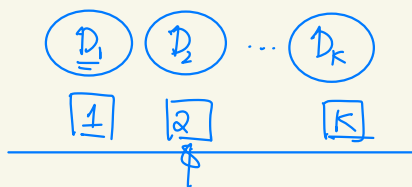
each arm $a \in [K]$ has an unknown distribution $\mathcal{D}_a$ with mean $\mu(a)$

such that $r_1(a), r_2(a), \dots, r_T(a)$ are i.i.d. samples of $\mathcal{D}_a$

Goal:

① Best-Arm Identification (BAI)

after T rounds, find $a^* = \underset{a}{\operatorname{argmax}} \mu(a)$



② Regret minimization

$$\begin{aligned} \text{pseudo-regret} \quad \Delta \text{Reg}_T &= \max_{a \in [K]} \mathbb{E} \left[\sum_{t=1}^T r_t(a) - \sum_{t=1}^T r_t(a_t) \right] \\ &= T \cdot \mu(a^*) - \sum_{t=1}^T \mu(a_t) \end{aligned}$$

Adversarial MAB: Exp 3 $\rightarrow \mathcal{O}(\sqrt{TK \log k})$

Stochastic MAB

worst-case bound

instance-dependent bound

for each arm $a \in [k]$:

① sub-optimality gap: $\Delta_a = \mu(a^*) - \mu(a)$

② number of times arm a is pulled within t rounds:

$$n_t(a) = \sum_{s=1}^t \mathbb{1}\{a_s = a\}$$

$$\text{Reg}_T = T \cdot \mu(a^*) - \sum_{t=1}^T \mu(a_t)$$

$$= \sum_{t=1}^T \mu(a^*) - \mu(a_t)$$

$$= \sum_{t=1}^T \Delta_{a_t} = \sum_{t=1}^T \sum_{a \in [k]} \Delta_a \cdot \mathbb{1}\{a = a_t\}$$

$$= \sum_{a \in [k]} \sum_{t=1}^T \Delta_a \cdot \mathbb{1}\{a = a_t\}$$

$$= \sum_{a \in [k]} \Delta_a \left(\sum_{t=1}^T \mathbb{1}\{a = a_t\} \right) = \sum_{a \in [k]} \Delta_a \cdot n_T(a)$$

$$\text{Reg}_T = \sum_{a \in [k]} \Delta_a \cdot n_T(a)$$

Algorithm: ETC

(Explore-then-Commit)

1	2	...	k
✓	✓		✓
✓	✓		✓
✓	✓		✓
D_1	D_2		D_k

T_0/k samples per distribution.

① Do explore for first T_0 rounds,
by pulling each arm for T_0/K times.

② Do exploit for the rest $T - T_0$ rounds,
by always pulling $\hat{a} = \operatorname{argmax}_{a \in [K]} \hat{\mu}_{T_0}(a)$

Strategic Exploration!

$$\hat{\mu}_{T_0}(a) = \frac{1}{T_0} \cdot \sum_{t=1}^{T_0} \mathbb{1}\{a_t = a\} \cdot r_t(a_t)$$

Thm (Regret Bound of ETC)

Suppose $\forall t \in [T], a \in [K], 0 \leq r_t(a) \leq 1$.

$$\Rightarrow \text{Reg}_T \leq \sum_{a \in [K]} \left(\underbrace{\frac{T_0}{K}}_{\text{explore}} + \underbrace{2T \cdot \exp\left(-\frac{T_0 \Delta_a^2}{2K}\right)}_{\text{commit}} \right) \Delta_a$$

Tune T_0 : ① Known Δ_a : $e^{-u} \leq \frac{1}{u}$ for $u > 0$

$$\begin{aligned} \sum_a \Delta_a \cdot \exp\left(-\frac{T_0 \Delta_a^2}{2K}\right) &\leq \sum_a \Delta_a \frac{2K}{T_0 \Delta_a^2} \\ &= \frac{2K}{T_0} \cdot \sum_a \frac{1}{\Delta_a} \leq \frac{2K}{T_0} \cdot \frac{\sqrt{K}}{\min_{a \neq a^*} \Delta_a} \lesssim \frac{S_\Delta}{T_0} \end{aligned}$$

$$\text{Let } S_\Delta = \sum_{a \neq a^*} \Delta_a$$

$$\Rightarrow \text{Reg}_T \lesssim \frac{T_0}{K} S_\Delta + T \cdot \frac{S_\Delta}{T_0} = \mathcal{O}\left(\sqrt{T \cdot S_\Delta}\right) \text{ by } T_0 = \mathcal{O}\left(\sqrt{\frac{KT}{S_\Delta}}\right)$$

② Unknown $\{\Delta_a\}$ $x \cdot e^{-\frac{cmx^2}{2}} \leq \frac{C}{\sqrt{m}}$ for $x \geq 0$

$\uparrow$ $\uparrow$ $\uparrow$
 $\frac{1}{2}$ $\frac{T_0}{K}$ Δ_a^2

$$\Rightarrow \text{Reg}_T \lesssim K \cdot \underline{m} + C \cdot T \cdot \frac{1}{\sqrt{m}}$$

$$= T_0 + \frac{CTK}{2\sqrt{T_0}} + \frac{CTK}{2\sqrt{T_0}}$$

$$= O(K^{2/3} T^{2/3}) \quad \text{when } T_0 \asymp K^{2/3} T^{2/3}$$

In general: suppose we have an algorithm A with sample complexity $1/\epsilon^2 \Rightarrow$ ETC regret $\asymp T^{2/3}$.

Proof: $\text{Reg}_T = \sum_{a \in [K]} \Delta_a(\underbrace{n_T(a)}_{\text{r.v.}})$

Explore-then-Commit

$$\begin{aligned} \underline{n_T(a)} &= \frac{T_0}{K} + (T - T_0) \cdot \underline{\Pr(\hat{a} = a)} \\ &\leq \frac{T_0}{K} + (T - T_0) \cdot \Pr(\hat{\mu}_{T_0}(a) \geq \hat{\mu}_{T_0}(a^*)) \end{aligned}$$

$\hat{a} = \underset{a \in [K]}{\text{argmax}} \left[\hat{\mu}_{T_0}(a) \right]$

Claim: when $\left[\hat{\mu}_{T_0}(a) \geq \hat{\mu}_{T_0}(a^*) \right]$ then one of the following

two events must happen

$$\left\{ \begin{array}{l} \hat{\mu}_{T_0}(a) \geq (\mu(a) + \mu(a^*)) / 2 \\ \hat{\mu}_{T_0}(a^*) \leq (\mu(a) + \mu(a^*)) / 2 \end{array} \right.$$

$a^* = \arg \max_{a \in [k]} \mu(a)$

Proof: (Contradiction) $\hat{\mu}_{T_0}(a) < \frac{\mu(a) + \mu(a^*)}{2} < \hat{\mu}_{T_0}(a^*) \quad \square$

By Union Bound,

$$n_T(a) \leq \frac{T_0}{K} + (T - T_0) \cdot \left(\Pr \left(\hat{\mu}_{T_0}(a) \geq \frac{\mu(a) + \mu(a^*)}{2} \right) + \Pr \left(\hat{\mu}_{T_0}(a^*) \leq \frac{\mu(a) + \mu(a^*)}{2} \right) \right)$$

Hoeffding: $X_i \in [0, 1], \bar{X} = \frac{1}{m} \sum_{i=1}^m X_i$

$$\Pr(\bar{X} - \mathbb{E}[\bar{X}] \geq \varepsilon) \leq \exp(-2m\varepsilon^2)$$

$$\Pr(\bar{X} - \mathbb{E}[\bar{X}] \leq -\varepsilon) \leq \exp(-2m\varepsilon^2)$$

$$\frac{\mu(a) + \mu(a^*)}{2}$$

$$= \mu(a) + \frac{\mu(a^*) - \mu(a)}{2}$$

$$= \mu(a) + \frac{\Delta_a}{2}$$

$$\Pr \left(\hat{\mu}_{T_0}(a) \geq \frac{\mu(a) + \mu(a^*)}{2} \right) = \Pr \left(\hat{\mu}_{T_0}(a) - \mu(a) \geq \frac{\Delta_a}{2} \right) \\ \leq \exp \left(-2 \frac{T_0}{K} \cdot \left(\frac{\Delta_a}{2} \right)^2 \right)$$

Similarly, $\Pr \left(\hat{\mu}_{T_0}(a^*) \leq \frac{\mu(a) + \mu(a^*)}{2} \right) \leq \exp \left(- \frac{T_0 \Delta_a^2}{2K} \right)$

$$\Rightarrow \text{Reg}_T = \sum_{a \in [k]} \Delta_a n_T(a) \leq \sum_{a \in [k]} \Delta_a \left(\frac{T_0}{K} + 2T \cdot \exp \left(- \frac{T_0 \Delta_a^2}{2K} \right) \right) \quad \square$$

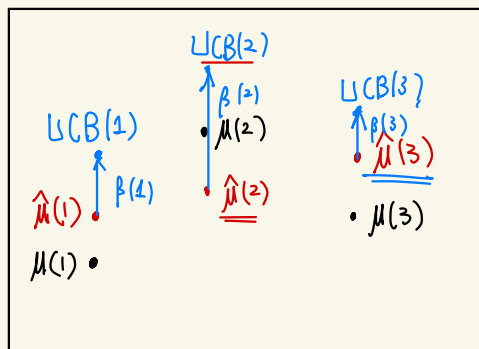
Recall: Explore-then-Commit algorithm for stochastic MAB

→ Need strategic exploration!

A UCB (Upper Confidence Bound):

A with high probability,

$$\underline{\mu(a)} \leq \underline{UCB(a)} = \hat{\mu}(a) + \beta(a)$$



empirical mean
↓
exploitation

bonus term (uncertainty)
↓
exploration

trade-off

$$\star a_t = \underset{a \in [K]}{\operatorname{argmax}} UCB_{t-1}(a)$$

LCB in offline RL
LCB in online RL

Principle: Optimism in Face of Uncertainty (OFU)

→ encourages exploration

Empirical average: $\hat{\mu}_t(a) = \frac{1}{n_t(a)} \cdot \sum_{s=1}^t \mathbb{1}\{a_s = a\} \cdot \underline{\mu(a_s)}$

$$\mathbb{E}[\hat{\mu}_t(a)] = \mu(a) \rightarrow \text{concentration}$$

Lemma (Estimation Error / Confidence Set)

with probability $\geq 1 - \frac{2K}{T}$: $|\mu(a) - \hat{\mu}_t(a)| \leq \sqrt{\frac{\log T}{n_t(a)}} \quad \forall a, t.$

Proof: $P_T(|\bar{X} - E[\bar{X}]| \geq \varepsilon) \leq 2 \exp(-2m\varepsilon^2) \quad x_i \in [0,1]$

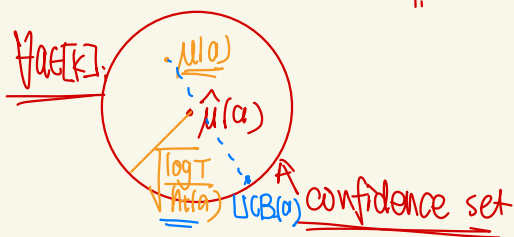
$\Leftrightarrow \text{w.p.} \geq 1-\delta: |\bar{X} - E[\bar{X}]| \leq \sqrt{\frac{1}{2m} \cdot \log \frac{2}{\delta}}$

Here: $|\hat{\mu}_t(a) - \mu(a)| \leq \sqrt{\frac{1}{2n_t(a)} \cdot \log \frac{2}{\delta}} \quad \text{w.p.} \geq 1-\delta$

Union Bound: w.p. $\geq 1-\delta: \forall a \in [K], t \in [T],$

$|\hat{\mu}_t(a) - \mu(a)| \leq \sqrt{\frac{1}{2n_t(a)} \cdot \log \frac{2KT}{\delta}} \quad \delta = \frac{2K}{T}$

$\Rightarrow \text{w.p.} \geq 1 - \frac{2K}{T}: \forall a, t: |\hat{\mu}_t(a) - \mu(a)| \leq \sqrt{\frac{\log T}{n_t(a)}} \quad \square$



$\mu(a) \leq \hat{\mu}_t(a) + \sqrt{\frac{\log T}{n_t(a)}} = \text{UCB}_t(a)$

bonus $\beta_t(a)$

Theorem (Regret Bound of UCB) Suppose $r_t(a) \in [0,1]$

w.p. $\geq 1 - \frac{2K}{T}: \text{Reg}_T \leq \sum_{a: \Delta_a > 0} \frac{4 \log T}{\Delta_a} + \Delta_a = O\left(\sum_{a: \Delta_a > 0} \frac{\log T}{\Delta_a}\right)$

Proof: $\text{Reg}_T = \sum_{a \in [K]} \Delta_a \cdot \pi_T(a)$

At round t : $a_t = \arg \max_{a \in [K]} \text{UCB}_{t-1}(a)$

w.p. $\geq 1 - \frac{2K}{T}$ $\Delta_{a_t} = \mu(a^*) - \mu(a_t) \leq \underline{\text{UCB}_{t-1}(\underline{a^*})} - \mu(a_t)$

$$\begin{aligned} & \leq \text{UCB}_{t-1}(a_t) - \mu(a_t) \\ | \text{UCB}_{t-1}(a_t) - \hat{\mu}_{t-1}(a_t) | & \leq 2 \cdot \frac{\sqrt{\log(t-1)}}{\sqrt{n_{t-1}(a_t)}} \leq 2 \sqrt{\frac{\log T}{n_{t-1}(a_t)}} \\ | \mu(a_t) - \hat{\mu}_{t-1}(a_t) | & \end{aligned}$$

Let t be the last time arm a is selected.

$$\Delta_a \leq 2 \sqrt{\frac{\log T}{n_{t-1}(a)}} = 2 \sqrt{\frac{\log T}{n_T(a) - 1}} \quad \text{independent to } t$$

$$\Rightarrow n_T(a) \leq 1 + \frac{4 \log T}{\Delta_a^2}$$

$$\Rightarrow \text{Reg}_T = \sum_{a \in [K]} \Delta_a n_T(a) \leq \sum_{a: \Delta_a > 0} \Delta_a \left(4 \frac{\log T}{\Delta_a^2} + 1 \right) \quad \square$$

$$\text{Reg}_T = O\left(\sum_{a: \Delta_a > 0} \frac{\log T}{\Delta_a}\right)$$

$\Delta_a \downarrow : \text{Reg}_T \uparrow$ (BAI : it's hard to distinguish the optimal arm from the suboptimal one)

Regret Minimization : always pull arm $a \rightarrow \text{Reg}_T = \underline{\Delta_a T}$

$$\Rightarrow \text{Reg}_T \leq \min \left\{ \sum_{a: \Delta_a > 0} \frac{4 \log T}{\Delta_a} + \Delta_a, \max_{a \in [K]} \underline{\Delta_a T} \right\}$$

Distribution-free Bound:

$$\eta_T(a) \leq 1 + \frac{4 \log T}{\Delta_a^2}$$

$$\text{Reg}_T = \sum_{a \in [K]} A_a \cdot \eta_T(a) = \sum_{a: A_a \leq \Delta} A_a \cdot \eta_T(a) + \sum_{a: A_a > \Delta} A_a \cdot \eta_T(a)$$

splitting
threshold TBD

$$\leq T \Delta + \sum_{a: A_a > \Delta} A_a \left(1 + \frac{4 \log T}{\Delta_a^2} \right)$$

$$= T \Delta + \sum_{a: A_a > \Delta} \left(\frac{4 \log T}{\Delta_a} + A_a \right)$$

$$\leq T \Delta + \frac{4 K \log T}{\Delta} + \sum_{a \in [K]} A_a$$

$$\text{By } \Delta = 2 \sqrt{(K \log T) / T}$$

$$\Rightarrow \text{Reg}_T \leq 2 \sqrt{T K \log T} + \sum_{a \in [K]} A_a = O(\sqrt{T K \log k})$$

$\leq K$
same as Exp 3

Claim: UCB algorithm is

nearly minimax optimal (up to some logarithmic factor)

Theorem (Lower Bound of MAB) For any algorithm A .

there exists a sequence of stochastic rewards such that

$$\inf_A \sup_{D_1, \dots, D_K} \mathbb{E} [\text{Reg}_T] = \Omega(\sqrt{TK})$$

- Remark:
1. UCB algorithm can achieve minimax optimal. (Bandits Ch 9)
 2. There exists high-probability lower bound. (Bandits Ch 17)

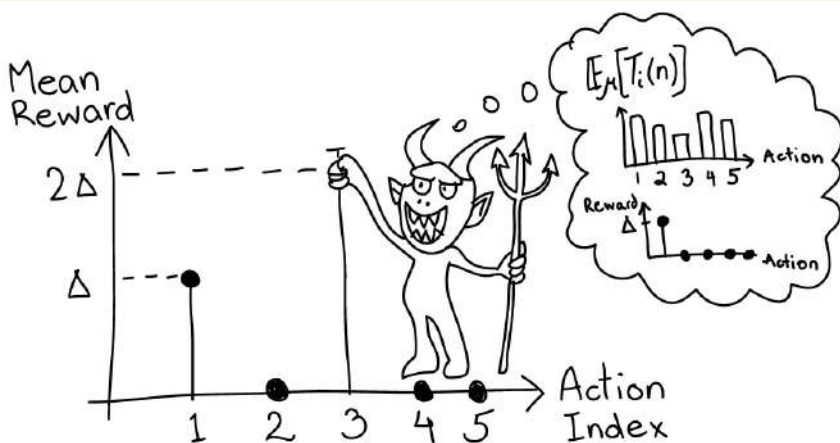


Figure 15.1 The idea of the minimax lower bound. Given a policy and one environment, the evil antagonist picks another environment so that the policy will suffer a large regret in at least one environment.

c.f. On Lower Bounds for Regret in Reinforcement Learning.

$$\inf_{\underline{A}} \sup_{\underline{D}_1, \dots, \underline{D}_K} \underline{E} [\text{Reg}_T] \geq \underline{\Omega}(\sqrt{KT})$$

$$\text{Reg}_T = T \cdot \underline{\mu(a^*)} - \sum_{t=1}^T \underline{\mu(a_t)}$$

Adversarial
MAB

↓ reduction ★

Stochastic
MAB

Intuition : T iterations : $\exists a \in [K] \text{ s.t. } \underline{n_T(a)} \leq \frac{T}{K}$

Concentration, estimation error $O(\frac{1}{\sqrt{n}})$

hard to distinguish:

$$\text{Ber}(\frac{1}{2}) \xrightarrow{(K-1)} \text{Ber}(\frac{1}{2} + \sqrt{\frac{K}{T}}) \rightarrow T \cdot \frac{K-1}{K} \cdot \sqrt{\frac{K}{T}} \asymp \sqrt{KT} \text{ Regret.}$$

Classic world

$$\underline{\underline{\Gamma_t(a)}} \begin{cases} \sim \underline{\underline{\text{Ber}(\delta)}} & a \neq a^* \\ \sim \underline{\underline{\text{Ber}(\delta+\varepsilon)}} & a = a^* \end{cases}$$

$\varepsilon > 0$ small

Auxiliary world:

uninformed agent

$$\underline{\underline{\tilde{\Gamma}_t(a)}} \begin{cases} \sim \underline{\underline{\text{Ber}(\delta)}} & a \neq a^* \\ \sim \underline{\underline{\text{Ber}(\delta)}} & a = a^* \end{cases}$$

< Classic world: $\underline{\underline{a_t}} \sim \underline{\underline{\Pi_t}}(\underline{\underline{H_t}})$ policy

$r_t = r(a_t)$

$$\underline{\underline{H_t}} = \{ \underline{\underline{a_1}}, \underline{\underline{\Gamma_1}}, \underline{\underline{a_2}}, \dots, \underline{\underline{a_{t+1}}}, \underline{\underline{\Gamma_{t+1}}} \} \quad \underline{\underline{\Lambda_t(a)}}$$

Auxiliary world:

$$\underline{\underline{\tilde{a}_t}} \sim \underline{\underline{\Pi_t}}(\underline{\underline{\tilde{H}_t}})$$

$$\underline{\underline{\tilde{H}_t}} = \{ \underline{\underline{\tilde{a}_1}}, \underline{\underline{\tilde{\Gamma}_1}}, \underline{\underline{\tilde{a}_2}}, \dots, \underline{\underline{\tilde{a}_{t+1}}}, \underline{\underline{\tilde{\Gamma}_{t+1}}} \} \quad \underline{\underline{\tilde{\Lambda}_t(a)}}$$

$$\underline{\underline{E[\text{Reg}_T]}} = \underline{\underline{E[T \mu(a^*) - \sum_{t=1}^T \mu(a_t)]}}$$

$$= E \left[\sum_{a \in [K]} \underline{\underline{\Delta_a}} \cdot \underline{\underline{n_T(a)}} \right]$$

$$\Delta_a = \mu(a^*) - \mu(a) = \delta + \varepsilon - \delta = \varepsilon$$

$$= E \left[\sum_{a \neq a^*} \underline{\underline{\varepsilon}} \cdot \underline{\underline{n_T(a)}} \right]$$

$$= \varepsilon \cdot E \left[T - \underline{\underline{n_T(a^*)}} \right]$$

$$\sum_{a \in [K]} n_T(a) = T$$

$$= \varepsilon \cdot E \left[T - \underline{\underline{\tilde{n}_T(a^*)}} \right] + E \left[\underline{\underline{\tilde{n}_T(a^*)}} - n_T(a^*) \right]$$

$$= \varepsilon T \left(1 - \frac{1}{K} \right) + \underline{\underline{E[\tilde{n}_T(a^*) - n_T(a^*)]}}$$

purely uniform

$$\underline{\underline{\tilde{\Pi}_T(a)}} = \frac{T}{K}$$

(in general, pigeonhole)

$$\forall f: \Omega \rightarrow [0,1], \quad |E_P[f] - E_{P'}[f]|$$

$$\leq \sup_{g: \Omega \rightarrow [0,1]} |E_P[g] - E_{P'}[g]| = \|P - P'\|_{TV}$$

$$f = \frac{\eta_T(a)}{T} \in [0, 1]$$

Canonical Bandit Model

Classic world $(P;)$

randomness from $(r_1, \dots, r_T) := r_1^T$

at timestep t :

$$Pr\left(\underbrace{r_t^T}_{\text{future reward}} = \underbrace{z_t^T}_{\text{history}} \mid \mathcal{H}_t\right)$$

Auxiliary world $\tilde{P}$: $Pr(\tilde{r}_t^T = z_t^T \mid \tilde{\mathcal{H}}_t)$

$$T \cdot \left| E_P \left[\frac{\eta_T(a^*)}{T} \right] - E_{\tilde{P}} \left[\frac{\eta_T(a^*)}{T} \right] \right|$$

$$\leq T \cdot \|P - \tilde{P}\|_{TV}$$

$$\leq T \cdot \sqrt{\frac{1}{2} KL(\tilde{P} \| P)}$$

$$\Rightarrow \underline{E[\text{Reg}_T]} \geq \varepsilon T \left(1 - \frac{1}{K} - \sqrt{\frac{1}{2} KL(\tilde{P} \| P)} \right)$$

Goal: Bound $\underline{KL}(\tilde{P} \| P)$ ($KL(\tilde{P}(\underline{z}_1^T) \| P(z_1^T))$)

$\uparrow$
total variation distance

$$① \|P - P'\|_{TV} = \sup_A |P(A) - P'(A)|$$

$$② \|P - P'\|_{TV} = \frac{1}{2} \int_X (p(x) - p'(x)) dx = \frac{1}{2} \|P - P'\|_1$$

③ Pinsker's inequality:

$$\|P - P'\|_{TV} \leq \sqrt{\frac{1}{2} KL(P \| P')}$$

$$> \underline{P} \& \hat{P}$$

$$\begin{aligned} KL(P \| P') &= \sum_x p(x) \cdot \log \frac{p(x)}{p'(x)} \end{aligned}$$

Claim: Chain rule of KL divergence

$$KL(\tilde{P}(\mathbf{Z}^T) \parallel P(\mathbf{Z}^T)) = \sum_{t=1}^T KL(\tilde{P}(z_t | \mathcal{H}_t) \parallel P(z_t | \mathcal{H}_t))$$

Proof: $P(\mathbf{Z}^T) = P(z_1, \dots, z_T)$

$$= P(z_1) \cdot P(z_2 | \mathcal{H}_2) \cdot \dots \cdot P(z_T | \mathcal{H}_T)$$

$$\tilde{P}(\mathbf{Z}^T) = \tilde{P}(z_1) \cdot \tilde{P}(z_2 | \mathcal{H}_2) \cdot \dots \cdot \tilde{P}(z_T | \mathcal{H}_T)$$

$$\Rightarrow KL(\tilde{P} \parallel P) = \sum_{\mathbf{Z}^T} \tilde{P}(\mathbf{Z}^T) \cdot \log \frac{\tilde{P}(\mathbf{Z}^T)}{P(\mathbf{Z}^T)}$$

$$= \sum_{t=1}^T \sum_{\mathbf{Z}^T} \tilde{P}(\mathbf{Z}^T) \cdot \log \frac{\tilde{P}(z_t | \mathcal{H}_t)}{P(z_t | \mathcal{H}_t)}$$

□

$\mathbf{Z}_{1:t-1}, \mathbf{Z}_{t+1:T}$ marginalize out

$$\Rightarrow \underline{KL(\tilde{P} \parallel P)} = \sum_{t=1}^T KL(\tilde{P}(z_t | \mathcal{H}_t) \parallel P(z_t | \mathcal{H}_t))$$

reward distribution

when $\tilde{a}_t = a^*$: $\tilde{P}(z_t | \mathcal{H}_t) \neq P(z_t | \mathcal{H}_t)$ at timestep t (aux & classic world)

when $\tilde{a}_t = a^*$: $\tilde{P}(z_t | \mathcal{H}_t) = P(z_t | \mathcal{H}_t) \Rightarrow KL(\cdot \parallel \cdot) = 0$

$$= \sum_{t=1}^T \underbrace{Pr(\tilde{a}_t = a^*)}_{\text{auxiliary world}} \cdot KL(\underbrace{Ber(\delta)}_{\text{auxiliary world}} \parallel \underbrace{Ber(\delta + \epsilon)}_{\text{auxiliary world}})$$

$$\begin{aligned}
&= \sum_{t=1}^T \cdot \frac{1}{K} \cdot \left(\delta \log \frac{\delta}{\delta + \varepsilon} + (1-\delta) \log \frac{1-\delta}{1-\delta-\varepsilon} \right) \\
&= \frac{T}{K} \cdot \left(\delta \log \frac{\delta}{\delta + \varepsilon} + (1-\delta) \cdot \log \frac{1-\delta}{1-\delta-\varepsilon} \right)
\end{aligned}$$

Proposition : (bound of KL divergence) $\forall 0 \leq \delta \leq \frac{1}{2}, \varepsilon \leq 1-2\delta$.

$$\delta \log \frac{\delta}{\delta + \varepsilon} + (1-\delta) \cdot \log \frac{1-\delta}{1-\delta-\varepsilon} \leq \frac{\varepsilon^2}{\delta}$$

$$\Rightarrow \mathbb{E}[\text{Reg}_T] \geq \varepsilon T \cdot \left(1 - \frac{1}{K} - \sqrt{\frac{\varepsilon^2}{2\delta} \cdot \frac{T}{K}} \right)$$

$$\geq \sqrt{\frac{\delta K}{8T}} \cdot T \cdot \left(1 - \frac{1}{K} - \frac{1}{4} \right) \quad \varepsilon^2 = \frac{\delta K}{8T}$$

$$\text{for } K \geq 2 \geq \frac{1}{12} \sqrt{\delta K T} = \frac{1}{24} \sqrt{K T} \quad \delta = \frac{1}{4}$$

Martingale Concentration $X_1, \dots, X_n$ $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ $\mu = \mathbb{E}[X_i]$

Recall : (Hoeffding) $\Pr(|\bar{X} - \mu| \geq \varepsilon) \leq 2 \exp\left(-\frac{2n^2 \varepsilon^2}{\sum_{i=1}^n \underline{C_i^2}}\right)$

(Bernstein) $\Pr(|\bar{X} - \mu| \geq \varepsilon) \leq 2 \exp\left(-\frac{n \varepsilon^2 / 2}{\underline{\sigma^2 + b \varepsilon / 3}}\right)$

independence $\Rightarrow$ inter-dependency

Martingale (fair gambling game)

def(1) A stochastic process $\{\underline{X}_n\}_{n \geq 0}$ is a martingale

$$\text{if } \forall n \geq 0: \mathbb{E}[\underline{X}_{n+1} | \underline{X}_0, \dots, \underline{X}_n] = \underline{X}_n$$

current
history up to now

(2) A stochastic process $\{\underline{Z}_n\}_{n \geq 0}$ is a martingale

wrt. $\{\underline{X}_n\}_{n \geq 0}$ if $\forall n \geq 0: \mathbb{E}[\underline{Z}_{n+1} | \underline{X}_0, \dots, \underline{X}_n] = \underline{Z}_n$

(3) Given a sequence $\{\underline{X}_n\}_{n \geq 0}$: $\underline{F}_n = \sigma(\underline{X}_0, \dots, \underline{X}_n)$

$$\underline{F}_0 \subseteq \underline{F}_1 \subseteq \underline{F}_2 \dots \subseteq \underline{F}_n \subseteq \underline{F}_{n+1} \subseteq \dots$$

minimum σ -algebra

All information up to round n

We say $\{\underline{F}_n\}_{n \geq 0}$ is a filtration

A stochastic process $\{\underline{Z}_n\}_{n \geq 0}$ is a martingale

wrt. $\{\underline{F}_n\}_{n \geq 0}$ if

① $\forall n \geq 0$. $\underline{Z}_n$ is $\underline{F}_n$ -measurable

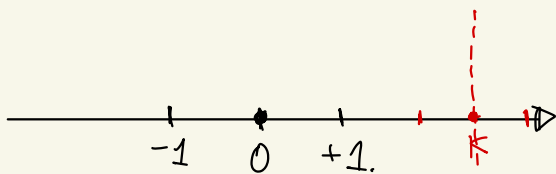
② (martingale condition)

$$\mathbb{E}[\underline{Z}_{n+1} | \underline{F}_n] = \underline{Z}_n$$



Similarly, $\{Z_n\}_{n \geq 0}$ $\begin{cases} \text{super-martingale: } \underline{E[Z_{n+1} | \mathcal{F}_n]} \leq Z_n \\ \text{sub-martingale: } \underline{E[Z_{n+1} | \mathcal{F}_n]} \geq Z_n \end{cases}$

e.g. (1-d random walk)



$$X_n \in \{-1, +1\} \cdot \left(\frac{X_{n+1}}{2} \sim \text{Ber}\left(\frac{1}{2}\right) \right) \quad \begin{matrix} \xleftarrow{\text{wp } \frac{1}{2}} \\ \xrightarrow{\text{wp } \frac{1}{2}} \end{matrix}$$

$Z_n = \sum_{i=1}^n X_i$ is a martingale, since

① Z_n is $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ -measurable

$$\textcircled{2} \ E[Z_{n+1} | \mathcal{F}_n] = E[\underline{Z_n} + X_{n+1} | \mathcal{F}_n]$$

$$= Z_n + \underbrace{E[X_{n+1} | \mathcal{F}_n]}_{=0} = Z_n.$$

Martingale Difference: $E[\underline{Z_{n+1}} - \underline{Z_n} | \mathcal{F}_n] = 0$

Azuma's Inequality: Let $Z_0, Z_1, \dots, Z_n$ be a martingale

such that $Z_i - Z_{i-1} \in [a_i, b_i]$. Let $C_i = b_i - a_i$.

$\rightarrow \sum_{i=1}^n (Z_i - Z_{i-1}) \triangleq \sum_{i=1}^n X_i$

Martingale Difference

$\Rightarrow \Pr(|\underline{Z_n} - Z_0| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n C_i^2}\right)$

$\Leftrightarrow \text{w.p. } \geq 1 - \delta : |Z_n - Z_0| \leq \sqrt{2 \cdot \left(\sum_{i=1}^n C_i^2\right) \cdot \log \frac{2}{\delta}}$

Proof, (in Hoeffding) $\mathbb{E}[e^{\sum_{i=1}^n \varepsilon_i}] = \mathbb{E}\left[\prod_{i=1}^n e^{\varepsilon_i}\right]$ ✓

independence $\prod_{i=1}^n \mathbb{E}[e^{\varepsilon_i}]$

Here, we need to leverage
Martingale property!

↳ Hoeffding's Lemma

$$\begin{aligned}\mathbb{E}\left[\prod_{i=1}^n e^{\varepsilon_i}\right] &= \mathbb{E}\left[\mathbb{E}\left[\prod_{i=1}^n e^{\varepsilon_i} \mid \underline{x_0, \dots, x_{n-1}}\right]\right] \\ &= \mathbb{E}\left[\left(\prod_{i=1}^{n-1} e^{\varepsilon_i}\right) \cdot \mathbb{E}[e^{\varepsilon_n} \mid x_0, \dots, x_{n-1}]\right]\end{aligned}$$

By Martingale property: $\mathbb{E}[\underbrace{\varepsilon_n}_{\text{martingale difference}} \mid x_0, \dots, x_{n-1}] = 0$

→ Hoeffding's Lemma (conditional expectation version):

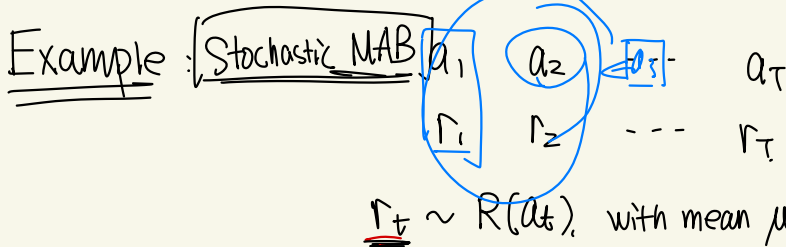
$$\mathbb{E}[e^{\varepsilon_n} \mid x_0, \dots, x_{n-1}] \leq e^{-\varepsilon^2(b-a)^2/8}$$

Freedman's Inequality (Martingale version of Bernstein's)

predictable variance process, $V_T = \sum_{t=1}^T \mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}]$
 $\mathbb{E}$ martingale difference
 $|X_t| \leq b$

Let $S_T = \sum_{t=1}^T X_t$

$\forall \varepsilon > 0$: $\Pr(S_T \geq \varepsilon \text{ and } V_T \leq \sigma^2) \leq \exp\left(-\frac{\varepsilon^2/2}{\sigma^2 + b\varepsilon/3}\right)$



previously: pseudo-regret $\underline{\text{Reg}}_T = T \cdot \mu(a^*) - \sum_{t=1}^T \mu(a_t)$

$\mathcal{F}_t = \sigma(a_1, r_1, a_2, r_2, \dots, a_{t-1}, r_{t-1}, a_t, r_t)$

Regret Decomposition Lemma

Now: bound $S_T = \sum_{t=1}^T r_t - \mu(a_t)$

a martingale ! $\rightarrow$ Azuma's inequality

$\mathbb{E}[S_t - S_{t-1} \mid \mathcal{F}_{t-1}]$

$= \mathbb{E}[r_t - \mu(a_t) \mid \mathcal{F}_{t-1}]$ $\geq$ law of total expectation

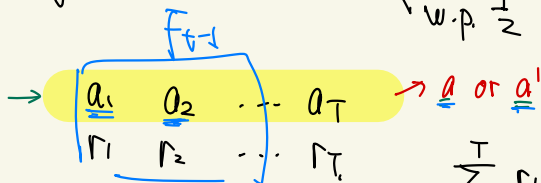
$= \mathbb{E}[\mathbb{E}[r_t - \mu(a_t) \mid \mathcal{F}_{t-1}, a_t]]$

$a_t \mid \mathcal{F}_{t-1}$

$= \mathbb{E}[\mathbb{E}[r_t - \mu(a_t) \mid a_t]] = \mathbb{E}[0] = 0$

MAB Example 1. $\{a, a'\}$

Algorithm: toss a coin: $\begin{cases} \text{w.p. } \frac{1}{2} : a_t = a \text{ for all } t \\ \text{w.p. } \frac{1}{2} : a_t = a' \text{ for all } t \end{cases}$



Goal: need to bound

$\sum_{t=1}^T r_t - \mu(a_t)$ $\checkmark$ $\frac{\mu(a) + \mu(a')}{2}$

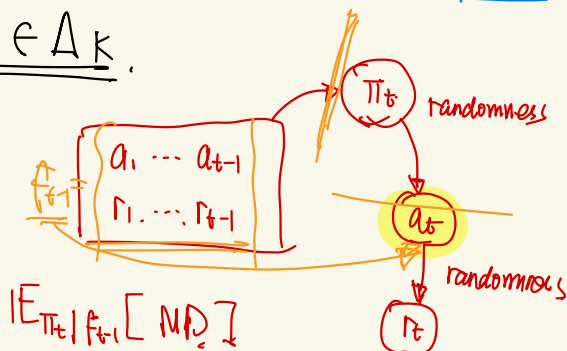
$\sum_{t=1}^T (r_t - \mathbb{E}[r_t])$ $\times$

$$\sum_{t=1}^T r_t - \boxed{\text{not unique!}}$$

MAB Example 2. $\forall t$:

Algorithm: $(a_1, r_1, \dots, a_{t-1}, r_{t-1}) \mapsto \pi_t \in \Delta_K$

$$\left\langle \begin{array}{l} \sum_{t=1}^T r_t - \mu(a_t) \\ \sum_{t=1}^T r_t - \mathbb{E}[R(a)] \\ \quad \quad \quad a \sim \pi_t \end{array} \right\rangle \checkmark$$



concentration?

$$\mathbb{E}[r_t - \mu(a_t) \mid \mathcal{F}_{t-1}] = 0$$

$$\mathbb{E}[r_t - \frac{\mu(a) + \mu(a')}{2} \mid \mathcal{F}_{t-1}] \neq 0 ?$$

conditional expectation!

(Online) Ridge Regression

At round $t = 1, 2, \dots, T$:

nature chooses

$$\boxed{x_t \in \mathbb{R}^d}$$

arbitrary
(adversarial)

reveals a noisy label $y_t \in [0, R]$

$$\text{s.t. } y_t = x_t^T \theta^* + \varepsilon_t$$

ε_t zero-mean noise.

Goal: estimate θ^*

s.t. given any $x \in \mathbb{R}^d$,

we can accurately predict $y(x) = x^T \theta^*$.

$$\underline{y} = \underline{x^T \theta^*} + \varepsilon + \textcircled{\varepsilon_{\text{bias}}}$$

- < noisy ✓
- < noiseless
- < well-specified ✓
- < mis-specified
- < fixed-design ✓
- < random-design

$x_1, \dots, x_n \rightarrow$ design matrix

$$\underline{x} \sim p$$

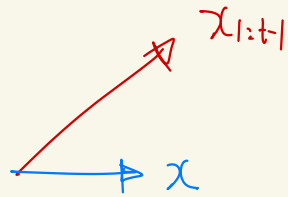
c.f., Analysis of random design ridge regression. Daniel

Assume: $\|x_t\|_2 \leq 1$, $\|\theta^*\|_2 \leq B$.

Design matrix $X = \begin{pmatrix} x_1 & x_2 & \dots & x_T \\ | & | & & | \end{pmatrix} \in \mathbb{R}^{d \times T}$

$\hat{f}_{t-1}$ = $\sigma(x_1, \dots, x_{t-1}, y_1, \dots, y_{t-1})$

↳ based on filtration $\hat{f}_{t-1}$ we choose an arbitrary x_t .



Ridge Regression, at round t.

① $\hat{\theta}_t = \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \underbrace{\sum_{i=1}^{t-1} (x_i^\top \theta - y_i)^2 + \|\theta\|_2^2}_{L_t(\theta)}$ to ensures non-singularity.

② $\hat{y}_t(x) = x^\top \hat{\theta}_t$

Regularized LS

Define covariance matrix $\Lambda_t = \underline{\mathbf{I}} + \sum_{i=1}^{t-1} x_i x_i^\top$.

$\Rightarrow \hat{\theta}_t = \Lambda_t^{-1} \cdot \sum_{i=1}^{t-1} x_i y_i$

$\hat{y}_t(x) = x^\top \Lambda_t^{-1} \sum_{i=1}^{t-1} x_i y_i$

$\nabla L_t(\theta) \big|_{\theta = \hat{\theta}_t} = 0$

"Sample Complexity" : given $\hat{f}_{t-1}$, how accurate can we predict?

Note: error $\not\rightarrow 0$ since nature can choose poor designed

$x_{1:t}$ $\rightarrow$ unidentifiability

Theorem (One-shot prediction of ridge regression)

$\forall t$: w.p. $\geq 1-\delta$: $|\hat{y}_t(x) - y(x)| \leq \|x\|_{\Lambda_t^{-1}} \cdot O\left(R \cdot \sqrt{d + \log \frac{1}{\delta}}\right)$

• Relation to sample size: $\frac{1}{\sqrt{n}}$? $\Rightarrow \Lambda_t = I + \sum_{i=1}^{t-1} x_i x_i^T \sim t$

• $\|x\|_{\Lambda_t^{-1}} = \sqrt{x^T \Lambda_t^{-1} x}$

$\rightarrow \text{span}\{x_1, \dots, x_{t-1}\}$

if $x \in \text{span}\{x_1, \dots, x_{t-1}\}$: good prediction.

if $x \notin \text{span}\{x_1, \dots, x_{t-1}\}$: bad prediction.

"effective sample size" = how many data points align with x .

Proposition: w.p. $\geq 1-\delta$: $\left\| \sum_{i=1}^{t-1} x_i \underbrace{(\varepsilon_i)}_{\text{residual}} \right\|_{\Lambda_t^{-1}} \leq O\left(R \cdot \sqrt{d + \log \frac{1}{\delta}}\right)$

Self-normalized concentration for vector-valued martingale.

Proof of Theorem: $\underbrace{|\hat{y}_t(x) - y(x)|}_{\text{best linear predictor}} \quad y = x^\top \theta^* + \varepsilon$
 $\boxed{y(x) = x^\top \theta^*}$

$$= \left| x^\top \Lambda_t^{-1} \sum_{i=1}^{t-1} x_i \underbrace{y_i}_{\text{noisy label}} - x^\top \theta^* \right|$$

$$\leq \left| x^\top \Lambda_t^{-1} \sum_{i=1}^{t-1} x_i (y_i - x_i^\top \theta^*) \right| \quad (1)$$

$$+ \left| \underbrace{x^\top \Lambda_t^{-1} \sum_{i=1}^{t-1} x_i (x_i^\top \theta^*)}_{\text{noiseless label}} - \underbrace{x^\top \theta^*}_{\text{noiseless label}} \right| \quad (2)$$

prediction as if we see noiseless label

$$(1) \left| x^\top \Lambda_t^{-1} \sum_{i=1}^{t-1} x_i (y_i - x_i^\top \theta^*) \right|$$

$$= \left| x^\top \Lambda_t^{-\frac{1}{2}} \underbrace{\Lambda_t^{-\frac{1}{2}} \sum_{i=1}^{t-1} x_i \varepsilon_i}_{\text{self-normalized concentration}} \right|$$

$$\leq \|x^\top \Lambda_t^{-\frac{1}{2}}\|_2 \cdot \underbrace{\left\| \Lambda_t^{-\frac{1}{2}} \sum_{i=1}^{t-1} x_i \varepsilon_i \right\|_2}_{\text{self-normalized concentration}}$$

$$= \|x\|_{\Lambda_t^{-1}} \cdot \underbrace{\left\| \sum_{i=1}^{t-1} x_i \varepsilon_i \right\|_{\Lambda_t^{-1}}}_{\text{self-normalized concentration}}$$

$$\leq \|x\|_{\Lambda_t^{-1}} \cdot \underbrace{O\left(R \sqrt{d + \log \frac{1}{\delta}}\right)}_{\text{self-normalized concentration}}$$

$$(2) \left| x^\top \Lambda_t^{-1} \sum_{i=1}^{t-1} x_i (x_i^\top \theta^*) - x^\top \theta^* \right|$$

$$= \left| x^\top \Lambda_t^{-\frac{1}{2}} \underbrace{\Lambda_t^{-\frac{1}{2}} \left(\sum_{i=1}^{t-1} x_i x_i^\top \right) \theta^*}_{\Lambda_t - I} - x^\top \Lambda_t^{-\frac{1}{2}} \underbrace{\Lambda_t^{-\frac{1}{2}} \Lambda_t \theta^*}_{\Lambda_t - I} \right|$$

$$= |x^T \Lambda_t^{-\frac{1}{2}} \Lambda_t^{-\frac{1}{2}} I \cdot \theta^*| \leq \|x\|_{\Lambda_t^{-1}} \cdot \|\theta^*\|_{\Lambda_t^{-1}}$$

$$\left[\Lambda_t = I + \sum_{i=1}^{t-1} x_i x_i^T \succeq I \right]$$

$$\Rightarrow \Lambda_t^{-1/2} \succeq I$$

$$\leq \|x\|_{\Lambda_t^{-1}} \cdot \underbrace{\|\theta^*\|_2}_{\text{constant}} \leq \underbrace{(B \cdot)}_{\text{constant}} \|x\|_{\Lambda_t^{-1}}$$

$$\Rightarrow |\hat{y}_t(x) - y(x)| \leq \|x\|_{\Lambda_t^{-1}} \cdot \mathcal{O}\left(R \sqrt{d + \log \frac{1}{\delta}}\right)$$

Regret Analysis of Online Ridge Regression:

$$\text{Reg}_T = \sum_{t=1}^T |\hat{y}_t(x_t) - y(x_t)|$$

$$T \uparrow (T > \underline{d})$$

$$\leq \sum_{t=1}^T \|x_t\|_{\Lambda_t^{-1}} \cdot \mathcal{O}\left(R \sqrt{d + \log \frac{1}{\delta}}\right)$$

$$\text{Reg}_T = \mathcal{O}(\sqrt{T \log T})$$

$$= \tilde{\mathcal{O}}(\sqrt{T})$$

$$\leq \mathcal{O}\left(R \sqrt{d + \log \frac{1}{\delta}}\right) \cdot \sqrt{T} \cdot \sqrt{\sum_{t=1}^T x_t^T \Lambda_t^{-1} x_t}$$

Lemma (Elliptical Potential Lemma)

$$\Lambda_t = I + \sum_{i=1}^{t-1} x_i x_i^T$$

$$\sum_{t=1}^T \underline{x_t^T \Lambda_t^{-1} x_t} \leq \underbrace{2 \log \det(\Lambda_{T+1})}_{\text{volume}} \leq 2d \log(T+1)$$

Proof: $\boxed{Z \leq 2 \log(HZ) \text{ for } Z \in [0,1]}$ $\Sigma_t = \{ \theta : \|\theta\|_{\Lambda_t}^2 \leq 1 \}$

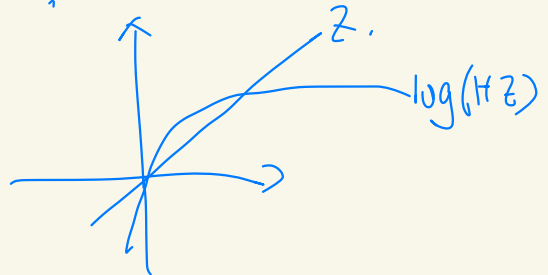
is Ellipsoid

$$\Lambda_t = I + \sum_{i=1}^{t-1} x_i x_i^T \quad (\Lambda_0 = I)$$

$$\Rightarrow \Lambda_t^{-1} \preceq I,$$

$$\Rightarrow x_t^T \Lambda_t^{-1} x_t \leq 1.$$

$$\Rightarrow x_t^T \Lambda_t^{-1} x_t \leq \underline{2 \log(1 + x_t^T \Lambda_t^{-1} x_t)}$$



$$\det(\Lambda_{t+1}) = \det(\Lambda_t + x_t x_t^T)$$

$$= \det(\Lambda_t^{1/2} (I + \Lambda_t^{-1/2} x_t x_t^T \Lambda_t^{-1/2}) \Lambda_t^{1/2})$$

$$\det(AB) = \det(A) \det(B)$$

$$= \det(\Lambda_t) \cdot \det(I + \Lambda_t^{-1/2} x_t x_t^T \Lambda_t^{-1/2})$$

$$= \det(\Lambda_t) \cdot (1 + x_t^T \Lambda_t^{-1} x_t)$$

$$\Rightarrow 1 + x_t^T \Lambda_t^{-1} x_t = \frac{\det(\Lambda_{t+1})}{\det(\Lambda_t)}$$

$A \in \mathbb{R}^{m \times n}$

$$\det(I_m + AA^T) = \det(I_n + A^T A)$$

$$\det(I + vv^T)$$

$$= \det(I + V^T V)$$

$$= 1 + \|v\|_2^2$$

$$\det(A) = \prod_{i=1}^d \sigma_i(A)$$

$$\Rightarrow \sum_{t=1}^T x_t^T \Lambda_t^{-1} x_t$$

$$\leq 2 \sum_{t=1}^T \log(1 + x_t^T \Lambda_t^{-1} x_t)$$

$$\leq 2 \sum_{t=1}^T \log \frac{\det(\Lambda_{t+1})}{\det(\Lambda_t)}$$

$$\xrightarrow{\text{telescoping}} 2 \cdot \log \frac{\det(\Lambda_{T+1})}{\det(\Lambda_0)}$$

$$\Lambda_0 = I \Rightarrow \det(\Lambda_0) = 1.$$

$$\det(\Lambda_{T+1}) = \prod_{i=1}^d \sigma_i(\Lambda_{T+1})$$

$$\leq \left(\sigma_{\max}(\Lambda_{T+1}) \right)^d$$

$$= \left(\max_{u: \|u\|_2=1} u^T \left(\underbrace{I}_{\leq 1} + \underbrace{\sum_{t=1}^T x_t x_t^T}_{\leq T} \right) u \right)^d \leq (T+1)^d.$$

$$\Rightarrow \sum_{t=1}^T x_t^T \Lambda_t^{-1} x_t \leq 2 \log \det(\Lambda_{T+1})$$

$$\leq 2d \log(T+1). \quad \square$$

$$(I + VV^T) \cdot \underbrace{V}_{\text{eigenvektor}} = \underbrace{(1 + \|V\|_2^2)}_{(1 + \|V\|_2^2)} \cdot V.$$

$$\left(\underline{I + VV^T} \right) \cdot \underbrace{V^\perp}_{\text{eigenvektor}} = V^\perp \quad (1)$$

Self-Normalized Concentration for Vector-Valued Martingales

[Abbas-Tadkori, 2011]

Recall: w.p. $\geq 1 - \delta$: $\| \sum_{i=1}^{t-1} \underline{x}_i \underline{\varepsilon}_i \|_{\underline{\Lambda}_t^{-1}} = O\left(R \cdot \sqrt{d + \log \frac{1}{\delta}} \right)$

$$\underline{\Lambda}_t = \mathbf{I} + \sum_{i=1}^{t-1} \underline{x}_i \underline{x}_i^T$$

$$\underline{\varepsilon}_i = \underline{y}_i - \underline{w}_i^T \underline{x}_i$$

$$\underline{y}_i \in [0, R]$$

Theorem

$\{ \underline{F}_t \}_{t \geq 0}$ filtration

$$\underline{F}_t = \sigma(\underline{X}_1, \dots, \underline{X}_{t+1}, \eta_1, \dots, \eta_t)$$

$\{ \eta_t \}_{t \geq 1}$ $\left\{ \begin{array}{l} \underline{F}_t \text{-measurable} \\ R\text{-sub-Gaussian (conditionally)} \end{array} \right.$

$\in \mathbb{R}^d$

$$\forall \lambda \geq 0: \mathbb{E}[e^{\lambda \eta_t} | \underline{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 R^2}{2}\right)$$

$\{ \underline{X}_t \}_{t \geq 1}$: $\underline{F}_{t-1}$ -measurable.

Let $\underline{V}_t = \mathbf{V} + \sum_{s=1}^t \underline{X}_s \underline{X}_s^T$. $\underline{S}_t = \sum_{s=1}^t \eta_s \underline{X}_s$. ($\mathbf{V}$ is PD matrix in $\mathbb{R}^{d \times d}$)

$\Rightarrow \forall \delta > 0$. with probability $\geq 1 - \delta$, $\forall t \geq 0$.

$$\| \underline{S}_t \|_{\underline{V}_t^{-1}}^2 \leq 2R^2 \cdot \log\left(\frac{\det(\underline{V}_t)^{1/2} \cdot \det(\mathbf{V})^{-1/2}}{\delta}\right)$$

def (stopping time) $\{ \underline{X}_t \}_{t \geq 0}$

$$\log \frac{\det(\underline{V}_t)}{\det(\mathbf{V})} = O(d \log t)$$

$\downarrow$
R.V. $\underline{T}$ is a stopping time w.r.t. $\{ \underline{X}_t \}_{t \geq 0}$.

s.t. $\{ \underline{T} \leq t \} \subseteq \underline{F}_t$ (whether stopping at time t is determined by $\underline{X}_0, \dots, \underline{X}_t$)

OST (Optional Stopping Theorem)

$$\mathcal{F}_t = \sigma(X_0, \dots, X_t)$$

Let $\{Y_t\}$ is a martingale w.r.t. $\{\mathcal{F}_t\}$; τ is a stopping time w.r.t. $\{\mathcal{F}_t\}$

$$\underline{E[Y_\tau]} = E[Y_0]$$

If $\tau < \infty$ a.s.

< sub-martingale: $E[Y_{t+1} | \mathcal{F}_t] \geq Y_t \Rightarrow E[Y_\tau] \geq E[Y_0]$

super-martingale: $E[Y_{t+1} | \mathcal{F}_t] \leq Y_t \Rightarrow E[Y_\tau] \leq E[Y_0]$

Lemma

Let τ be a stopping time w.r.t. $\{\mathcal{F}_t\}_{t \geq 0}$

$$\forall \delta > 0 \text{ w.p. } \geq 1 - \delta: \underline{\|S_\tau\|_{\bar{V}_\tau^{-1}}^2} \leq 2R^2 \cdot \log\left(\frac{\det(\bar{V}_\tau)^{1/2} \cdot \det(V)^{-1/2}}{\delta}\right)$$

Proof of theorem, Bad events (probability $\leq \delta$)

$$\forall t \geq 0: \underline{B_t(\delta)} = \left\{ \omega \in \Omega : \underline{\|S_t\|_{\bar{V}_t^{-1}}^2} > 2R^2 \cdot \log\left(\frac{\det(\bar{V}_t)^{1/2} \det(V)^{-1/2}}{\delta}\right) \right\}$$

Goal: control

$$\underline{\bigcup_{t \geq 0} B_t(\delta)}$$

$$\bar{V}_t = V + \sum_{s=1}^t X_s X_s^T$$

$$\underline{S_t} = \sum_{s=1}^t \eta_s X_s$$

$$\{\tau(\omega) \leq t\} \in \mathcal{F}_t$$

$$\min \phi = \infty$$

Claim: $\tau(\omega)$ is a stopping time.

$$\Rightarrow \bigcup_{t \geq 0} B_t(\delta) = \{\omega : \tau(\omega) < \infty\}$$

$$\boxed{\Pr\left(\bigcup_{t \geq 0} B_t(\delta)\right)} = \Pr(\underline{\tau} < \infty)$$

$$= \Pr(\underline{\tau} < \infty, \underline{\|S_\tau\|_{\bar{V}_\tau^{-1}}^2} > 2R^2 \cdot \log \frac{\det(\bar{V}_\tau)^{1/2} \det(V)^{-1/2}}{\delta})$$

$$\leq \Pr(\underline{\|S_\tau\|_{\bar{V}_\tau^{-1}}^2} > 2R^2 \log \frac{\det(\bar{V}_\tau)^{1/2} \det(V)^{-1/2}}{\delta}) = \delta$$

(by Lemma) $\square$

Proposition. $\lambda \in \mathbb{R}^d$. Let $\underline{M_t^\lambda} = \exp\left(\sum_{s=1}^t \left(\frac{\eta_s \langle \lambda, X_s \rangle}{R} - \frac{1}{2} \langle \lambda, X_s \rangle^2\right)\right)$

then, $\underline{M_t^\lambda}$ is almost surely well-defined. and $\underline{\mathbb{E}[M_t^\lambda]} \leq 1$

Proof. η_s is $\mathbb{R}$ -sub-Gaussian, $\underline{\mathbb{E}[e^{\alpha \eta_s} | \mathcal{F}_{s-1}]} \leq e^{\frac{\alpha^2 R^2}{2}} \quad \forall \alpha > 0$

$$\alpha = \frac{\langle \lambda, X_s \rangle}{R} \Rightarrow \mathbb{E}\left[\exp\left(\frac{\eta_s \langle \lambda, X_s \rangle}{R}\right) \mid \mathcal{F}_{s-1}\right] \leq \exp\left(\frac{1}{2} \langle \lambda, X_s \rangle^2\right)$$

$$\underline{D_s^\lambda} = \mathbb{E}\left[\exp\left(\frac{\eta_s \langle \lambda, X_s \rangle}{R} - \frac{1}{2} \langle \lambda, X_s \rangle^2\right) \mid \mathcal{F}_{s-1}\right] \leq 1.$$

$$\underline{M_t^\lambda} = \underline{M_{t-1}^\lambda} \cdot \underline{D_t^\lambda}$$

$$\Rightarrow \underline{\mathbb{E}[M_t^\lambda | \mathcal{F}_{t-1}]} = \mathbb{E}[\underline{M_{t-1}^\lambda} \cdot \underline{D_t^\lambda} | \mathcal{F}_{t-1}]$$

$$= \underline{M_{t-1}^\lambda} \cdot \underbrace{\mathbb{E}[D_t^\lambda | \mathcal{F}_{t-1}]}_{\leq 1} \leq \underline{M_{t-1}^\lambda}$$

$\Rightarrow \{M_t^\lambda\}_{t \geq 1}$. Super-martingale.

$$\boxed{E[M_t^\lambda] \leq E[M_0^\lambda] \text{ by OST}} \quad \xrightarrow{1} ?$$

$$\begin{aligned} \underline{\forall t \geq 0:} \quad \underline{E[M_t^\lambda]} &= E[E[M_t^\lambda | \mathcal{F}_{t-1}]] \\ &\leq \underline{E[M_{t-1}^\lambda]} \leq \dots \leq E[M_0^\lambda] = 1 \end{aligned}$$

Martingale Convergence.

$\{Y_t\}_{t \geq 0}$ is a non-negative super-martingale.

$\Rightarrow$ almost surely. $Y_t \downarrow \underline{Y_\infty}$

$$\& E[Y_\infty] \leq E[Y_0]$$

$\mathcal{F}_\infty$ -measurable
(tail σ -algebra)

Let $\underline{Q_t = M_{\min\{t, \tau\}}^\lambda}$



$\underline{M_\tau^\lambda}$

More precisely:

$$\boxed{\liminf_{t \rightarrow \infty} Q_t = M_\tau^\lambda}$$

Fatou's Lemma: $E[\liminf_{t \rightarrow \infty} Z_t] \leq \liminf_{t \rightarrow \infty} E[Z_t]$

$$\Rightarrow E[M_\tau^\lambda] = E[\liminf_{t \rightarrow \infty} Q_t] \leq \liminf_{t \rightarrow \infty} E[Q_t] \leq 1$$

□

Recall: Lemma $\|S_t\|_{V_t^{-1}}^2 \leq 2R^2 \cdot \log\left(\frac{\det(V_t)^{1/2} \det(V)^{-1/2}}{\sigma}\right)$.

Proof: wlog. $R=1$.

Let $V_t = \sum_{s=1}^t X_s X_s^T$.

$$\begin{cases} \sum_{s=1}^t \eta_s \langle \lambda, X_s \rangle = \langle \lambda, S_t \rangle \\ \sum_{s=1}^t \langle \lambda, X_s \rangle^2 = \lambda^T \left(\sum_{s=1}^t X_s X_s^T \right) \lambda \\ = \lambda^T V_t \lambda = \|\lambda\|_{V_t}^2 \end{cases}$$

By proposition,

$$M_t^\lambda = \exp\left(\langle \lambda, S_t \rangle - \frac{1}{2} \|\lambda\|_{V_t}^2\right)$$

$$\mathbb{E}[M_t^\lambda] \leq 1$$

Mixture

$$\lambda \sim \mathcal{N}(0, V^{-1})$$

Self-Normalized Processes

P & K & L.

Let $M_t = \mathbb{E}[M_t^\lambda | \mathcal{F}_\infty]$.

$\Rightarrow \mathbb{E}[M_t] = \mathbb{E}[\mathbb{E}[M_t^\lambda | \mathcal{F}_\infty]] = 1$. tail σ -algebra.

$$f(\lambda) = \frac{1}{c(V^{-1})} \exp\left(-\frac{1}{2} \|\lambda\|_{V^{-1}}^2\right), \text{ where } c(V^{-1}) = \int_{\lambda} \exp\left(-\frac{1}{2} \|\lambda\|_{V^{-1}}^2\right) d\lambda$$

$\Rightarrow M_t = \int_{\mathbb{R}^d} \exp(\langle \lambda, S_t \rangle - \frac{1}{2} \|\lambda\|_{V_t}^2) \cdot f(\lambda) d\lambda$

$-\frac{1}{2} \|\lambda - V_t^{-1} S_t\|_{V_t}^2 + \frac{1}{2} \|S_t\|_{V_t^{-1}}^2$

$= (2\pi)^{\frac{d}{2}} \cdot \det(V^{-1})^{\frac{1}{2}}$

★

$$= \exp\left(\frac{1}{2} \cdot \|S_t\|_{V_t^{-1}}^2\right) \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2} \|\lambda - V_t^{-1} S_t\|_{V_t}^2\right) f(\lambda) d\lambda$$

$$= \frac{1}{C(V^{-1})} \cdot \exp\left(\frac{1}{2} \cdot \|S_t\|_{V_t^{-1}}^2\right) \cdot \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2} \left(\|\lambda - V_t^{-1} S_t\|_{V_t}^2 + \|\lambda\|_V^2 \right)\right) d\lambda$$

$$\|x - a\|_P^2 + \|x\|_Q^2 = \|x - (P+Q)^{-1}Pa\|_{P+Q}^2 + \|a\|_P^2 - \|Pa\|_{(P+Q)^{-1}}^2$$

$$\begin{aligned} \Rightarrow \| \lambda - V_t^{-1} S_t \|_{V_t}^2 + \| \lambda \|_V^2 &= \| \lambda - (\underline{V_t + V})^{-1} S_t \|_{V_t + V}^2 + \| \underline{V_t^{-1} S_t} \|_{V_t}^2 - \| S_t \|_{(V_t + V)^{-1}}^2 \\ &= \| \lambda - \bar{V}_t^{-1} S_t \|_{\bar{V}_t}^2 + \| S_t \|_{\bar{V}_t^{-1}}^2 - \| S_t \|_{\bar{V}_t^{-1}}^2 \end{aligned}$$

$$\begin{aligned} M_t &= \frac{1}{C(V^{-1})} \cdot \exp\left(\frac{1}{2} \|S_t\|_{V_t^{-1}}^2\right) \cdot \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2} \|\lambda - \bar{V}_t^{-1} S_t\|_{\bar{V}_t}^2\right) d\lambda \\ &= \frac{C(\bar{V}_t^{-1})}{C(V^{-1})} \cdot \exp\left(\frac{1}{2} \|S_t\|_{\bar{V}_t^{-1}}^2\right) \cdot \underbrace{\tilde{\Lambda} \sim \mathcal{N}(\bar{V}_t^{-1} S_t, \bar{V}_t^{-1})}_{\downarrow C(\bar{V}_t^{-1})} \\ &= \frac{\det(\bar{V}_t)^{-1/2}}{\det(V)^{-1/2}} \cdot \exp\left(\frac{1}{2} \|S_t\|_{\bar{V}_t^{-1}}^2\right) \end{aligned}$$

Since $\mathbb{E}[M_\tau] \leq 1$. We have

$$\begin{aligned} \Pr\left(\|S_\tau\|_{\bar{V}_\tau^{-1}}^2 > 2 \cdot \log \frac{\det(\bar{V}_\tau)^{1/2} \det(V)^{-1/2}}{\delta}\right) &\leq \mathbb{E}\left[\frac{\exp\left(\frac{1}{2} \|S_\tau\|_{\bar{V}_\tau^{-1}}^2\right)}{\frac{1}{\delta} \cdot \det(\bar{V}_\tau)^{1/2} \det(V)^{-1/2}}\right] \\ &= \delta \cdot \mathbb{E}[M_t] \leq \delta. \quad \square \end{aligned}$$

Linear (Contextual) Bandits

LinUCB

C.f. Yasin Abbasi-Yadkori, David Pal, and Csaba Szepesvári.

Improved Algorithms for Linear Stochastic Bandits (2011)

• Each arm is associated with a feature vector $x \in \mathcal{X} = \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$

• At round $t=1, 2, \dots, T$:

1) the player first chooses an arm $X_t \in \mathcal{X}$

2) and then the arm reveals a reward $r_t \in \mathbb{R}$.

context $\rightarrow X_t$
action

Linear Bandits: $\boxed{r_t(x) = x^\top \theta^* + \eta_t}$

θ^* : unknown parameter s.t. $\|\theta^*\|_2 \leq B$
 η_t : R -sub-Gaussian.

Goal: minimize regret $\text{Reg}_T = T \cdot \underline{\mu(a^*)} - \sum_{t=1}^T \mu(a_t)$

$$= T \cdot \max_{x \in \mathcal{X}} x^\top \theta^* - \sum_{t=1}^T x_t^\top \theta^*.$$

Algorithm: LinUCB

At round $t=1, 2, \dots, T$:

1) $X_t = \arg\max_{x \in \mathcal{X}}$

$\underline{\mu(B_{t-1}(x))}$ $\xleftarrow{\text{optimistic estimation of}}$ $x^\top \theta^*$
 $\mu(x) = x^\top \theta^*$
 $\uparrow$
estimate θ^* ★

2) observes Γ_t . update the estimation $\hat{\underline{\theta}}_t$ ($\hat{\underline{\mu}}_t$)

3) update $\underline{\text{LCB}}_t(x)$ by new estimation

Estimation : Ridge Regression (Regularized LS)

$$\hat{\underline{\theta}}_t = \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \sum_{s=1}^{t-1} (\underbrace{X_s^T \theta}_{\uparrow} - \underbrace{r_s}_{\uparrow})^2 + \underbrace{\|\theta\|_2^2}_{\downarrow}$$

$$\Rightarrow \hat{\underline{\theta}}_t = \underline{\Lambda}_t^{-1} \sum_{s=1}^{t-1} X_s \cdot r_s \quad \text{where } \underline{\Lambda}_t = \underline{I} + \sum_{s=1}^{t-1} X_s X_s^T$$

Direct Matrix Multiplication : $O(d^3)$ computation

Incremental update: $\hat{\theta}_{t-1} \rightarrow \hat{\theta}_t$

(rank-1 update)

$O(d^2)$ computation.

$$\text{Let } \underline{K}_t = \underline{\Lambda}_t^{-1}$$

$$\underline{P}_t = \sum_{s=1}^t X_s X_s^T$$

$\Rightarrow$

$$K_t = \frac{P_{t-1} X_t}{1 + X_t^T P_{t-1} X_t}$$

$$\hat{\theta}_t = \hat{\theta}_{t-1} + \underline{K}_t (r_t - \underline{X}_t^T \hat{\theta}_{t-1})$$

$$\underline{P}_t = \underline{P}_{t-1} - \underline{K}_t X_t^T P_{t-1}$$

Sherman-Morrison
- Woodbury formula

Ridge Regression $\rightarrow$ one-shot estimation error w.p. $\geq 1-\delta$

$$\forall x: \underbrace{\left| x^T \hat{\theta}_t - x^T \theta^* \right|}_{\text{effective sample size}} \leq \underbrace{\|x\|_{\Lambda_t^{-1}}}_{\text{self-normalized concentration}} \cdot \underbrace{\mathcal{O}\left(R \sqrt{d + \log \frac{t}{\delta}}\right)}_{\text{self-normalized concentration}}$$

confidence set:

w.p. $\geq 1 - \delta$: $\underbrace{x^T \theta^*}_{\mu(x)} \leq \text{UCB}(x)$

LinUCB Algorithm:

Principle: Optimism in Face of Uncertainty bonus term.

Confidence Set:

$$\underbrace{|\mu(x) - \hat{\mu}_t(x)|}_{\text{Concentration analysis}} \leq \frac{\beta_{t+1} \|x\|_{\Lambda_t^{-1}}}{\beta_t(x)}$$

Construct UCB

$$\text{UCB}_t(x) = \hat{\mu}_t(x) + \beta_t(x)$$

② Concentration

Greedy action selection:

$$x_{t+1} = \arg \max_{x \in \mathcal{X}} \text{UCB}_t(x)$$

③ Arm Selection

Submit (x_{t+1}, r_{t+1})

Estimation by Regularized LS:

$$\hat{\theta}_t = \arg \min_{\theta} \sum_{s=1}^{t-1} (x_s^T \theta - r_s)^2 + \|\theta\|_2^2$$

$$\hat{\mu}_t(x) = x^T \hat{\theta}_t$$

$$r_t = x_t^T \theta^* + \eta_t$$

Learning History

$$F_t = \{(x_1, r_1), \dots, (x_t, r_t)\}$$

① Estimation < Statistical efficiency
Computational efficiency

Theorem (Regret Bound of LinUCB) w.p. $\geq 1 - \frac{1}{T}$,

$$\text{Reg}_T \lesssim \underline{R \cdot \sqrt{d \log T} \cdot \sqrt{T d \log(T+1)}} = \underline{\tilde{O}(d\sqrt{T})}$$

Proof: Let $X^* = \underset{x \in \mathcal{X}}{\operatorname{argmax}} x^T \theta^*$,

By estimation error of Ridge Regression, w

$$\text{w.p.} \geq 1 - \frac{\delta}{2}: \forall t \in [T], \quad \underline{X_*^T \theta_*} \leq \underline{X_*^T \hat{\theta}_t + \beta_{t-1} \|X_*\|_{\Lambda_t^{-1}}}$$

$$\text{w.p.} \geq 1 - \frac{\delta}{2}: \forall t \in [T], \quad X_t^T \theta_* \geq X_t^T \hat{\theta}_t - \beta_{t-1} \|X_t\|_{\Lambda_t^{-1}}$$

$$\Rightarrow \forall t \in [T], \quad \underbrace{X_*^T \theta_*}_{\text{true mean}} - X_t^T \theta_*$$

$$= \left(\underbrace{X_*^T \hat{\theta}_t + \beta_{t-1} \|X_*\|_{\Lambda_t^{-1}}}_{\text{estimated mean}} \right) - \left(X_t^T \hat{\theta}_t - \beta_{t-1} \|X_t\|_{\Lambda_t^{-1}} \right)$$

By Greedy action selection $X_t = \underset{x \in \mathcal{X}}{\operatorname{argmax}} \text{UCB}_{t-1}(x)$

$$\star \quad \underline{X_*^T \hat{\theta}_t + \beta_{t-1} \|X_*\|_{\Lambda_t^{-1}}} \leq \underline{X_t^T \hat{\theta}_t + \beta_{t-1} \|X_t\|_{\Lambda_t^{-1}}}$$

$$\Rightarrow X_*^T \theta_* - X_t^T \theta_* \leq \underline{(X_*^T \hat{\theta}_t - X_t^T \hat{\theta}_t)} + \beta_{t-1} \cdot (\|X_*\|_{\Lambda_t^{-1}} + \|X_t\|_{\Lambda_t^{-1}})$$

$$\leq \underline{2 \cdot (\beta_{t-1}) \|X_t\|_{\Lambda_t^{-1}}}$$

$$\Rightarrow \underline{\text{Reg}_T} = \left[\sum_{t=1}^T X_*^T \theta_* - X_t^T \theta_* \right] \quad \beta_{t-1} \leq \beta_t$$

$$\leq 2 \cdot \beta_T \cdot \underbrace{\sum_{t=1}^T \|\chi_t\|_{\Lambda_t^{-1}}}_{\star}$$

$$\leq 2 \beta_T \cdot \sqrt{T} \cdot \sqrt{\sum_{t=1}^T \chi_t^T \Lambda_t^{-1} \chi_t}$$

↪ Elliptical Potential

$$\leq 2 \beta_T \cdot \sqrt{T} \cdot \sqrt{2d \log T}$$

$$= O\left(R \cdot \sqrt{d \log \frac{T}{\delta}} \cdot \sqrt{T} \cdot \sqrt{2d \log T}\right) = \tilde{O}(\sqrt{d T}) \quad \square$$

Tabular PAC-RL

< Rmax Exploration (Brafman et al, 2003)
UCB-VI (Azar et al, 2017)

PAC (Probably Approximately Correct)

$$\Pr\left(\underbrace{|\hat{h} - h^*|}_{\text{approximately}} \leq \underbrace{\varepsilon}_{\text{probably}}\right) \geq \underbrace{1 - \delta}_{\text{probably}}$$

⇒ Sample Complexity: w.p. $\geq 1 - \delta$, $|\hat{h} - h^*| \leq \varepsilon$

in RL: w.p. $\geq 1 - \delta$ $\underbrace{J_M(\hat{\pi}) - J_M^*}_{\neq} \leq \underbrace{\varepsilon \cdot V_{\max}}_{\neq}$

$$\underbrace{J(\pi)}_{\in [0, V_{\max}]} = \mathbb{E}_{\pi} \left[\sum_{h=1}^{\infty} \gamma^{h-1} r_h \right]$$

$$\Rightarrow T = \text{poly}\left(|S|, |A|, \frac{1}{1-\gamma}, \frac{1}{\varepsilon}, \frac{1}{\delta}\right)$$

within T iteration to give a PAC guarantee.

* Sample Complexity $\leftrightarrow$ Regret

E³: Near-optimal reinforcement learning in polynomial time. (2002)

Rmax: R-max — a general polynomial time algorithm for near-optimal reinforcement learning (2003)

Rmax Exploration

$$M = (S, A, P, \underline{R}, \gamma, d_0)$$

deterministic in $[0, \underline{R_{\max}}]$

① n(s,a): visitation count to each (s,a).

② n(s,a,s'): # of times we observe transition tuples (s,a,s')

Explore-then-Terminate.

→ threshold parameter m.

$$\underline{K} = \{ (s,a) : \underline{n(s,a)} = \underline{m} \}$$

Algorithm: 1. Build an MDP $\hat{M}_K$.

$$\hat{P}_K(s'|s,a) = \begin{cases} \frac{n(s,a,s')}{n(s,a)} & \text{if } (s,a) = K. \\ \mathbb{1}_{\{s'=s\}} & \text{o.w. (self loop)} \end{cases}$$

$$\hat{R}_K(s,a) = \begin{cases} R(s,a) & \text{if } (s,a) = K. \\ \underline{R_{\max}} & \text{o.w.} \end{cases}$$

Optimism in face of Uncertainty

2. Do planning on $\hat{M}_K$ (VI, PI) $\rightarrow \pi_{\hat{M}_K}^*$

3. Roll out a trajectory $s_1, a_1, \dots, s_H, a_H$ by $\pi_{\hat{M}_K}^*$

4. If any observed (s_h, a_h) , $h \in [H]$ has $n(s_h, a_h) < M$
 $\Rightarrow n(s_h, a_h) += 1$. $n(s_h, a_h, s_{h+1}) += 1$.

Claim 1 (Optimism) $\forall \pi: S \rightarrow A: \underbrace{J_{M_K}(\pi)} \geq \underbrace{J_M(\pi)}$

where M_K is the expected version of $\hat{M}_K$.

$$P_K(s'|s, a) = \begin{cases} P(s'|s, a) & \text{if } (s, a) \in K \\ \mathbb{1}_{\{s'=s\}} & \text{o.w.} \end{cases} \quad R_K(s, a) = \hat{R}_K(s, a)$$

	<u>M</u>	<u><u>M_K</u></u>	<u><u>$\hat{M}_K$</u></u>
<u>known</u> (<u>K</u>)	$= M$	$= M$	$\approx M$
unknown	$= M$	self-loop	self-loop

Proof: wlog. consider deterministic $\pi: S \rightarrow A$.

$$V_M^\pi(s) = R(s, \pi) + \gamma \cdot \sum_{s'} P(s'|s, \pi) \cdot V_M^\pi(s')$$

$$V_{M_K}^\pi(s) = R_K(s, \pi) + \gamma \cdot \sum_{s'} P_K(s'|s, \pi) \cdot V_M^\pi(s')$$

$$\text{If } (s, \pi) \in K : V_M^\pi(s) = V_{M_K}^\pi(s)$$

$$\text{If } (s, \pi) \notin K : V_{M_K}^\pi(s) = R_{\max} + \gamma \cdot V_M^\pi(s)$$

$$\Rightarrow V_{M_K}^\pi(s) = \frac{R_{\max}}{1-\gamma} = V_{\max} \geq V_{M_K}^\pi(s) \quad \square$$

Claim 2: (concentration) Let $\hat{p}$ be the empirical version of $P(s, a)$ based on m i.i.d samples of $s' \sim P(s, a)$

$$\text{w.p. } \geq 1 - \delta : \|\hat{p} - P(s, a)\|_1 \leq \sqrt{\frac{2}{m} \cdot \log \frac{2 \cdot (2^{|S|} - 2)}{\delta}} = \eta$$

Remark: ~~independence~~ martingale concentration.

Claim 3: (Value difference) Suppose M_1 , M_2 only differs

in dynamics, $\|\underline{V_{M_1}^*} - \underline{V_{M_2}^*}\|_\infty \leq \frac{V_{\max}}{1-\gamma} \cdot \max_{(s,a) \in S \times A} \|P_1(s,a) - P_2(s,a)\|_1$

$\hookrightarrow \epsilon_p$

Proof: $\|V_{M_1}^* - V_{M_2}^*\|_\infty$

$$= \|V_{M_1}^* - \mathcal{T}_{M_2} V_{M_1}^* + \underbrace{(\mathcal{T}_{M_2} V_{M_1}^* - \mathcal{T}_{M_2} V_{M_2}^*)}_{\text{blue}}\|_\infty$$

$$\leq \underbrace{\|V_{M_1}^* - \mathcal{T}_{M_2} V_{M_1}^*\|_\infty}_{\downarrow} + \gamma \cdot \|V_{M_1}^* - V_{M_2}^*\|_\infty$$

$$\| \mathcal{T}_{M_1} V_{M_1}^* - \mathcal{T}_{M_2} V_{M_2}^* \|_\infty$$

$$\begin{aligned}
&= \max_{s,a} \left| \underbrace{R_1(s,a)}_{\text{supp}(s,a)} + \gamma \cdot \underbrace{\mathbb{E}_{s' \sim P_1(s,a)} [V_{M_1}^*(s')]}_{\text{supp}(s,a)} - \underbrace{R_2(s,a)}_{\text{supp}(s,a)} - \gamma \cdot \underbrace{\mathbb{E}_{s' \sim P_2(s,a)} [V_{M_1}^*(s')]}_{\text{supp}(s,a)} \right| \\
&= \gamma \cdot \max_{s,a} \left| \langle P_1(s,a) - P_2(s,a), \underline{V_{M_1}^*} \rangle \right| \quad \text{2 Hölder-} \\
&\leq \gamma \cdot V_{\max} \cdot \varepsilon_p
\end{aligned}$$

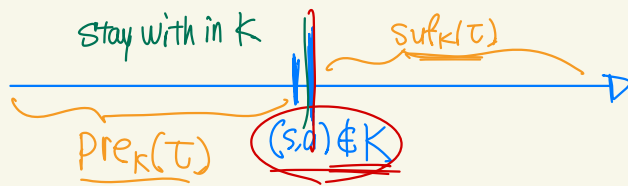
Cor (Simulation Lemma) Suppose M_1 and M_2 only differ in dynamics. $\forall \pi$: $\left| \underline{J_{M_1}(\pi)} - J_{M_2}(\pi) \right| \leq \frac{V_{\max}}{1-\gamma} \cdot \varepsilon_p$.

Claim 4: Suppose MDP (M_1, M_2) agrees exactly on K
transition reward

Let $\text{escape}_K(\tau) = 1$ $\{ \text{trajectory } \tau \text{ visits some } (s,a) \notin K \}$

$$\Rightarrow \left| \underline{J_{M_1}(\pi)} - J_{M_2}(\pi) \right| \leq V_{\max} \cdot \Pr_{M_1, \tau \sim \pi} [\underline{\text{escape}_K(\tau) = 1}]$$

Proof: for τ such that $\text{escape}_K(\tau) = 1$,



Cumulative reward $\rightarrow$

$R_M(\text{pre}_K(\tau))$	$R_M(\text{suff}_K(\tau))$
$\Pr_{M, \pi} [\underline{\text{pre}_K(\tau)}]$	$\Pr_{M, \pi} [\text{suff}_K(\tau)]$

$$J_{M_1}(\pi) = \sum_{\tau} Pr_{M_1, \pi}(\tau) \cdot R_{M_1}(\tau)$$

$$\begin{aligned} & \left[\sum_{\substack{\tau: \text{escape}(\tau)=1 \\ \text{pre}_K(\tau)}} \cancel{Pr_{M_1, \pi}(\tau)} \cdot (R_{M_1}(\text{pre}_K(\tau)) + R_{M_1}(\text{suf}_K(\tau))) \right] \quad \text{Cumulative reward} \\ & + \sum_{\tau: \text{escape}(\tau)=0} Pr_{M_1, \pi}(\tau) \cdot R_{M_1}(\tau) \end{aligned}$$

$\leq V_{\max}$

$$\begin{aligned} J_{M_2}(\pi) & \geq \sum_{\text{pre}_K(\tau)} Pr_{M_2, \pi}(\text{pre}_K(\tau)) \cdot (R_{M_2}(\text{pre}_K(\tau)) + 0) \\ & + \sum_{\tau: \text{escape}(\tau)=0} Pr_{M_2, \pi}(\tau) \cdot R_{M_2}(\tau) \end{aligned}$$

Since M_1, M_2 agrees exactly on K . $\rightarrow d_0(s_i) \cdot \pi(a_i | s_i) \cdot \underline{P_{M_2}(s_2 | s_1, a_1)} \cdot \underline{\pi(a_2 | s_2)} \dots$

for τ such that $\text{escape}(\tau)=0 \Rightarrow$ stay within K

$$\Rightarrow J_{M_1}(\pi) - J_{M_2}(\pi) \leq \sum_{\substack{\text{pre}_K(\tau) \\ \tau: \text{escape}(\tau)=1}} Pr_{M_1, \pi}(\text{pre}_K(\tau)) \cdot V_{\max}$$

Sample Complexity of $R_{\max}$

$$Pr_{M_1, \pi}[\text{escape}(\tau)=1] \cdot \square$$

$\pi_{M_F}^*$: $\angle$ ϵ -optimal

(OR) there is significant probability in increasing the counter of some $n(s, a)$ (for tuned m)

When $\pi_{\hat{M}_K}^*$ is ε -suboptimal:

$$\varepsilon V_{\max} < J_M(\pi_M^*) - J_M(\pi_{\hat{M}_K}^*)$$

$$\leq J_{\underline{M}_K}(\pi_M^*) - J_M(\pi_{\hat{M}_K}^*)$$

optimism

$$\leq J_{\underline{M}_K}(\pi_{\hat{M}_K}^*) - J_M(\pi_{\hat{M}_K}^*)$$

Claim 3

$$\leq J_{\hat{M}_K}(\pi_{\hat{M}_K}^*) + \frac{V_{\max}}{1-\gamma} \cdot \varepsilon_P(M_K, \hat{M}_K) - J_M(\pi_{\hat{M}_K}^*)$$

Simulation Lemma

$$\leq J_{\underline{M}_K}(\pi_{\hat{M}_K}^*) + \frac{2V_{\max}}{1-\gamma} \cdot \varepsilon_P(M_K, \hat{M}_K) - J_{\underline{M}_K}(\pi_{\hat{M}_K}^*)$$

Claim 4

$$\leq V_{\max} \cdot \Pr_{M, \pi_{\hat{M}_K}^*}[\text{escape}(\tau)=1] + \frac{2V_{\max}}{1-\gamma} \cdot \varepsilon_P(M_K, \hat{M}_K)$$

$$\Rightarrow \Pr_{M, \pi_{\hat{M}_K}^*}[\text{escape}(\tau)=1] \geq \varepsilon - \frac{2}{1-\gamma} \eta = \frac{\varepsilon}{2}$$

With

$$\eta = \frac{(1-\gamma)\varepsilon}{4}$$

$$\text{such that } \max_{s,a} \|P_K(s,a) - \hat{P}_K(s,a)\|_1 \leq \eta$$

$\Leftrightarrow$

$$= \sqrt{\frac{2|S|}{m}} \cdot \log \frac{1}{\delta}$$

$$\forall (s,a) \in K: \|P(s,a) - \hat{P}_K(s,a)\|_1 \leq \eta$$

$$\Rightarrow m = O\left(\frac{|S|}{\varepsilon^2(1-\gamma)^2} \cdot \log \frac{1}{\delta}\right)$$

$\Rightarrow$ If we set $m = O\left(\frac{|S|}{\varepsilon^2(1+\gamma)^2} \log \frac{1}{\delta}\right)$

then $\Pr_{M, \pi_{\hat{M}_K}^*} [\text{escape}(\tau) = 1] \geq \frac{\varepsilon}{2}$
 $\uparrow$
 τ visits some $(s, a) \in K$
 (on average)
 $\uparrow$
 improvement probability

In total: $|S| \cdot |A| \cdot m$ samples are needed.

$$\Rightarrow \underline{J_M(\pi_M^*) - J_M(\pi_{\hat{M}_K}^*)} \leq \frac{V_{\max}}{1-\gamma} \cdot \eta \cdot = \frac{\varepsilon}{4} \cdot V_{\max} \xrightarrow{\varepsilon\text{-optimal}}$$

Sample Complexity of $R_{\max}$: $O\left(\frac{|S| \cdot |A| \cdot m}{\varepsilon}\right) = O\left(\frac{|S|^2 |A|}{\varepsilon^2 (1+\gamma)^2} \cdot \log \frac{1}{\delta}\right)$

UCB-VI

c.f. Azar et al., 2017. Minimax Regret

Bounds for Reinforcement Learning

Recap: inhomogeneous, episodic MDP

$M = (S, A, \underset{\substack{\uparrow \\ \text{horizon}}}{H}, \underline{P}, \underline{R})$

$\underline{P} = \{P_h\}_{h=1}^H, P_h: S \times A \rightarrow \Delta(S)$

$\underline{R} = \{r_h\}_{h=1}^H, \underline{r}_h: S \times A \rightarrow \underline{[0, 1]}$

episode: $\underline{s_1, a_1, \dots, s_H, a_H, \underline{s_{H+1}}}$ Termination

(non-stationary) policy: $\pi: S \times [H] \rightarrow A$.

$$\pi = \{ \pi_h \}_{h=1}^H$$

$$\underline{V, Q} \in [0, H]$$

value function

$$\begin{cases} \underline{V}_h^\pi(s) = \mathbb{E}_\pi \left[\sum_{h'=h}^H r_{h'}(S_{h'}, \pi_{h'}(S_{h'})) \mid S_h = s \right] \\ Q_h^\pi(s, a) = r_h(s, a) + \mathbb{E}_\pi \left[\sum_{h'=h+1}^H r_{h'}(S_{h'}, \pi_{h'}(S_{h'})) \mid S_h = s, a_h = a \right] \end{cases}$$

Bellman equation: $\underline{Q}_h^\pi(s, a) = (r_h + \mathbb{P}_h \cdot \underline{V}_{h+1}^\pi)(s, a)$

$$\underline{V}_h^\pi(s) = \underline{Q}_h^\pi(s, \pi_h(s)) \quad \underline{V}_{H+1}^\pi(s) = 0$$

Bellman operator $\rightarrow$ Value Iteration (DP)

Bellman optimality equation: $\underline{Q}_h^*(s, a) = (r_h + \mathbb{P}_h \cdot \underline{V}_{h+1}^*)(s, a)$

$$\underline{V}_h^*(s) = \max_{a \in A} \underline{Q}_h^*(s, a) \quad \underline{V}_{H+1}^*(s) = 0$$

Goal: for K iterations: $\text{Reg}_K = \sum_{k=1}^K \left(\underline{V}_1^*(\underline{s}_1^k) - \underline{V}_1^{\pi^k}(s_1^k) \right)$

T: # of total samples we needed

$$T = KH$$

UCB (upper confidence bound): w.p. $\geq 1 - \delta$: $\underline{Q}_h^k \leq \text{UCB}_k$

Model-based (S, A, P, R, H)

Core: accurately estimate P

Algorithm :

$$\text{maintain } \begin{cases} N_h^k(s, a) = \sum_{i=1}^{k-1} \mathbb{1}\{S_h = s, a_h = a\} \\ N_h^k(s, a, s') = \sum_{i=1}^{k-1} \mathbb{1}\{S_h = s, a_h = a, S_{h+1} = s'\} \end{cases}$$

At round $k=1, 2, \dots, K$:

1. Estimate $\hat{P}_h^k(s' | s, a) = \frac{N_h^k(s, a, s')}{N_h^k(s, a)}$ with $\frac{0}{0} = 0$

2. Compute a bonus term $b_h^k(s, a) = O\left(H \sqrt{\frac{L}{N_h^k(s, a)}}\right)$ history-dependent

3. Run value iteration on $(S, A, \{\hat{P}_h^k\}_{h=1}^H, \{\Gamma_h + b_h^k\}_{h=1}^H, H)$

for $h=H, \dots, 1$:

$$\hat{V}_{H+1}^k(s) = 0$$

"Value-Targeted Regression" $\uparrow$ $\langle \hat{P}(\cdot | s, a), \hat{V}_{h+1}^k(\cdot) \rangle$

$$\hat{\Phi}_h^k(s, a) = \underbrace{\Gamma_h(s, a) + b_h^k(s, a)}_{\text{LCB}_k} + \underbrace{\hat{P}(s, a) \cdot \hat{V}_{h+1}^k}_{\text{clipping}}$$

$\uparrow$ $[0, H]$

VI:

$$\hat{V}_h^k(s) = \max_{a \in A} \hat{\Phi}_h^k(s, a)$$

4. Policy: $\pi^k = \{\pi_h^k\}_{h=1}^H$ s.t. $\pi_h^k(s) = \arg\max_{a \in A} \hat{\Phi}_h^k(s, a)$

for $h=1, \dots, H$: roll out policy π^k to generate a trajectory.

LGB $\rightarrow$ Optimism ★

(Lemma) $\forall k \in [K], h \in [H], (s, a) \in S \times A$;

$$\hat{Q}_h^k(s, a) \geq \underline{Q}_h^*(s, a)$$

$$\hat{V}_h^k(s) \geq V_h^*(s)$$

Proof: (Induction) Base: $\hat{V}_{H+1}^k(s) = V_{H+1}^*(s) = 0$.

Assume: $\forall h$: $\boxed{\hat{V}_{h+1}^k(s) \geq V_{h+1}^*(s)}$ (I.H.)

$$\hat{Q}_h^k(s, a) = \left[r_h(s, a) + b_h^k(s, a) + \hat{P}_h^k(s, a) \cdot \hat{V}_{h+1}^k \right]_{[0, H]}$$

$$Q_h^*(s, a) = r_h(s, a) + \underline{P_h(s, a) \cdot V_h^*}$$

When $\hat{Q}_h^k(s, a) = H \geq Q_h^*(s, a)$. ✓

When $\hat{Q}_h^k(s, a) = r_h(s, a) + b_h^k(s, a) + \hat{P}_h^k(s, a) \cdot \hat{V}_{h+1}^k$,

$$\hat{Q}_h^k(s, a) - Q_h^*(s, a)$$

$$= b_h^k(s, a) + \hat{P}_h^k(s, a) \cdot \hat{V}_{h+1}^k - P_h(s, a) \cdot V_{h+1}^*$$

$$\stackrel{(I.H.)}{\geq} b_h^*(s, a) + \hat{P}_h^k(s, a) \cdot \underline{V_{h+1}^*} - P_h(s, a) \cdot V_{h+1}^*$$

$$= b_h^*(s, a) + (\hat{P}_h^k(s, a) - P_h(s, a)) \cdot V_{h+1}^* \geq 0$$

$\Leftrightarrow$

$$\left| \underbrace{\widehat{P}_h^k(s,a)}_{\downarrow} - \underbrace{P_h(s,a)}_{\downarrow} \right| \cdot V_{h+1}^* \leq \underline{b_h^*(s,a)} \quad \checkmark$$

for all s, a, k, h .

$$E[\widehat{P}_h^k(s'|s,a)] = P_h(s'|s,a)$$

$$\widehat{P}_h^k(s'|s,a) = \frac{N_h^k(s,a,s')}{N_h^k(s,a)}$$

union bound

$$L = \log \frac{SAKH}{\delta}$$

Claim: w.p. $\geq 1-\delta$: $\forall \underline{s, a, k, h}$

$$\left| \widehat{P}_h^k(s,a) - P_h(s,a) \right| \cdot V_{h+1}^* \leq H \cdot \sqrt{\frac{L}{N_h^k(s,a)}}$$

We set $\underline{b_h^*(s,a)} = O\left(H \cdot \sqrt{\frac{L}{N_h^k(s,a)}}\right)$

$$\Rightarrow \underline{\widehat{Q}_h^k(s,a)} \geq \underline{Q_h^*(s,a)}$$

since $\widehat{V}_h^k(s,a) = \max_{a \in A} \widehat{Q}_h^k(s,a)$

$$V_h^*(s,a) = \max_{a \in A} Q_h^*(s,a)$$

$$\Rightarrow \underline{\widehat{V}_h^k(s,a)} \geq \underline{V_h^*(s,a)}$$

Induction! $\square$

Concentration Analysis

Claim 1: w.p. $\geq 1-\delta$. $\forall s, a, k, h$:

confidence set

$$\left| \left(\hat{P}_h^k(s, a) - P_h(s, a) \right) \cdot \underbrace{V_{h+1}^*}_{\substack{\Theta_h^* \\ \text{SAKH}}} \right| \lesssim H \cdot \sqrt{\frac{L}{N_h^k(s, a)}} \rightarrow \log \frac{\text{SAKH}}{\delta}$$

$$X_i = \mathbb{1} \{ (S_h^i, a_h^i) = (s, a) \} \cdot \underbrace{V_{h+1}^*(S_{h+1}^i)}_{\substack{\in [0, H] \\ \text{R.V.}}} \quad S_{h+1}^i \sim P_h(\cdot | s, a) \quad \text{X.i.d} \quad i \in [k]$$

→ we get $N_h^k(s, a)$ samples from $P_h(\cdot | s, a)$

of iterations so far

$$\Rightarrow \begin{cases} \hat{P}_h^k(s, a) \cdot \underline{V_{h+1}^*} = \frac{1}{N_h^k(s, a)} \sum_{i=1}^{N_h^k(s, a)} X_i \\ P_h(\cdot | s, a) \cdot V_{h+1}^* = \mathbb{E}[X_i] \end{cases}$$

⇒ ~~Hoeffding's inequality~~ : $\left| \left(\hat{P}_h^k(s, a) - P_h(s, a) \right) \cdot V_{h+1}^* \right| \leq H \cdot \sqrt{\frac{L}{N_h^k(s, a)}}$

Proof of Claim 1 : $X_i = \mathbb{1} \{ (S_h^i, a_h^i) = (s, a) \} \cdot V_{h+1}^*(S_{h+1}^i)$

$\{X_i\}_{i=1}^k$ is a martingale sequence ! $i \in [k]$

$\mathcal{H}_{h,i}$: history at time step h from iteration 1 to i .

$$\mathbb{E}[X_{i+1} | \mathcal{H}_{h,i}] = X_i$$

$$|X_i| \leq H$$

$$\sum_{i=1}^k |X_i - \mathbb{E}[X_i | \mathcal{H}_{h,i}]|^2 \leq N_h^k(s, a)$$

Azuma-Hoeffding's inequality :

→ bounded variance

$$\left| \sum_{i=1}^k (x_i - x_{i-1}) \right| \leq 2H \cdot \sqrt{N_h^k(s,a) \cdot \log \frac{1}{\delta}}$$

$$\sum_{i=1}^k \left(\mathbb{1}\{(s_h^i, a_h^i) = (s, a)\} \cdot \boxed{V_{h+1}^*(s_{h+1}^i)} - N_h^i(s,a) \cdot P_h(s,a) \cdot V_{h+1}^* \right)$$

$$\Rightarrow \left| \left(\hat{P}_h^k(s,a) - P_h(s,a) \right) \cdot \boxed{V_{h+1}^*} \right| \lesssim H \cdot \sqrt{\frac{\log \frac{1}{\delta}}{N_h^k(s,a)}} \quad \square$$

fixed (deterministic)

$\forall$ fixed $f: S \mapsto [0, H]$,

$$\left| \sum_{i=1}^k \mathbb{1}\{(s_h^i, a_h^i) = (s, a)\} \cdot f(s_{h+1}^i) - N_h^k(s,a) \cdot P_h(s,a) \cdot f \right| \lesssim \sqrt{\frac{H^2 L}{N_h^k(s,a)}}$$

Claim 2: w.p. $\geq 1 - \delta$: $\forall s, a, k, h$,

$$\left| \left(\hat{P}_h^k(s,a) - P_h(s,a) \right) \cdot f \right| \lesssim H \cdot \sqrt{\frac{SL}{N_h^k(s,a)}}$$

f may depend on history

Regret Analysis of UCB-VI

Recall: $\text{Reg}_k = \sum_{i=1}^k V_1^*(s_i^k) - V_1^{\pi^k}(s_i^k)$

$$\text{Reg}_k = \sum_{i=1}^k \underline{V_i^*}(s_i^k) - V_i^{\pi^k}(s_i^k)$$

$$= \sum_{i=1}^k Q_i^*(s_i^k, \underline{\pi_i^*}(s_i^k)) - V_i^{\pi^k}(s_i^k)$$

optimism $\hookrightarrow$

$$\leq \sum_{i=1}^k \hat{Q}_i^k(s_i^k, \underline{\pi_i^*}(s_i^k)) - V_i^{\pi^k}(s_i^k)$$

greedy of π^k

$$\leq \sum_{i=1}^k \hat{Q}_i^k(s_i^k, a_i^k) - V_i^{\pi^k}(s_i^k)$$

(clipping)

$$\hat{Q}_h^k(s, a) \leq r_h(s, a) + b_h^k(s, a) + \hat{P}_h^k(s, a) \cdot \hat{V}_{h+1}^k$$

$$= r_h(s, a) + b_h^k(s, a) + \hat{P}_h(s, a) \cdot \hat{V}_{h+1}^k + \underbrace{(\hat{P}_h^k(s, a) - \hat{P}_h(s, a)) \cdot \hat{V}_{h+1}^k}_{\text{concentration analysis}}$$

By Claim 2,

history dependent

$$\left| (\hat{P}_h^k(s, a) - \hat{P}_h(s, a)) \cdot \hat{V}_{h+1}^k \right| \leq H \cdot \frac{\sqrt{S \cdot L}}{\sqrt{N_h^k(s, a)}} = \underline{\sqrt{S} \cdot b_h^k(s, a)}$$

concentration analysis

$$\hat{Q}_h^k(s, a) \leq r_h(s, a) + (\sqrt{S} + 1) \cdot b_h^k(s, a) + \underline{\hat{P}_h(s, a) \cdot \hat{V}_{h+1}^k}$$

$$\Rightarrow \hat{Q}_1^k(s_1^k, a_1^k) - \underline{V_1^{\pi^k}(s_1^k)}$$

$$\leq \cancel{r_1(s_1^k, a_1^k)} + (\sqrt{S} + 1) \cdot b_1^k(s_1^k, a_1^k) + \hat{P}_1(s_1^k, a_1^k) \cdot \underline{\hat{V}_2^k}$$

$$- \hat{P}_1(s_1^k, a_1^k) \cdot \underline{V_2^{\pi^k}} - \cancel{r_1(s_1^k, a_1^k)} \rightarrow \text{Bellman equation}$$

$$= (\sqrt{S}+1) \cdot \underline{b_1^k(s_1^k, a_1^k)} + \underbrace{IP_1(s_1^k, a_1^k) \cdot (\hat{V}_2^k - V_2^{\pi_k})}_{\text{concentration}} \\ + \underbrace{(\hat{V}_2^k(s_2^k) - V_2^{\pi_k}(s_2^k))}_{\hat{Q}_2^k(s_2^k, a_2^k)} - (\hat{V}_2^k(s_2^k) - V_2^{\pi_k}(s_2^k))$$

$$\frac{IP_1(s_1^k, a_1^k) \cdot (\hat{V}_2^k - V_2^{\pi_k})}{(\hat{V}_2^k - V_2^{\pi_k})(s_2^k)} \hookrightarrow s_2^k \sim IP_1(\cdot | s_1^k, a_1^k)$$

$$\Delta \underline{M_h^k} = \underline{IP_h(s_h^k, a_h^k) \cdot (\hat{V}_{h+1}^k - V_{h+1}^{\pi_k})} - (\hat{V}_{h+1}^k - V_{h+1}^{\pi_k}) \cdot \underline{(s_{h+1}^k)}$$

$$\Rightarrow \underline{\hat{Q}_1^k(s_1^k, a_1^k)} - V_1^{\pi_k}(s_1^k) \leq \underline{\hat{Q}_2^k(s_2^k, a_2^k)} - V_2^{\pi_k}(s_2^k) \\ + \underline{M_1^k} + (\sqrt{S}+1) \cdot b_1^k(s_1^k, a_1^k)$$

$$\leq \dots \leq \sum_{h=1}^H M_h^k + (\sqrt{S}+1) \cdot \sum_{h=1}^H b_h^k(s_h^k, a_h^k)$$

$$\Rightarrow \text{Reg}_k \leq \sum_{k=1}^K \hat{Q}_1^k(s_1^k, a_1^k) - V_1^{\pi_k}(s_1^k)$$

$$\leq \underbrace{\sum_{k=1}^K \sum_{h=1}^H M_h^k}_{(1)} + (\sqrt{S}+1) \cdot \underbrace{\sum_{k=1}^K \sum_{h=1}^H b_h^k(s_h^k, a_h^k)}_{(2)}$$

①: $\{M_h^k\}_{(k,h) \in [K] \times [H]}$ is a martingale difference sequence,

Since $\mathbb{E}[M_n^K | \mathcal{H}_{1:h, 1:k}] = 0$

$$|M_n^K| \leq 2H$$

fix $\hat{V}_{h+1}^k$ π_k fixed

$\Rightarrow$ By Azuma-Hoeffding, $\sum_{k=1}^K \sum_{h=1}^H M_n^K \leq 2H \cdot \sqrt{2KH \log \frac{2}{\delta}}$

② $\sum_{k=1}^K \sum_{h=1}^H b_n^K(s_h^K, a_h^K) = O\left(H \cdot \sqrt{L} \cdot \sum_{k=1}^K \sum_{h=1}^H \frac{1}{\sqrt{N_h^K(s_h^K, a_h^K)}}\right)$

$\sum_{h,s,a} \sum_{i=1}^{N_h^K(s,a)} \frac{1}{\sqrt{i}}$

★ Pigeonhole principle

(s, a, h)

$\leq 2 \sum_{h,s,a} \sqrt{N_h^K(s,a)}$

$1, 2, \dots, N_h^K(s,a)$

(by $\sum_{i=1}^N \frac{1}{\sqrt{i}} \leq \int_0^N \frac{1}{\sqrt{x}} dx = 2 \cdot \sqrt{x} \Big|_{x=0}^N = 2\sqrt{N}$)

$\sum_{h,s,a} \sqrt{N_h^K(s,a)} \leq \sqrt{HSA} \cdot \sqrt{\sum_{h,s,a} N_h^K(s,a)}$

$= \sqrt{HSA \cdot T}$

$\rightarrow$ total # of visits (samples)

$T = K \cdot H$

$\Rightarrow \sum_{k=1}^K \sum_{h=1}^H b_n^K(s_h^K, a_h^K) = O\left(\sqrt{SA \cdot H^3 T L}\right)$

$\Rightarrow$ Regret Bound of UCB-VI.

$$\text{Reg}_K \lesssim \sqrt{H^2 T} + \sqrt{S} \cdot \sqrt{SAH^3 T L} = \underline{\underline{SH^2 \sqrt{AKL}}}$$

* Sample Complexity: submit $\hat{\pi} = \text{Unif}[\pi_{1:k}]$

$\Rightarrow$ To achieve an ε -optimal solution,

$$\begin{aligned} V_1^*(s_1) - V_1^{\hat{\pi}}(s_1) &= \frac{1}{K} \sum_{k=1}^K V_1^*(s_1) - V_1^{\pi_k}(s_1) \\ &= \frac{\text{Reg}_K}{K} = \frac{SH^2 \sqrt{AL}}{\sqrt{K}} \leq \varepsilon \end{aligned}$$

$$\Rightarrow K = \frac{S^2 H^4 A L}{\varepsilon^2} \text{ iterations}$$

$$\Rightarrow T = KH = \frac{S^2 H^5 A L}{\varepsilon^2} \text{ samples}$$

trajectory-level mixture

1. Sample $i \sim \text{Unif}[k]$
2. Roll-out the whole trajectory by π_i

Suboptimal (in S)

Concentration Analysis (cont'd)

Recall: for any fixed $\underline{f} : S \mapsto [0, H]$:

$$\text{w.p. } \geq 1 - \delta : \left| \left(\hat{\underline{P}}_n^k(s, a) - \underline{P}_n(s, a) \right) \cdot \underline{f} \right| \lesssim H \cdot \sqrt{\frac{L}{N_n^k(s, a)}}$$

In optimism: $\underline{f} = \underline{V}_{n+1}^*$

$b_n^k(s, a)$

$$\underline{\underline{f}} = \hat{V}_{h+1}^k \quad (\text{history-dependent})$$

Claim 2: w.p. $\geq 1-\delta$. $\forall k, h.s.a.$ $\forall f: S \mapsto [0, H]$

$$\underline{\underline{|(\hat{P}_h^f(s,a) - P_h(s,a)) \cdot f| \lesssim H \cdot \sqrt{\frac{8L}{N_h^f(s,a)}}}}$$

Technical thing: covering

Covering

Goal: uniform convergence

X_i i.i.d. sample

(Hoeffding)

$$\sup_{f \in \mathcal{F}} \left| \mathbb{E}[f(X)] - \frac{1}{n} \sum_{i=1}^n f(X_i) \right|$$

$$\forall f \in \mathcal{F}: \underline{\underline{w.p. \geq 1-\delta: |\mathbb{E}[f(X)] - \frac{1}{n} \sum_{i=1}^n f(X_i)| \leq \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}}}$$

$$\Rightarrow \underline{\underline{w.p. \geq 1-\delta: \forall f \in \mathcal{F}: |\mathbb{E}[f(X)] - \frac{1}{n} \sum_{i=1}^n f(X_i)| \leq \sqrt{\frac{1}{2n} \log \frac{2|\mathcal{F}|}{\delta}}}}$$

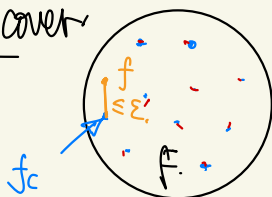
def (ε -cover) $\mathcal{F}_\varepsilon$ is an ε -cover of $\mathcal{F}$:

if $\forall f \in \mathcal{F}. \exists \underline{\underline{f_c}} \in \mathcal{F}_\varepsilon. \underline{\underline{\|f - f_c\|_\infty \leq \varepsilon}}$

covering
number

Let $N_\varepsilon(\mathcal{F})$ be the size of the smallest ε -cover

$$w.p. \geq 1-\delta: \forall f_c \in \mathcal{F}_\varepsilon: \left| \mathbb{E}[f(X)] - \frac{1}{n} \sum_{i=1}^n f(X_i) \right| \leq \sqrt{\frac{1}{2n} \log \frac{2N_\varepsilon}{\delta}}$$



$$\forall f \in \mathcal{F}: \quad \left| \mathbb{E}[f(x)] - \frac{1}{n} \sum_{i=1}^n f(x_i) \right|$$

$$\exists f \in \mathcal{F}_\varepsilon: \quad \left| \mathbb{E}[f(x)] - \frac{1}{n} \sum_{i=1}^n f(x_i) \right| \rightarrow \text{estimation error (in finite cover)}$$

$$\underbrace{\|f - f_c\|_\infty \leq \varepsilon}_{\text{approximation error}} + \left| \mathbb{E}[f(x)] - \mathbb{E}[f_c(x)] \right| + \left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \frac{1}{n} \sum_{i=1}^n f_c(x_i) \right|$$

$$\leq \underbrace{\sqrt{\frac{1}{2n} \log \frac{2N_\varepsilon}{\delta}}}_{\text{①}} + \underbrace{2\varepsilon}_{\text{②}} \quad \Delta.$$

$$\varepsilon \uparrow \quad \text{②} \uparrow \quad \text{①} \downarrow$$

$$\varepsilon \downarrow \quad \text{②} \downarrow \quad \text{①} \uparrow$$

$\rightarrow$ trade-off $\rightarrow$ tune ε

$\log N_\varepsilon$: metric entropy (typically: grow with ε very slowly

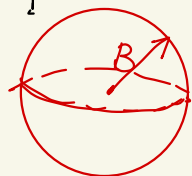
Typically: $\varepsilon = \sqrt{\frac{1}{2n} \log \frac{2N_\varepsilon}{\delta}}$ e.g. $\left(\log \frac{1}{\varepsilon} \right)$

Example: covering number of a linear class

$$\mathcal{F} = \{x \mapsto \phi(x)^T \theta : \|\theta\|_2 \leq B, \theta \in \mathbb{R}^d\}$$

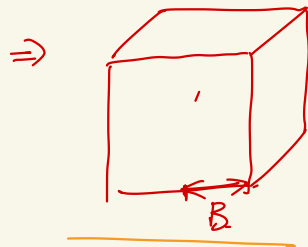
$$\|\phi(x)\|_2 \leq 1$$

$$N_\varepsilon(\mathcal{F}) = N_\varepsilon(\text{Ball}_2(B))$$



$$\text{Ball}_2(B) \rightarrow \text{Ball}_\infty(B)$$

$$(\|\theta\|_2 \leq B \Rightarrow \|\theta\|_\infty \leq B)$$



A hand-drawn 1D number line with a central point. To the left, there is a shaded interval of length \$\epsilon'\$, indicated by a bracket and a double arrow. To the right, there is an interval of length \$B\$, also indicated by a bracket and a double arrow. The diagram is followed by the expression \$\Rightarrow \frac{2B}{\epsilon'} + 1\$.

$$\Rightarrow \frac{2B}{\epsilon'} + 1$$

per-coordinate

$$\|\theta - \theta_c\|_2 \leq \sqrt{d} \cdot \|\theta - \theta_c\|_\infty$$

$$\Rightarrow \mathcal{N}_\epsilon = \left(\frac{2B}{\epsilon'} + 1 \right)^d$$

$$\leq \sqrt{d} \cdot \epsilon' = \epsilon$$

$$= \left(\frac{2B\sqrt{d}}{\epsilon} + 1 \right)^d$$

↑
 ϵ -covering number
of an d -dimensional
 L_2 -ball with radius B

$$\|f - f_c\|_\infty = \max_x |\phi(x)^\top (\theta - \theta_c)|$$

$$\leq \max_x \|\phi(x)\|_2 \cdot \|\theta - \theta_c\|_2$$

$$\leq \epsilon$$

$$\Rightarrow \log \mathcal{N}_\epsilon \lesssim d \cdot \log \frac{B\sqrt{d}}{\epsilon}$$

Back to concentration analysis: $f: \underline{S} \rightarrow [0, H], (|\mathbb{R}^{|\mathcal{S}|}|)$

$$\rightarrow \epsilon\text{-covering number of } \mathcal{F} : |\mathcal{N}_\epsilon| \leq \left(1 + \frac{2H\sqrt{S}}{\epsilon} \right)^S$$

$$\Rightarrow \log |\mathcal{N}_\epsilon| \lesssim \underline{S \cdot \log H\sqrt{S}/\epsilon}$$

$$\underline{\forall f \in F: |\hat{P}_h^k(s, a) - P_h(s, a)| \cdot f}$$

$$\leq |\hat{P}_h^k(s, a) - P_h(s, a)| \cdot f_c + 2 \cdot \|f - f_c\|_\infty$$

where $f_c \in \underline{F_\varepsilon}$ (fixed ε -net of F)

$$\lesssim H \cdot \sqrt{\frac{1}{N_h^k(s, a)} \log \frac{SAKH |W_\varepsilon|}{\delta}} + \underline{2\varepsilon}$$

$$\lesssim H \cdot \sqrt{\frac{\delta}{N_h^k(s, a)} \log \frac{SAKH}{\delta}} \quad (\text{by } \varepsilon = O\left(H \cdot \sqrt{\frac{SL}{N_h^k(s, a)}}\right) !)$$

Hoeffding-type concentration $\rightarrow$ Azuma's ineq.

Bernstein-type concentration $\rightarrow$ Freedman's ineq

Claim 3: w.p. $\geq 1 - \delta$; $\forall k, h, s, a$. $\forall f: S \rightarrow [0, H]$;

$$|\hat{P}_k^h(s, a) - P_h(s, a)| \cdot f \leq \underbrace{E_{s' \sim P_h(s, a)} [f(s')]}_H + \frac{\varepsilon_h^k(s, a)}{\xi_h^k(s, a)}$$

Concentration:

$$|\hat{P}_h^k(s' | s, a) - P_h(s' | s, a)| \leq \sqrt{\frac{2 P_h(s' | s, a) \cdot L}{N_h^k(s, a)}} + \frac{3L}{N_h^k(s, a)} \quad \left(\frac{3H^2 \delta \cdot L}{N_h^k(s, a)} \right)$$

$$f: V_1 - V_2$$

AM-QM inequality

$$|(\hat{P}_h^k(s, a) - P_h(s, a)) \cdot \hat{V}_{h+1}^k|$$

→ Claim 3

$$= |(\hat{P}_h^k(s, a) - P_h(s, a)) \cdot (\hat{V}_{h+1}^k - V_{h+1}^*)|$$

$$+ |(\hat{P}_h^k(s, a) - P_h(s, a)) \cdot \underbrace{V_{h+1}^*}_{\text{fixed}}| \rightarrow \text{Claim 1}$$

Bound, $\mathbb{E}_{S' \sim P_h(s, a)} [\hat{V}_{h+1}^k(s') - V_{h+1}^*(s')]$

$$\hat{V}_{h+1}^k(s') - V_{h+1}^*(s') \leq \frac{e}{4} \cdot \mathbb{E} \left[\sum_{h'=h+1}^H \left(\underbrace{b_{h'}^k(S_{h'}, a_{h'})}_{S_{h+1}=s'} + \underbrace{\frac{1}{\xi_{h'}}^k(S_{h'}, a_{h'})}_{S_{h+1}=s'} \right) \right]$$

$$\Rightarrow \text{Reg}_k \lesssim \sum_{k=1}^K \sum_{h=1}^H M_h^k + \underbrace{b_h^k(S_h^k, a_h^k)}_{\text{Martingale Difference}} + \underbrace{\frac{1}{\xi_h^k(S_h^k, a_h^k)}}_{\text{logarithmic}}$$

$$\sum_{i=1}^{N_h^k} \frac{1}{i}$$

Martingale Difference

$$\sum_k \sum_h b_h^k = O(\sqrt{SAH^3 TL})$$

$$\sum_k \sum_h \frac{1}{\xi_h^k} = O(H^2 SL \cdot \sum_k \sum_h \frac{1}{N_h^k}) = O(H^2 SL \cdot \sum_{h, S, A} \sum_{i=1}^{N_h^k} \frac{1}{i})$$

$$= O(H^3 S^2 A \log k)$$

NOT dominant term!

c.f. Jin et al. 2020. Provably Efficient Reinforcement Learning with Linear Function Approximation

(Jiang et al. 2017)

Diagram illustrating the decomposition of the tensor product of two representations:

$$\underline{S} \otimes \underline{A} = \underline{S} \otimes \underline{\Phi} \oplus \underline{\Psi}$$

The diagram shows three boxes representing representations. The first box is labeled $\underline{S} \otimes \underline{A}$. The second box is labeled $\underline{S} \otimes \underline{\Phi}$, with a blue circle around the $\underline{\Phi}$ and a blue arrow pointing to it from the text $\phi(s, a) \in \mathbb{R}^d$. The third box is labeled $\underline{\Psi}$, with a blue arrow pointing to it from the text $\psi(s)$. The decomposition is indicated by an equals sign and a plus sign.

$$R(s, a) = \phi(s, a)^T \Theta_R \quad (\|\phi(s, a)\|_2 \leq 1)$$

$$\begin{aligned} \langle \quad & \underline{P_h(\cdot | s, a)} = \underline{\phi(s, a)^T \mu_h(\cdot)} \\ & \underline{\Gamma_h(s, a) = \phi(s, a)^T \theta_h} \end{aligned}$$

(assume known deterministic reward)

$Q_h^\pi(s, a)$ is linear in $\phi(s, a)$

$$\exists \{W_h^\pi\}_{h=1}^H \quad \text{s.t.} \quad \underline{Q_h^\pi(s,a) = \phi(s,a)^\top W_h^\pi}.$$

Bellman Equation:
$$Q_h^\pi(s,a) = \underbrace{r_h(s,a)}_{f'} + \underbrace{V_h^\pi(s,a)}_{f} \cdot V_{h+1}^\pi$$

$$= \phi(s,a)^\top \theta_h + \sum_{s'} \phi(s,a)^\top \mu_h(s') \cdot V_{h+1}^\pi(s')$$

$$= \phi(s,a)^\top \cdot \left(\theta_h + \sum_{s'} \mu_h(s') \cdot V_{h+1}^\pi(s') \right)$$

Annotations: $f' \nleftrightarrow f$, f : linear class, W_h^π

NOT necessarily linear

$$\underline{Q_h^*(s,a)} = \underbrace{r_h(s,a)}_{\text{target}} + \underbrace{V_{h+1}^*}_{\text{feature}}$$

$$\underbrace{\phi(s,a)^T}_{\text{feature}} \underbrace{W_h^*}_{\text{feature}}$$

$$V_{h+1}^*(s') = \max_{a \in A} Q_{h+1}^*(s', a')$$

Bellman Completeness:
 $\forall f \in F: \mathcal{T}f \in F.$

$$\forall f \in \mathbb{R}^{SA}: \uparrow f \in F$$

Algorithm: UCB-LSVI (Upper-Confidence Bound, Least-Square Value Iteration)

At round $k=1, 2, \dots, K$:

$$\hat{V}_{H+1}^k \equiv 0$$

At timestep $h=H, \dots, 1$:

$$\hat{W}_h^k = (\Delta_h^k)^{-1} \cdot \sum_{i=1}^{k-1} \phi(s_h^i, a_h^i) \left(\underbrace{r_h(s_h^i, a_h^i)}_{\text{noise}} + \hat{V}_{h+1}^k(s_{h+1}^i) \right)$$

$$\text{where } \Delta_h^k = I + \sum_{i=1}^{k-1} \phi(s_h^i, a_h^i) \cdot \phi(s_h^i, a_h^i)^T$$

$$\tilde{Q}_h^k(s, a; \hat{V}_{h+1}^k) = \phi(s, a)^T \hat{W}_h^k$$

UCB

$$\hat{Q}_h^k(s, a) = \left[\tilde{Q}_h^k(s, a; \hat{V}_{h+1}^k) + \underbrace{b_h^k(s, a)}_{\text{noise}} \right]$$

$$\hat{V}_h^k(s) = \max_{a \in A} \hat{Q}_h^k(s, a)$$

$$\text{Submit } \pi^k(s) = \operatorname{argmax}_{a \in A} \hat{Q}_h^k(s, a)$$

Ridge Regression

$$y_k = \underbrace{x_k^T \theta}_{\text{noise}} + \underbrace{\varepsilon_k}_{\text{noise}}$$

$$\hat{\theta}_k = \operatorname{argmin}_{\theta} \sum_{i=1}^k (y_i - x_i^T \theta)^2 + \lambda \|\theta\|_2^2$$

$$= \underbrace{\Delta_k^{-1}}_{\text{noise}} \sum_{i=1}^k \underline{x_i} y_i$$

$$\text{where } \Delta_k = I + \sum_{i=1}^k x_i x_i^T$$

and roll out π^* to collect trajectory $(S_1^k, a_1^k, \dots, S_h^k, a_h^k)$

Concentration (estimation error) in Ridge Regression: $\forall x: \begin{cases} y(x) = x^T \theta^* \\ \hat{y}_k(x) = x^T \hat{\theta}_k^* \end{cases}$

Confidence set

$$|\hat{y}_k(x) - y(x)| \leq \|x\|_{\Lambda_k^{-1}} \cdot \mathcal{O}(H \sqrt{d + \log \frac{1}{\delta}})$$

effective
sample size

$$x_i = \phi(s_i^h, a_i^h)$$

$$\underline{y}_i = r_h(s_i^h, a_i^h) + V(s_{h+1}^i) \quad \text{for } V = \boxed{V_{h+1}^k}$$

(history-dependent)

For any fixed V : w.p. $\geq 1 - \delta$: $\forall (s, a)$

$$\left| \underbrace{\tilde{Q}_h^k(s, a; V)}_{\hat{y}_k(s, a)} - \underbrace{\Gamma_h(s, a)}_{y(s, a)} - \underbrace{P_h(s, a) \cdot V}_{\dots} \right| \leq \underbrace{\| \phi \|_{\Lambda_k^{-1}} \cdot \mathcal{O}(H \sqrt{d})}_{\dots}$$

"Bonus term": $b_h^k(s, a) = \| \phi(s, a) \|_{\Lambda_k^{-1}} \cdot \mathcal{O}(H \sqrt{d + \log \frac{1}{\delta}})$

$$V_{w, \Lambda} = \max_a \left(\underbrace{w^T \phi(s, a)}_{\dots} + \underbrace{\beta \cdot \sqrt{\phi(s, a)^T \Lambda^{-1} \phi(s, a)}}_{\dots} \right)_{[0, H]}$$

β constant

Covering of $\underline{V} = \left\{ \underline{V}_{w, \Lambda} : \| \underline{w} \|_2 \leq \underline{B}, \sigma_{\min}(\underline{\Lambda}) \geq 1 \right\}$

$$\|\hat{w}_h^k\|_2 = \|(\Lambda_h^k)^{-1} \cdot \sum_{i=1}^{k-1} \phi_h^i(r_h^i + \hat{V}_{h+1}^k(s_{h+1}^i))\|_2$$

$$\Lambda_h^k \preceq I \Rightarrow \sum_{i=1}^{k-1} \|\phi_h^i \cdot (r_h^i + \hat{V}_{h+1}^k(s_{h+1}^i))\|_2 \leq K(H+1) = \underline{B}$$

1. Create an $\alpha/2$ -cover of $\{w: \|w\|_2 \leq B\}$.

$$\rightarrow \log N_w = \tilde{O}(d)$$

2. Create an $\alpha^2/4$ -cover of "linear" class

$$\{\phi^T \Lambda^{-1} \phi : \sigma_{\min}(\Lambda) \geq 1\}$$

$$\rightarrow \log N_\Lambda = \tilde{O}(d^2)$$

$$\begin{array}{l} \text{ball in } \mathbb{R}^d; \\ N_\epsilon = \left(1 + \frac{2B\sqrt{d}}{\epsilon}\right)^d \\ \text{linear in } \{\phi_i \phi_j\}_{i,j} \subseteq \mathbb{R}^{d^2}. \end{array}$$

3. Let Λ_c be the covering center of $\{\phi^T \Lambda^{-1} \phi : \sigma_{\min}(\Lambda) \geq 1\}$

$$\Rightarrow \alpha/2\text{-cover of } \{\sqrt{\phi^T \Lambda^{-1} \phi} : \sigma_{\min}(\Lambda) \geq 1\}$$

$$|\sqrt{\phi^T \Lambda^{-1} \phi} - \sqrt{\phi^T \Lambda_c^{-1} \phi}| \leq \sqrt{|\phi^T \Lambda^{-1} \phi - \phi^T \Lambda_c^{-1} \phi|} \leq \sqrt{\frac{\alpha^2}{4}} = \frac{\alpha}{2}$$

4. Additive Composition. $\rightarrow \alpha$ -cover $\left\| \begin{array}{c} \text{shaded box} \\ \leq \frac{\alpha}{2} \leq \frac{\alpha}{2} \rightarrow \leq \alpha \end{array} \right\|$

$\min\{H, \cdot\}$ composition

$\max_a(\cdot)$ composition

non-expansion in $\|\cdot\|_\infty$

$$|\max_a f(s,a) - \max_a f'(s,a)|$$

$$\leq \max_a |f(s,a) - f'(s,a)|$$

$$\Rightarrow \log N_{\nu} = \tilde{O}(d^2)$$

Let $\mathcal{V}_{\alpha}$ be a finite α -cover of $\mathcal{V}$ (contains all $\hat{V}_{h+1}^k$)

$$\text{w.p.} \geq 1 - \delta : \forall \underline{V_c} \in \underline{\mathcal{V}_{\alpha}} : \forall (s, a)$$

$$\left| \underbrace{\tilde{Q}_h^k(s, a; \underline{V_c})}_{\hat{y}_k(x)} - \underbrace{r_h(s, a)}_{y(x)} - \underbrace{P_h(s, a) \cdot \underline{V_c}}_{y(x)} \right| \leq \|\phi(s, a)\| (\Lambda_h^k)^{-1} \cdot \tilde{O}\left(H \cdot \sqrt{d + \log \frac{N_{\nu}}{\delta}}\right)$$

$\forall V \in \mathcal{V}$:

$$\left| \tilde{Q}_h^k(s, a; V) - r_h(s, a) - P_h(s, a) \cdot V \right| \leq \underbrace{\|\phi(s, a)\| (\Lambda_h^k)^{-1} \cdot \tilde{O}\left(H(d + \sqrt{\log \frac{1}{\delta}})\right)}_{\tilde{O}\left(H(d + \sqrt{\log \frac{1}{\delta}})\right)}$$

$$\leq \|\phi(s, a)\| (\Lambda_h^k)^{-1} \cdot \tilde{O}\left(H(d + \sqrt{\log \frac{1}{\delta}})\right) \leftarrow \text{estimation error}$$

$$+ \left| \tilde{Q}_h^k(s, a; V) - \tilde{Q}_h^k(s, a; \underline{V_c}) \right| \quad \text{①} \leftarrow \text{approximation error}$$

$$+ \left| P_h(s, a) \cdot \underline{V} - P_h(s, a) \cdot \underline{V_c} \right| \quad \text{②}$$

$$\text{②} \leq \|V - V_c\|_{\infty} \leq \alpha$$

$$\text{①} = \left| \underbrace{\phi(s, a)^T}_{\leq 1} \cdot \underbrace{(\Lambda_h^k)^{-1}}_{\leq I} \cdot \sum_{i=1}^{k-1} \underbrace{\phi(s_h^i, a_h^i)}_{\leq 1} \cdot \underbrace{(V(s_{h+1}^i) - V_c(s_{h+1}^i))}_{\leq 1} \right|$$

$$\leq (k-1) \alpha$$

By $\alpha = O\left(\frac{H}{\kappa} \left(d + \sqrt{\log \frac{1}{\delta}}\right)\right) \Rightarrow \underline{\text{w.p. } \geq 1 - \delta}$:

Confidence Set $\left| \underline{\hat{Q}}_h^k(s, a) - \underline{r}_h(s, a) - \underline{P}_h(s, a) \cdot \hat{V}_{h+1}^k \right|$ history-dependent

$\leq \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}} \cdot \tilde{O}\left(H \left(d + \sqrt{\log \frac{1}{\delta}}\right)\right)$
(addition $\sqrt{d}$ factor)

Bonus term:

$\underline{b}_h^k(s, a) = \beta \cdot \sqrt{\phi(s, a)^\top (\Lambda_h^k)^{-1} \phi(s, a)}$ where $\beta = \tilde{O}\left(H \left(d + \sqrt{\log \frac{1}{\delta}}\right)\right)$

Claim: (Optimism) $\forall h, k, s, a$: $\begin{cases} \hat{Q}_h^k(s, a) \geq Q_h^*(s, a) \\ \hat{V}_h^k(s, a) \geq V_h^*(s, a) \end{cases}$
w.p. $\geq 1 - \delta$.

Proof: $\hat{Q}_h^k(s, a) = \left[\tilde{Q}_h^k(s, a) + \underline{b}_h^k(s, a) \right]_{[0, H]}$
I.H. $\hat{V}_{h+1}^k \geq V_{h+1}^*$

If $\hat{Q}_h^k(s, a) = H \geq Q_h^*(s, a)$ (trivially)

If $\hat{Q}_h^k(s, a) = \tilde{Q}_h^k(s, a) + b_h^k(s, a)$

$\geq r_h(s, a) + P_{h+1}(s, a) \cdot \hat{V}_{h+1}^k$ 2 Confidence Set

$\geq r_h(s, a) + P_{h+1}(s, a) \cdot V_{h+1}^*$ $\triangleleft$ I.H.

$= \underline{Q}_h^*(s, a)$

□

Regret Analysis of LCB-LSVI.

$$\begin{aligned}
 \text{Reg}_k &= \sum_{i=1}^k \underline{V_i^*(s_i^k) - V_i^{\pi^k}(s_i^k)} \\
 &= \sum_{i=1}^k \underline{Q_i^*(s_i^k, \pi_i^*(s_i^k)) - V_i^{\pi^k}(s_i^k)} \\
 &\leq \sum_{i=1}^k \underline{\hat{Q}_i^k(s_i^k, \pi_i^*(s_i^k)) - V_i^{\pi^k}(s_i^k)} \\
 &\leq \sum_{i=1}^k \underline{\hat{Q}_i^k(s_i^k, a_i^k) - V_i^{\pi^k}(s_i^k)}
 \end{aligned}$$

⌋ Optimism

⌋ Greedy:
 $a_i^k = \arg \max_a Q_i^k(s_i^k, a)$

Bellman Equation

$$\begin{aligned}
 &\hat{Q}_n^k(s_n^k, a_n^k) - V_n^{\pi^k}(s_n^k) \\
 &\leq \phi(s_n^k, a_n^k)^T \hat{W}_n^k + b_n^k(s_n^k, a_n^k) - \underbrace{r_n(s_n^k, a_n^k) - \mathbb{P}_n^k(s_n^k, a_n^k) \cdot V_{n+1}^{\pi^k}}_{\text{Bellman Equation}} \\
 &\leq \cancel{r_n(s_n^k, a_n^k)} + \mathbb{P}_n^k(s_n^k, a_n^k) \cdot \hat{V}_{n+1}^k + \underline{b_n^k(s_n^k, a_n^k)} \\
 &\quad + \underline{b_n^k(s_n^k, a_n^k)} - \cancel{r_n(s_n^k, a_n^k)} - \mathbb{P}_n^k(s_n^k, a_n^k) \cdot V_{n+1}^{\pi^k} \\
 &= 2b_n^k(s_n^k, a_n^k) + \mathbb{P}_n^k(s_n^k, a_n^k) \cdot (\hat{V}_{n+1}^k - V_{n+1}^{\pi^k}) \\
 &\quad + \underline{\hat{Q}_{n+1}^k(s_{n+1}^k, a_{n+1}^k) - V_{n+1}^{\pi^k}(s_{n+1}^k)} - (\hat{V}_{n+1}^k(s_{n+1}^k) - V_n^{\pi^k}(s_{n+1}^k)) \\
 &\Rightarrow \text{Reg}_k \leq \sum_{i=1}^k \sum_{n=1}^H \left(\underline{b_n^k} + \underline{M_n^k} \right)
 \end{aligned}$$

$\underline{M_n^k}$: Martingale difference

Azuma-Hoeffding: $\sum_{k=1}^K \sum_{h=1}^H M_h^k \lesssim H \cdot \sqrt{HK \cdot \log \frac{1}{\delta}}$. $\mathbb{E}[M_h^k | \mathcal{H}_h^k] = 0$. → $[HK]$

Final step: bounding accumulative bonus (dominate term)

$$\sum_{k=1}^K \sum_{h=1}^H \underline{b}_h^k(s_h^k, a_h^k) = \sum_{k=1}^K \sum_{h=1}^H \beta \cdot \sqrt{\phi_h^{k \top} (\Lambda_h^k)^{-1} \phi_h^k}$$

$$= \beta \cdot \sum_{h=1}^H \cdot \sqrt{K} \cdot \sqrt{\sum_{k=1}^K \phi_h^{k \top} (\Lambda_h^k)^{-1} \phi_h^k}$$

$$= \beta \cdot H \cdot \sqrt{K} \cdot \sqrt{d \log K}$$

$$\leq O(H \cdot d \cdot H \cdot \sqrt{K} \cdot \sqrt{d \log K})$$

$$\sum_K x_K^T \Lambda_K^{-1} x_K$$

$$\leq \sqrt{2d \log(K+1)}$$

(Elliptical Potential Lemma)

$$\rightarrow \text{Reg}_K \leq \tilde{O}(\sqrt{d^3 H^3 T L})$$

$\uparrow$
 HK
 (total # of samples)

log-factor.



Sample Complexity: $\tilde{O}(d^3 H^4 / \epsilon^2)$

Note: NOT a minimax optimal result!

Lower Bound: $\Omega(\sqrt{d H^2 T})$

$\left\{ \begin{array}{l} d \text{ factor} \\ H \text{ factor} \end{array} \right.$

Bernstein-type concentration.

Value-Targeted Regression

in linear MDP: least-square
(ridge regression)

$$I_{Ph}(s'|s,a) = \underbrace{\mu^*(s')}_{\substack{\text{size } d \\ \text{size } d}} \phi(s,a)$$

Linear Mixture MDP

$$I_{Ph}(s'|s,a) = \langle \phi(s'|s,a), \underbrace{\Theta_h^*}_{\substack{\text{size } d \\ \text{size } d}} \rangle$$

linear MDP $\subseteq$ Linear mixture MDP

$$\phi(s'|s,a) = \phi(s,a) \odot \mu^*(s')$$

Note: they are not comparable !

LCRL-VTR

(Upper Confidence Reinforcement Learning
- Value Target Regression)

Model-based SL framework:

$$\hat{Q}_h^k, \hat{V}_h^k : \left| \underbrace{I_{Ph}(s_h^k, a_h^k) \cdot \hat{V}_{h+1}^k}_{\text{unbiased estimate: } \hat{V}_{h+1}^k(s_{h+1}^k)} \right|$$

$$\text{oracle: } p(d) \rightarrow = \langle \sum_{s'} \phi(s'|s_h^k, a_h^k) \cdot \hat{V}_{h+1}^k(s'), \Theta_h^* \rangle$$

$$\triangleq \langle \underbrace{\phi_{\hat{V}_{h+1}^k}(s_h^k, a_h^k)}_{\text{want to estimate}}, \Theta_h^* \rangle$$

C.f.1. Ayoub et al. 2020. Model-based reinforcement learning with value-targeted regression.

C.f.2. Zhou et al. 2021. Nearly minimax optimal reinforcement learning for linear mixture mdp's.

$$\phi(s'|s,a) \in \mathbb{R}^d$$

$$\phi_i(s'|s,a) = \underline{P_i(s'|s,a)}$$

$$I_{Ph}(s'|s,a)$$

$$= \sum_{i=1}^d P_i(s'|s,a) \cdot (\Theta_h^*)_i$$

$$\underline{\text{Model-based } \underline{\mathcal{P}}_i(\underline{M})}$$

→ Ridge Regression: $x_k: \phi(\hat{V}_{h+1}^k(s_h^k, a_h^k)) \Rightarrow \boxed{y_k | x_k}$

$y_k: \hat{V}_{h+1}^k(s_{h+1}^k) \in [0, H]$

$\hat{\theta}_h^k = \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \|\theta\|_2^2 + \sum_{i=1}^{K-1} \left(\langle \phi_{\hat{V}_{h+1}^i}(s_h^i, a_h^i), \theta \rangle - \hat{V}_{h+1}^i(s_{h+1}^i) \right)^2$

$= (\Lambda_h^k)^{-1} \cdot \sum_{i=1}^{K-1} \phi_{\hat{V}_{h+1}^i}(s_h^i, a_h^i) \cdot \hat{V}_{h+1}^i(s_{h+1}^i)$

where $\Lambda_h^k = I + \sum_{i=1}^{K-1} \phi_{\hat{V}_{h+1}^i}(s_h^i, a_h^i) \cdot \phi_{\hat{V}_{h+1}^i}(s_h^i, a_h^i)^T$

Value-Targeted Regression (VTR)

By one-shot estimation error of ridge regression:

$|\hat{y}_k(x) - y(x)|$

$\hat{P}_h^k(s, a) - P_h(s, a)$

$= | \langle \phi_{\hat{V}_{h+1}^k}(s, a), \hat{\theta}_h^k - \theta_h^* \rangle | \leq \| \phi_{\hat{V}_{h+1}^k}(s, a) \|_{(\Lambda_h^k)^{-1}} \cdot \underbrace{\left[\mathcal{O}\left(H \sqrt{d \log \frac{1}{\delta}}\right) \right]}_{\text{covering}}$

No need for covering! (save a $\sqrt{d}$ dependence)

β_h^k

LSVI-UCB:

$x_k: \phi(s_h^k, a_h^k)$

$y_k: P_h(s_h^k, a_h^k) + \hat{V}_{h+1}^k(s_{h+1}^k)$

$y_k | x_k$

history-dependent

covering

Optimistically: Value iteration

$\hat{Q}_h^k(s, a) = \left[r_h(s, a) + \langle \phi_{\hat{V}_{h+1}^k}(s, a), \hat{\theta}_h^k \rangle + \hat{b}_h^k(s, a) \right]_{[0, H]}$

$\hat{V}_h^k(s, a) = \max_{a \in A} \hat{Q}_h^k(s, a)$

$C_h^k = \left\{ \theta: \|\hat{\theta}_h^k - \theta\|_{\Lambda_h^k} \leq \beta_h^k \right\}$

w.p. $\geq 1 - \delta$:

$\theta_h^* \in C_h^k$

$\hat{b}_h^k(s, a) = \| \phi_{\hat{V}_{h+1}^k}(s, a) \|_{(\Lambda_h^k)^{-1}} \cdot \beta_h^k$

Regret Analysis: $\text{Reg}_k = \sum_{i=1}^k V_i^*(s_i^k) - V_i^{\pi^k}(s_i^k)$

$$\leq \sum_{i=1}^k \widehat{V}_i^k(s_i^k) - V_i^{\pi^k}(s_i^k)$$

↓ optimality

$$\widehat{V}_n^k(s_n^k) - V_n^{\pi^k}(s_n^k) \leq \widehat{V}_{n+1}^k(s_{n+1}^k) - V_{n+1}^{\pi^k}(s_{n+1}^k)$$

$$+ \underbrace{\mathbb{P}_n(s_n^k, a_n^k) (\widehat{V}_{n+1}^k - V_{n+1}^{\pi^k}) - (\widehat{V}_{n+1}^k - V_{n+1}^{\pi^k})(s_{n+1}^k)}_{\rightarrow M_{n,k}}$$

$$+ 2 b_n^k(s_n^k, a_n^k)$$

→ $M_{n,k}$

↑
Martingale
Difference

$$\Rightarrow \text{Reg}_k \leq \sum_{k=1}^K \sum_{n=1}^H \underline{M}_{n,k} + \underline{2 b_n^k(s_n^k, a_n^k)}$$

$$\lesssim \beta \cdot \sum_{n=1}^H \cdot \sum_{k=1}^K \|\phi_{\widehat{V}_{n+1}^k}(s, a)\|_{(\Lambda_n^k)^{-1}}$$

$$\lesssim \beta \cdot \underbrace{H \cdot \sqrt{K}}_{O(H \cdot \sqrt{K})} \cdot \underbrace{\sqrt{\sum_{k=1}^K \phi^T(\Lambda_n^k)^{-1} \phi}}_{\leftarrow \text{elliptical potential}}$$

$$\lesssim d H^2 \sqrt{K} = d H^{\frac{3}{2}} \sqrt{K} \quad d \log K$$

In general: $\underline{\Theta}$: parameter of the state-to-value predictor
(which implies a transition model class)

$$\underline{\text{VTR}}: \widehat{\Theta} = \underset{\Theta \in \mathbb{R}^d}{\text{argmin}} \sum_i (\underline{x_i(\Theta)} - y_i)^2 \quad \left\{ \begin{array}{l} x(\Theta) = \mathbb{E}_0[\widehat{V}(s') | s, a] \\ y = \widehat{V}(s') \end{array} \right.$$

$$C = \left\{ \Theta : |x(\Theta) - x(\widehat{\Theta})| \leq \beta(\Theta) \right\} \quad \text{w.h.p. } \Theta^* \in C.$$

Eluder Dimension

UCRL-VTR⁺ [Zhou et al. 2021]

Variance control

$$\text{Reg}_K = \tilde{O}\left(\sqrt{d^2 H^2 + d H^3} \cdot \sqrt{T}\right)$$

Weighted Ridge Regression

linear Bandits: $r_t = \langle \underline{\theta}^*, a_t \rangle + \varepsilon_t$.

$$\left\{ \begin{array}{l} |\varepsilon_t| \leq \underline{R} \rightarrow \frac{R}{\underline{\sigma}_t} \\ \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}, a_t] = 0. \\ \text{Var}[\varepsilon_t | \mathcal{F}_{t-1}, a_t] \leq \underline{\sigma}_t^2 \end{array} \right. \quad (\text{heterogeneous})$$

$$\hat{\theta}_t = \underset{\theta}{\text{argmin}} \|\theta\|_2^2 + \sum_{i=1}^{t-1} \frac{(\langle \theta, a_i \rangle - r_i)^2}{\underline{\sigma}_i^2}$$

$$\hat{a}_i = \frac{a_i}{\underline{\sigma}_i}, \quad \hat{r}_i = \frac{r_i}{\underline{\sigma}_i}, \quad \hat{\varepsilon}_i = \frac{\varepsilon_i}{\underline{\sigma}_i} \rightarrow \text{upper bound of } \sigma_i^2. \rightarrow \text{Var}[\hat{\varepsilon}_i | \cdot] \leq \left(\frac{\sigma_i}{\underline{\sigma}_i}\right)^2 \leq 1.$$

Self-normalized concentration:

Hoeffding-type: $\tilde{O}(R\sqrt{d})$

Bernstein-type: $\tilde{O}(\sigma\sqrt{d} + \underline{R})$

$$\rightarrow \text{Reg}_T \lesssim R\sqrt{dT} + d \cdot \sqrt{\sum_{t=1}^T \sigma_t^2}$$

$\Downarrow$

Linear Mixture MDP:

$$\Delta \quad \text{Var}[\hat{V}_{h+1}^k(s') | s, a] = \underbrace{\left[\left\langle \phi_{\hat{V}_{h+1}^k}(s, a), \theta_h^* \right\rangle \right]}_{\substack{\text{VTR: } \tilde{\theta}_h^k \\ \text{(Ridge Regression)}}} - \underbrace{\left[\left\langle \phi_{\hat{V}_{h+1}^k}(s, a), \hat{\theta}_h^k \right\rangle \right]}_{\substack{\text{VTR: } \hat{\theta}_h^k \\ \text{(Weighted Ridge Regression)}}}^2$$

$\downarrow$
 $(\hat{\sigma}_h^k)^2$ (optimistic variance)
 $\downarrow$

Weighted Ridge Regression

Exploration with General Function Approximation

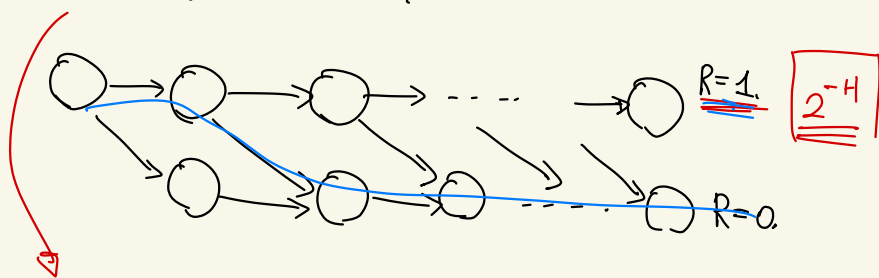
c.f. Jiang et al. 2017. Contextual decision processes with low bellman rank are pac-learnable

3 core challenges in RL:

$\left\{ \begin{array}{l} \text{temporal credit assignment.} \\ \text{generalization} \quad \checkmark \quad (\text{long-term planning}) \\ \text{exploration} \quad \star \quad \checkmark \end{array} \right.$

$\left\{ \begin{array}{l} \text{model-based} \\ \text{value-based} \\ \text{policy-based} \end{array} \right.$

Tabular: Strategic exploration.



$$\begin{aligned} \nabla J(\pi) &= \mathbb{E} \left[\frac{R(t) - \sum_h \nabla J(\pi)}{\tau} \right] \end{aligned}$$

$R_{\max}$
UCB-VI

$\Rightarrow$ Principle: optimism in face of uncertainty.

PAC-RL: find a policy π s.t. w.p. $\geq 1 - \delta$: $J(\pi^*) - J(\pi) \leq \epsilon$

Function Approximation:

$\mathcal{F}$ s.t. $Q^* \in \mathcal{F}$.

$$\mathbb{E}_{\pi} \left[\sum_{h=1}^H \Gamma_h \right]$$

Model Selection

Sample complexity:

$\text{poly}(|\mathcal{A}|, H, \log |\mathcal{F}|, \frac{1}{\epsilon}, \frac{1}{\delta})$

log-cardinality

$\rightarrow$ log covering-number

$\rightarrow$ VC dimension

Lower Bound

Bellman error at (S_h, a_h) :

$$f: \mathbb{R}^{S \times A} \rightarrow [0, H]$$

$$f(\underline{S}_h, \underline{a}_h) - \mathbb{E}_{\substack{r_h, S_{h+1} \\ \pi}} [r_h + \max_{a \in A} f(S_{h+1}, a)]$$

Average Bellman error at step h : function f roll-in policy π

$$\mathcal{E}_h(f, \pi) = \mathbb{E}_{\substack{S_h \sim d^\pi \\ a_1:h \sim \pi \\ a_h \sim \pi_f}} [f(S_h, a_h) - r_h - \max_{a \in A} f(S_{h+1}, a)]$$

$a_h, a_{h+1} \sim \pi_f$

• $\mathcal{E}_h(Q^*, \pi) = 0$ for all π, h .

$$\pi_f(s) = \operatorname{argmax}_{a \in A} f(s, a)$$

• $J(\pi) = \mathbb{E}_{\pi} [\sum_{h=0}^H r_h] = \mathbb{E}_{S \sim d_0} [V^\pi(s)] = \mathbb{E}_{S \sim d_0, a \sim \pi} [Q^\pi(s, a)]$

$J_f(\pi) = \mathbb{E}_{S \sim d_0, a \sim \pi} [f(s, a)]$

★ $J_f(\pi_f) - J(\pi_f) = \sum_{h=1}^H \mathcal{E}_h(f, \pi_f)$ ★

performance difference $\rightarrow$ Average Bellman Error

Bellman Error Telescoping

$$\mathbb{E} [f(s_1, a_1) + r_1 - f(s_1, a_2) + f(s_2, a_2) - r_2 - f(s_2, a_1) + \dots + f(s_H, a_H) - r_H]$$

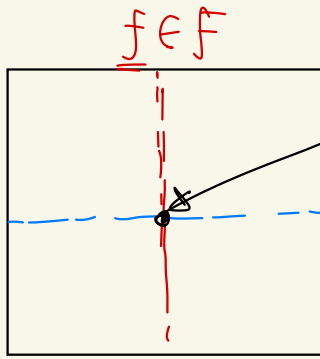
$a_h \sim \pi_f$
(on-policy)

$$\frac{J_f(\pi) - J(\pi)}{1 - \gamma} = \frac{1}{1 - \gamma} \mathbb{E}_{a \sim \pi} [f - \gamma \pi f]$$

effective horizon Bellman error

Bellman error
matrix $|F| \times |F|$.

$\pi \in \Pi_f$



$$\begin{aligned} \underline{\underline{\mathcal{E}_h(f, \pi)}} \\ = \mathbb{E} \left[\underline{\underline{f(S_h, a_h) - r_h}} \right. \\ \left. \begin{matrix} a_1: h-1 \sim \pi \\ a_h \sim \pi_{f_h} \end{matrix} - \max_{a \in A} f(S_{h+1}, a) \right] \end{aligned}$$

def (Bellman rank) M : uniform upper bound on the rank of matrices $\underline{\underline{\mathcal{E}_h(f, \pi)}}_{\pi, f}$ over $h \in [H]$

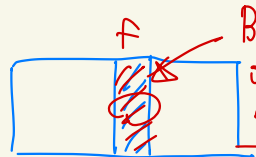
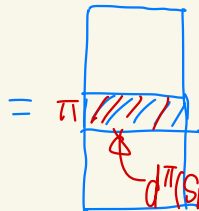
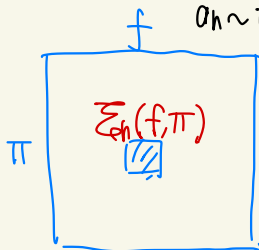
$$\exists \underline{\underline{L_h}}: F \rightarrow \mathbb{R}^M, \quad \underline{\underline{\Sigma_h}}: F \rightarrow \mathbb{R}^M, \quad \text{s.t.}$$

$$\forall f, f' \in F: \quad \underline{\underline{\mathcal{E}_h(f, \pi_{f'})}} = \langle \underline{\underline{L_h(f')}} , \underline{\underline{\Sigma_h(f)}} \rangle$$

$$\text{and } \|L_h(f')\|_2 \cdot \|\Sigma_h(f')\|_2 \leq \gamma.$$

e.g. Tabular MDP. Bellman rank $\leq \# \text{ of states}$.

$$\mathcal{E}_h(\underline{\underline{f}}, \pi) = \mathbb{E}_{\substack{a_1: h-1 \sim \pi \\ a_h \sim \pi_{f_h}}} \left[\underline{\underline{f(S_h, a_h) - r_h - \max_{a \in A} f(S_{h+1}, a)}} \right]$$



Bellman error of f on each state

$d^\pi(S_h)$: distribution over states induced by π

OLIVE (Optimism Led Iterative Value-function Elimination)

Recall: average Bellman error

Version-space elimination
(consistent)

$$\underline{\Sigma_h(f, \pi)} = \mathbb{E}_{\substack{a_{1:H} \sim \pi \\ a_h \sim \pi_f}} [f(\underline{s_h}, a_h) - r_h - \max_{a \in A} f(\underline{s_{h+1}}, a)]$$



↳ telescoping:

$$\underbrace{J_f(\pi_f)}_{f(s_1, \pi_f(s_1))} - \underbrace{J(\pi_f)}_{Q^{\pi_f}(s_1, \pi_f(s_1))} = \sum_{h=1}^H \Sigma_h(f, \pi_f)$$

$$\underbrace{[\Sigma_h(f, \pi)]}_{\pi \in \pi_f, f \in F}$$

↳ rand M matrix: Bellman Rank

① Tabular, Linear, Reactive POMDP

Function class: F . $[Q^*] \in F$

② Algorithm (informatic-theoretic)

Algorithm: $F_1 = F$

Explore-or-Terminate

for $t=1, 2, \dots$

• choose

$$f_t = \operatorname{argmax}_{f \in F_t}$$

$$J_f(\pi_f)$$

$$\pi_t = \pi_{f_t}$$

• estimate $J(\pi_t)$:

If $J(\pi_t) \geq J_{f_t}(\pi_t)$: return π_t

$$\begin{aligned} J(\pi_t) &\geq J_{f_t}(\pi_{f_t}) \\ &= \max_{f \in F_t} J_f(\pi_f) \\ &\geq J_{Q^*}(\pi^*) \\ &= J(\pi^*) + \sum_{h=1}^H \Sigma_h(Q^*, \pi^*) \\ &= J(\pi^*) \end{aligned}$$

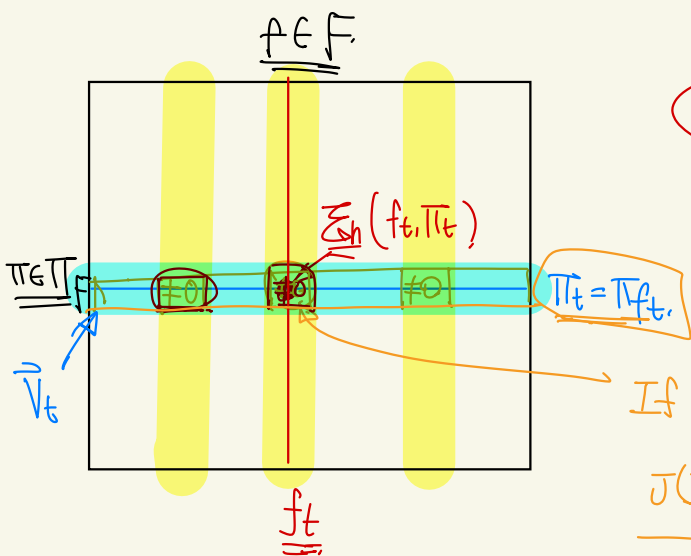
(π_t : optimal)

- estimate $\underline{\Sigma_h(f, \pi_t)}$ for all $\underline{f}, h$.
 - eliminate f . st. $\underline{\Sigma_h(f, \pi_t) \neq 0} \quad \forall h$.
- $$\Rightarrow \underline{\underline{f_{t+1} = \{f \in f_t : \underline{\Sigma_h(f, \pi_t) = 0}\}}}$$

$$\Sigma_h(f_t, \pi_t) \neq 0$$

Claim 1: Q^* will never be eliminated! ($\Sigma_h(Q^*, \underline{\pi}) = 0$)
 $\Rightarrow \underline{Q^* \in f_t} \quad \forall t$.

Claim 2: Poly rounds for eliminating all $h \in [H]$ ★



$$\text{Bellman Rank} \leq M$$

$\Rightarrow$ At most M linearly independent rows!

$$\text{If } \underline{\Sigma_h(f_t, \pi_t) = 0}$$

$$\underline{J(\pi_t) = J_{f_t}(\pi_t) + \sum_{h=1}^H \Sigma_h(f_t, \pi_t)}$$

$$= J_{f_t}(\pi_t) \rightarrow \text{return } \pi_t$$

At round t : f_t has not been eliminated at all previous $t-1$ rounds.

$$\underline{V_1[f_t] = V_2[f_t] = \dots = V_{t-1}[f_t] = 0}$$

$$V_t[f_t] \neq 0$$

$\Rightarrow \underline{\underline{\vec{V}_t}}$ cannot be a linear combination of $\underline{\vec{V}_1}, \dots, \underline{\vec{V}_{t-1}}$.

⇒ we find a new "elimination" direction,

⇒ At most M rounds to eliminate for a h f H

OLIVE with statistical estimation

$$J(\pi^*) - J(\hat{\pi}) \leq H \cdot \underline{\underline{\varepsilon}}$$

$F_t = F$. At round $t=1, 2, \dots$:

• Optimism: $f_t = \underset{f \in F_t}{\operatorname{argmax}} J_f(\pi_t)$. $\pi_t = \pi_{f_t}$

need oracle.

On Oracle-Efficient
PAC RL with
Rich Observations
Dann et al. 2018

$$\tilde{\sum}_h(f_t, \pi_t) = \frac{1}{n_{eval}} \sum_{i=1}^{n_{eval}} \left(f_t(S_h^{(i)}, a_h^{(i)}) - r_h^{(i)} - f_t(S_{h+1}^{(i)}, a_{h+1}^{(i)}) \right)$$

fixed f_t .
 $\forall h$

MC: n_{eval} trajectories followed by π_t

• If $\sum_{h=1}^H \tilde{\sum}_h(f_t, \pi_t) \leq H \cdot \underline{\underline{\varepsilon_1}}$ terminate & output π_t .

$$\| J_{f_t}(\pi_t) - J(\pi_t) = \sum_{h=1}^H \tilde{\sum}_h(f_t, \pi_t) \approx \sum_{h=1}^H \tilde{\sum}_h(f_t, \pi_t) \leq H \cdot \varepsilon_1$$

• Pick any $h_t \in [H]$ such that $\tilde{\sum}_{h_t}(f_t, \pi_t) > \varepsilon_1$.

fixed h_t
 $\forall f$

existence is guaranteed
by Pigeonhole.

$\sum_{h_t} (f) \pi_t$ for all $f \in F$.

$$= \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{a_{h_t}^{(i)} = \pi_f(S_{h_t}^{(i)})\}}{|A|} \cdot \left(f(S_{h_t}^{(i)}, a_{h_t}^{(i)}) - r_{h_t}^{(i)} - f(S_{h_t+1}^{(i)}, \pi_f(S_{h_t+1}^{(i)})) \right)$$

IS (Importance Sampling)

$$a_{h_t}^{(i)} \sim \pi_f(S_{h_t}^{(i)})$$

MC: n trajectories followed by $\begin{cases} a_h^{(i)} = \pi_h^{(i)}(S_h^{(i)}) \quad \forall h \neq h_t. \\ \underline{a_{h_t}^{(i)} \sim \text{Uniform}(A)} \end{cases}$

IS $\rightarrow$ unbiased estimator

$\rightarrow$ 1 trajectory works for all $f \in \mathcal{F}$ simultaneously.

• Elimination: $\underline{\hat{F}_{t+1}} = \{ f \in \hat{F}_t : |\hat{\Sigma}(f, \pi_t)| \leq \phi \}$

Concentration Analysis

① $\forall$ fixed f_t (thus π_t): w.p. $\geq 1 - \delta$: $\forall h \in [H]$,

$$|\hat{\Sigma}_h(f_t, \pi_t) - \Sigma_h(f_t, \pi_t)| \lesssim H \cdot \underbrace{\sqrt{\frac{1}{n_{\text{eval}}} \cdot \log \frac{H}{\delta}}}_{\rightarrow \varepsilon_1}$$

$$n_{\text{eval}} \gtrsim \frac{H^2}{\varepsilon_1^2} \log \frac{H}{\delta} : \underline{\text{w.p. } \geq 1 - \delta} : |\hat{\Sigma}_h(f_t, \pi_t) - \Sigma_h(f_t, \pi_t)| \leq \varepsilon_1$$

② $\forall$ fixed h_t, π_t : w.p. $\geq 1 - \delta$: $\forall f \in \mathcal{F}$:

$$|\hat{\Sigma}_{h_t}(f, \pi_t) - \Sigma_{h_t}(f, \pi_t)| \lesssim \underline{H \cdot |A|} \cdot \sqrt{\frac{1}{n} \cdot \log \frac{|F|}{\delta}} \rightarrow \phi$$

Remark: IS (low variance)
 $\downarrow$
Bernstein

$$H \cdot \left(\sqrt{\frac{|A|}{n} \cdot \log \frac{|F|}{\delta}} + \frac{|A|}{n} \cdot \log \frac{|F|}{\delta} \right)$$

$$n \geq \frac{|A|^2 \cdot H^2}{\phi^2} \log \frac{H}{\delta} : \underline{\underline{\text{w.p.} \geq 1 - \delta}} : |\hat{\Sigma}_{\Phi_{ht}}(f, \pi_t) - \Sigma_{\Phi_{ht}}(f, \pi_t)| \leq \phi.$$

Claim 1: If $|\hat{\Sigma}_{\Phi_{ht}}(f, \pi_t) - \Sigma_{\Phi_{ht}}(f, \pi_t)| \leq \phi$,

then Q^* is never eliminated,

(Proof: $\Sigma_{\Phi_{ht}}(Q^*, \pi_t) = 0$ $\Rightarrow |\hat{\Sigma}_{\Phi_{ht}}(Q^*, \pi_t)| \leq \phi \quad \square$)

Claim 2: (explore-or-terminate) If $|\hat{\Sigma}_h(f_t, \pi_t) - \Sigma_h(f_t, \pi_t)| \leq \underline{\underline{\varepsilon_1}}$

then : either ^① $\hat{\Sigma}_{\Phi_{ht}}(f_t, \pi_t) \gtrsim \varepsilon_1$

$\rightarrow$ explore

or ^② $J(\pi_t) - J(\pi^*) \geq H \cdot \varepsilon_1$

$\rightarrow$ terminate.

Proof: Terminate when $\sum_{h=1}^H \hat{\Sigma}_h(f_t, \pi_t) \leq \underline{\underline{H \cdot \varepsilon_1}}$

① If NOT: $\exists h_t : \hat{\Sigma}_{\Phi_{ht}}(f_t, \pi_t) > \varepsilon_1$

$$\Rightarrow \Sigma_{\Phi_{ht}}(f_t, \pi_t) > \varepsilon_1 - \frac{\varepsilon_1}{2} = \frac{\varepsilon_1}{2}.$$

② $J(\pi_t)$ = $J_{f_t}(\pi_t) - \sum_{h=1}^H \hat{\Sigma}_h(f_t, \pi_t).$

$$\geq J_{f_t}(\pi_t) - \sum_{h=1}^H \hat{\Sigma}_h(f_t, \pi_t) - H \cdot \varepsilon_1$$

$$\geq \underline{\underline{J_{f_t}(\pi_t)}} - 2 \cdot H \cdot \varepsilon_1.$$

$$f_t = \underset{f \in \mathcal{F}_t}{\operatorname{argmax}} \underline{\underline{J_f(\pi_f)}}$$

$$\underline{\text{Optimism}} \geq \underline{J_Q^*(\pi^*)} - 2H \cdot \varepsilon_1 = J(\pi^*) - 2H \varepsilon_1. \quad \square$$

Claim 3, (Iteration Complexity) $\forall h$: If whenever $\underline{h_t = h}$,

$$|\Sigma_{h_t}(f_t, \pi_t)| \geq 6\sqrt{M} \cdot \phi$$

$$\Rightarrow \# \text{ of iterations that } \underline{h_t = h} \leq \underline{M \cdot \log\left(\frac{\Sigma}{2\phi}\right) / \log \frac{\Sigma}{3}}.$$

Sample Complexity of OLIVE:

$$\# \text{ of samples} = \# \text{ of iterations}$$

$$\times H \times (\underline{n_{\text{eval}}} + n)$$

By Claim 2. If NOT terminate,

$$\Sigma_{h_t}(f_t, \pi_t) \gtrsim \varepsilon_1 = \sqrt{M} \cdot \phi \iff \text{tune } \phi = \frac{\varepsilon_1}{\sqrt{M}}$$

$$H \cdot \left(\frac{H^2}{\varepsilon_1^2} \cdot \log \frac{H}{\delta} + \frac{\overset{|A|}{|A|^2} \cdot H^2}{\phi^2} \cdot \log \frac{|F|}{\delta} \right) \cdot \underline{M \cdot \log\left(\frac{\Sigma}{2\phi}\right)}$$

$$= \tilde{O}\left(M^2 \cdot H^3 \cdot |A|^2 \cdot \frac{1}{\varepsilon_1^2} \cdot \log \frac{|F| \cdot \Sigma}{\delta}\right)$$

$$\text{poly}(|A|, H, \log |F| \left(\frac{1}{\varepsilon_1}\right)^{\frac{1}{\delta}}, M)$$

Bellman factorization:

$$\Sigma_h(f, \pi_{f'})$$

$$= \langle L_h(f'), \Sigma_h(f) \rangle.$$

$$\text{s.t. } L_h(f'), \Sigma_h(f) \in \mathbb{R}^M.$$

$$\& \|L_h(f')\|_2 \cdot \|\Sigma_h(f)\|_2 \leq \Sigma.$$

Split the failure prob. δ
 $\delta/2 \left(M \cdot \log \frac{\Sigma}{2\phi} \right).$

Recall: In OLIVE Analysis: $\boxed{\forall h \in \mathcal{H}}$

(If) whenever $\underline{h_t} = h$: $\boxed{|\Sigma_{\underline{h_t}}(f_t, \underline{\pi_t})| \geq 6\sqrt{M} \cdot \phi}$

$\Rightarrow$ # of iterations that $\underline{h_t} = h \leq M \cdot \log\left(\frac{\mathcal{L}}{2\phi}\right) / \log\frac{5}{3}$.

Volumetric Argument,

For any given $\underline{h}$:

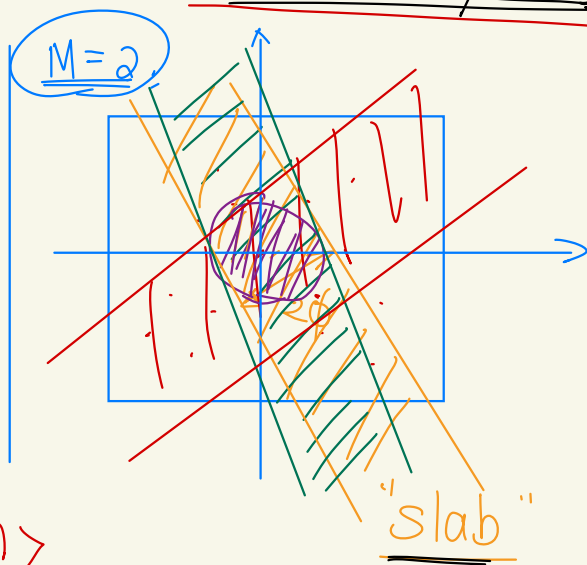
$\boxed{|\underline{i_1}, \underline{i_2}, \dots, \underline{i_L}, \dots, \underline{i_{T_h}}|}$

s.t. $\underline{T_h} = |\{t : \underline{h_t} = h\}|$

$$\Sigma_h(f, \pi_{f'}) = \langle \underline{U_h}(f'), \Sigma_h(f) \rangle$$

where $\underline{U_h}(f'), \Sigma_h(f) \in \mathbb{R}^M$.

$$\|\underline{U_h}(f')\|_2 \cdot \|\Sigma_h(f)\|_2 \leq \mathcal{L}$$



- inefficient exploration,
- efficient exploration

e.g. tabular MDP $\left\{ \begin{array}{l} \underline{U_h}(f'): \text{State distribution of } \pi_{f'} \\ \Sigma_h(f): \text{Bellman error across states} \\ \text{(measured by } f \text{)} \end{array} \right.$

Notations:

① $P_1, \dots, P_{T_h}$: $P_{i_t} = \mathcal{U}_h(f_{i_t})$
distribution optimistic function at round i_t .

② $\mathcal{U}(f_{i_0}), \mathcal{U}(f_{i_1}), \dots, \mathcal{U}(f_{i_{T_h}})$:

Version space $\mathcal{U}(f_{i_t}) = \{ \mathcal{E}_h(f) : f \in \mathcal{F}_{i_t} \}$

③ $\Psi = \sup_{f \in \mathcal{F}} \|\mathcal{U}_h(f)\|_2$, $\Phi = \sup_{f \in \mathcal{F}} \|\mathcal{E}_h(f)\|_2$

$\Psi \Phi \leq \frac{1}{2}$

④ $V_0, V_1, \dots, V_{T_h}$: $V_0 = \{v : \|v\|_2 \leq \Phi\}$

version space

$V_t = \{v \in V_{t-1} : |\langle P_t^T v \rangle| \leq 2\phi\}$

'slab'

⑤ $B_0, B_1, \dots, B_{T_h}$: B_t is the MVEE of V_t

Goal: UB T_h

ellipsoid:

(Minimum Volume Enclosed Ellipsoid)

$\mathcal{E}(A, c) = \{x \in \mathbb{R}^d : (x-c)^T A (x-c) \leq 1\}$

$\text{Vol}(\mathcal{E}(A, c)) = \frac{C_d}{\sqrt{\det(A)}}$

$C_d = \text{Vol}(\{x \in \mathbb{R}^d : \|x\|_2 \leq 1\})$
 $= \pi^{\frac{d}{2}} / \Gamma(\frac{d}{2} + 1)$



Optimization problem: $\min_{A \succ 0} \underline{-\log \det A.}$
 s.t. $(x_i - c)^T A (x_i - c) \leq 1$

Step 1. $\forall i \in [T_n]: \underline{\underline{U(f_{i_T}) \subseteq V_T}} \quad \checkmark$

$T \geq 1: \underline{\underline{\forall f \in F_{i_T}: \Rightarrow |\underline{\underline{\Sigma_h(f, \Pi_{f_{i_T}})}}| \leq 2\phi.}}$

$\langle \underline{\underline{P_T}}, \underline{\underline{\Sigma_h(f)}} \rangle = \langle \underline{\underline{U_h(f_{i_T})}}, \underline{\underline{\Sigma_h(f)}} \rangle$

$\Rightarrow \underline{\underline{\Sigma_h(f) \in V_T}} \quad \rightarrow \underline{\underline{\text{consistency}}}$

Step 2. $\exists \underline{\underline{v \in V_{T-1}}}$ such that $\underline{\underline{|P_T^T v| \geq 6\sqrt{M} \cdot \phi.}}$

"exploration direction"

$\Rightarrow (i_T - 1)$ round, f_{i_T} is valid

At round $\underline{\underline{i_T}}$: $\underline{\underline{f_{i_T}}}$ is chosen as an optimistic function.

$\underline{\underline{v = \Sigma_h(f_{i_T}) \in U(f_{i_{T-1}}) \subseteq U(f_{i_{T-1}}) \subseteq V_{T-1}.}}$

$\underline{\underline{|P_T^T v|}} = \underline{\underline{|\langle U_h(f_{i_T}), \underline{\underline{\Sigma_h(f_{i_T})}} \rangle|}} \quad (\text{monotonicity})$

$= \underline{\underline{|\Sigma_h(f_{i_T}, \Pi_{i_T})|}} \geq \underline{\underline{6\sqrt{M} \cdot \phi}} \quad (\text{by assumption})$



efficient exploration

Lemma (Todd, 1982) $E = \{w \in \mathbb{R}^d : w^T w \leq 1\}$

$$E_\beta = \{w \in E : |e_1^T w| \leq \beta\}$$

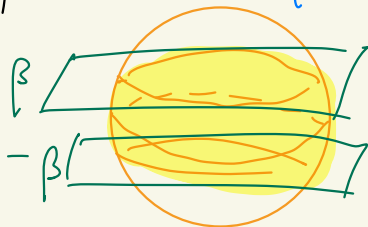
↳ slab

$$\frac{e_1^T w = \beta}{e_1^T w = -\beta}$$

$$E_+ = \{w \in \mathbb{R}^d : w^T (p(I - \sigma e e^T))^{-1} w \leq 1\}$$

is an MVEE of E_β if

$$\sigma = \frac{1 - d\beta^2}{1 - \beta^2} \quad \rho = \frac{d(1 - \beta^2)}{d - 1}$$



↓

$$\frac{\text{Vol}(E_\beta)}{\text{Vol}(E)} = \frac{\text{Vol}(E_+)}{\text{Vol}(E)} = \sqrt{\det(p(I - \sigma e e^T))}$$

↳ diagonal ev.

$$= \sqrt{\rho^{d-1} \cdot \rho(1 - \sigma)}$$

$$\begin{cases} e_i : \rho(1 - \sigma) \\ e_i^\perp : \rho \end{cases}$$

$$= \sqrt{d} \cdot \beta \cdot \left(\frac{d}{d-1}\right)^{\frac{d-1}{2}} \cdot (1 - \beta^2)^{\frac{d-1}{2}}$$

Lemma. closed & bounded $V \subseteq \mathbb{R}^d$. vector $p \in \mathbb{R}^d$

Let B be any enclosing ellipsoid of V . $B = \{v \in \mathbb{R}^d : v^T B^{-1} v \leq 1\}$

Suppose $\exists v \in V : |p^T v| \geq \kappa$



Let $\underline{B}_\gamma = \{v \in B : |p^T v| \leq \gamma\}$. If: $\frac{\gamma}{\kappa} \leq \frac{1}{\sqrt{d}}$

$\underline{B}_+$ be the MVEE of $\underline{B}_\gamma$.

$$\Rightarrow \frac{\text{Vol}(\underline{B}_+)}{\text{Vol}(B)} \leq \sqrt{d} \cdot \frac{\gamma}{\kappa} \cdot \left(\frac{d}{d-1}\right)^{\frac{d-1}{2}} \cdot \left(1 - \frac{\gamma^2}{\kappa^2}\right)^{\frac{d-1}{2}}$$

Proof: $v^T B^{-1} v \leq 1 \Rightarrow w^T w \leq 1, w = B^{-1/2} v$

ellipsoid
↓
unit ball

$$|p^T v| \leq \gamma \Rightarrow |e_1^T w| \leq \gamma/\alpha$$

R: Rotation Matrix

$$R \cdot B^{\frac{1}{2}} p = \alpha \cdot e_1$$

$$\Rightarrow \frac{\text{Vol}(\underline{B}_+)}{\text{Vol}(B)} = \frac{\text{Vol}(\text{MVEE}(\mathcal{I}_d \cap \{w : |e_1^T w| \leq \gamma/\alpha\}))}{\text{Vol}(\mathcal{I}_d)}$$

$$= \sqrt{d} \cdot \frac{\gamma}{\kappa} \cdot \left(\frac{d}{d-1}\right)^{\frac{d-1}{2}} \cdot \left(1 - \frac{\gamma^2}{\kappa^2}\right)^{\frac{d-1}{2}}$$

d-dim unit ball!

$$\alpha = \|R \cdot B^{\frac{1}{2}} p\|_2 = \|B^{\frac{1}{2}} p\|_2 = \sqrt{p^T B p} \rightarrow \kappa$$

$$\exists v: \kappa \leq |p^T v| = |p^T B^{\frac{1}{2}} \cdot B^{-\frac{1}{2}} v|$$

$$\leq \sqrt{p^T B p} \cdot \underbrace{\sqrt{v^T B^{-1} v}}_{\leq 1 \text{ (since } v \in B)} \leq \sqrt{p^T B p}$$

□

Proposition. $\frac{\gamma}{\kappa} = \frac{1}{3\sqrt{d}} \Rightarrow \frac{\text{Vol}(B^+)}{\text{Vol}(\underline{B})} \leq \frac{1}{3} \cdot e^{\frac{4}{9}} < 0.6$

Recall: Step 1: $\underline{V}_T \subseteq \underline{V}_{T-1} = \{v \in \underline{V}_{T-1} : |P_T^T v| \leq 2\phi\}$

Step 2: $\exists v \in \underline{V}_{T-1} : |P_T^T v| \geq \underline{6\sqrt{M}} \cdot \phi$

V_T : Centrally symmetric $\Rightarrow B_T$: Centrally symmetric

$d = M, \quad \kappa = 6\sqrt{M} \cdot \phi$
 $B = B_{T-1}, \quad \gamma = 2\phi$

$\Rightarrow \frac{\text{Vol}(B^+)}{\text{Vol}(B_{T-1})} \leq 0.6$

$B^+ = \text{MVEE}(V'_T), \quad V'_T = \{v \in B_{T-1} : |P_T^T v| \leq 2\phi\}$

$V_T = \{v \in V_{T-1} : |P_T^T v| \leq 2\phi\}$

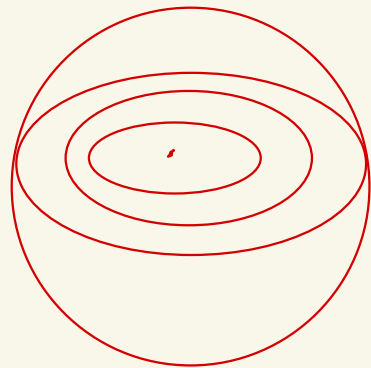
$\underline{V}_T \subseteq \underline{B}_{T-1}$
 $\Rightarrow V_T \subseteq V'_T$
 $\Rightarrow B_T \subseteq B^+$

$\Rightarrow \frac{\text{Vol}(B_T)}{\text{Vol}(B_{T-1})} \leq \frac{\text{Vol}(B^+)}{\text{Vol}(B_{T-1})} \leq 0.6$

Volume of B_T shrinks exponentially fast

Recall our goal is to upper bound T_h .

- ① B_0 is not too large
 ② B_{T_h} is not too small



$$B_0 = \text{MVEE}(\{v \in \mathbb{R}^M : \|v\|_2 \leq \Phi\})$$

$$= \underbrace{C_M}_{\text{Volume of } M\text{-dimensional unit ball}} \cdot (\Phi)^M$$

$$V_{T_h} \equiv \{v \in \mathbb{R}^M : |p_{\tau}^T v| \leq 2\phi\}$$

$$\equiv \{v \in \mathbb{R}^M : \sup_{p: \|p\|_2 \leq \underline{\Psi}} |p^T v| \leq 2\phi\}$$

$$\|p_{\tau}\|_2 \leq \underline{\Psi}$$

$$\equiv \{v \in \mathbb{R}^M : \|v\|_2 \leq 2\phi / \underline{\Psi}\} \quad \downarrow \text{Hölder's inequality.}$$

$$\forall v \text{ s.t. } \|v\|_2 \leq \frac{2\phi}{\underline{\Psi}}$$

$$|p^T v| \leq 2\phi \text{ for all feasible } p$$

$$\text{Vol: } C_M \cdot \left(\frac{2\phi}{\underline{\Psi}}\right)^M$$

$$\Rightarrow \frac{\text{Vol}(B_{T_h})}{\text{Vol}(B_0)} \geq \frac{C_M \cdot \left(\frac{2\phi}{\underline{\Psi}}\right)^M}{C_M \cdot (\Phi)^M} = \frac{(2\phi)^M}{(\underline{\Psi})^M} = \left(\frac{2\phi}{\underline{\Psi}}\right)^M$$

$$\prod_{t=1}^{T_h} \frac{\text{Vol}(B_t)}{\text{Vol}(B_{t+1})} \leq \underline{\underline{0.6}}^{T_h} \Rightarrow \underline{T_h} \approx \frac{M \cdot \log \frac{2\phi}{\underline{\Psi}}}{\log 0.6} \quad \square$$

Bellman Eluder Dimension

RL with general function approximation.

$\mathcal{F}$: realizability ($Q^* \in \mathcal{F}$)

low Bellman Rank (V-type)

$$\Sigma_h(f, \pi_{f'}) = \mathbb{E}_{\substack{a_{1:h} \sim \pi_{f'} \\ a_h \sim \pi_{f'}}} [f_h(S_h, a_h) - r_h - \max_{a \in \mathcal{A}} f_h(S_{h+1}, a)]$$

state distribution

$$= \langle \mathcal{L}_h(f'), \Sigma_h(f) \rangle$$

s.t. $\mathcal{L}_h(f'), \Sigma_h(f) \in \mathbb{R}^M$

$$\|\mathcal{L}_h(f')\|_2 \cdot \|\Sigma_h(f)\|_2 \leq \gamma.$$

• Q-type: $\Sigma_h(f, \pi_{f'}) = \mathbb{E}_{(S_h, a_h) \sim d \pi_{f'}} [(f_h - \gamma_h f_{h+1})(S_h, a_h)]$

new distribution

def (ε -independence between points) ↗ non-linear

$\mathcal{G}$ is a function class over $\mathcal{X}$. $x_1, \dots, x_n \in \mathcal{X}$.

$z \in \mathcal{X}$ is ε -independent to $\{x_1, \dots, x_n\}$ if $\exists \underline{g}_1, \underline{g}_2 \in \mathcal{G}$.

$$\sqrt{\sum_{i=1}^n (\underline{g}_1(x_i) - \underline{g}_2(x_i))^2} \leq \varepsilon.$$

$$\underline{g}_1(z) - \underline{g}_2(z) > \varepsilon.$$

new sample
↙
new information

"certifying" function.

c.f. Jin et al. 2021. Bellman eluder dimension: new rich classes of RL problems, and sample-efficient algorithms

Tabular MDP.

Linear MDP.

Generalized linear MDP.

LQR.

Reactive POMDP

Reactive PSR

Bellman optimality operator

Provably efficient RL with bounded ED. 2020

def (Eluder dimension) $\dim_E(\mathcal{G}, \varepsilon)$: the length of longest sequence

$\{x_1, \dots, x_n\} \subseteq \mathcal{X}$ s.t. x_i is ε -independent to $\{x_1, \dots, x_{i-1}\}$. $\forall i$

< Eluder dimension
VC dimension

r.f.: Russo and Van Roy, 2013

vector space $\dim(d)$

$\{x_1, \dots, x_d\}$ s.t. x_i is linearly independent to $\{x_1, \dots, x_{i-1}\}$

def (ε -independence between distributions)

$\mu \in \Delta_{\mathcal{X}}$ is ε -independent to $\{\mu_1, \dots, \mu_n\}$ w.r.t. $\mathcal{G}$. $\mu_i \in \Delta_{\mathcal{X}}$.

$$\text{if } \exists \underline{g} \in \mathcal{G} : \begin{cases} \sqrt{\sum_{i=1}^n (\mathbb{E}_{\mu_i}[\underline{g}])^2} = \varepsilon \\ |\mathbb{E}_{\mu}[\underline{g}]| > \varepsilon \end{cases}$$

def (Distributional Eluder Dimension) $\dim_{DE}(\mathcal{G}, \Pi, \varepsilon)$

length of longest distribution sequence $\{\pi_1, \dots, \pi_n\} \subseteq \Pi$

s.t. π_i is ε -independent to $\{\pi_1, \dots, \pi_{i-1}\}$. $\forall i$

$\dim_E(\mathcal{G}, \varepsilon) = \dim_{DE}(\mathcal{G} - \mathcal{G}, \Pi_{\sigma}, \varepsilon)$

$Edim \subseteq DEdim$

< $\mathcal{G} - \mathcal{G} = \{g_1 - g_2 : g_1, g_2 \in \mathcal{G}\}$

$\Pi_{\sigma} = \{\sigma_x(\cdot) : x \in \mathcal{X}\}$

def (Bellman Eluder Dimension)

$$(\mathbb{I} - \gamma_h) \mathcal{F} := \{ \underline{f_h - \gamma_h f_{h+1}} : f \in \mathcal{F} \}$$

function space of Bellman error

Inhomogeneous episodic MDP

$(\mathcal{S}, \mathcal{A}, H, P, R)$

$$P = \{P_h\}_{h=1}^H, \quad R = \{r_h\}_{h=1}^H.$$

$$\Pi = \{ \Pi_h \}_{h=1}^H$$

distribution over $\mathcal{S} \times \mathcal{A}$

$$\dim_{BE}(\mathcal{F}, \Pi, \varepsilon) = \max_{h \in [H]} \dim_{BE}((\mathbb{I} - \gamma_h) \mathcal{F}, \Pi_h, \varepsilon)$$

Remark: 1. Q-type BE dimension

→ V-type BE dimension

$$\underline{\mathbb{I}} < \frac{\Pi}{\Pi^*} < \mathbb{V} < \mathbb{Q}$$

2. distribution family $\Pi \subseteq \Delta(\mathcal{S} \times \mathcal{A})$

$$\textcircled{1} \Pi = \Pi_{\mathcal{F}} = \{ \Pi_{f,h} \}_{h=1}^H$$

occupancy measure of $\Pi_f : f \in \mathcal{F}$ at step h,

$$\textcircled{2} \Pi = \Pi_{\mathcal{O}} = \{ \Pi_{o,h} \}_{h=1}^H$$

$$\Pi_f = \{ \Pi_{f,h} \}_{h=1}^H, \quad \Pi_{f,h}(s) = \arg \max_{a \in \mathcal{A}} f_h(s, a)$$

$$\Pi_{o,h} = \{ \delta_{(s,a)}(\cdot) : s \in \mathcal{S}, a \in \mathcal{A} \}$$

low BE dimension: $\min_{\Pi \in \{ \Pi_{\mathcal{F}}, \Pi_{\mathcal{O}} \}} \dim_{BE}(\mathcal{F}, \Pi, \varepsilon)$ is small

Complexity

Measure

1. Simple examples with low complexity

2. Efficient algorithm (at least in information theory)

with low complexity

① Low Bellman Rank $\subseteq$ Low BE Dimension.

② Low Euler Dimension $\subseteq$ Low BE Dimension.

↳ Assumption 1. Bellman Completeness $Tf \in \mathcal{F}, \forall f \in \mathcal{F}, h.$

↓

DE dim of $(I - \tilde{T}_n)f$.

$$\Leftrightarrow \underline{\underline{\text{DEdim of } f.}}$$

$\Rightarrow$ Edim of J . Π_J

$$\Rightarrow \dim_{BE}(F, \Pi_{\mathcal{F}}, \varepsilon) \leq \max_{h \in [H]} \dim_E(F_h, \varepsilon) \quad \dots \dots \quad \left(\frac{\sum_{h \in [H]} f_h}{\sum_{h \in [H]} f_h} \right)$$

Completeness $\Rightarrow$ Realizability

$$(\{f \in f, \underbrace{\tau f \in f, \tau^2 f \in f}_{\vdots}, \dots, \underbrace{\tau^\infty f \in f}_{\mathbb{Q}^*})$$

③ Low BE dimension $\nsubseteq$ Low Bellman Rank
or Low Eluder Dimension

MAB with exp. many actions

low BE dimension.

low Bellman Rank
reactive POMDP
reactive PSR

kernel reactive Pomdp

- tabular MDP
- linear MDP
- LQR

low Eluder dimension

generalized linear MDP. kernel MDP.

General Function Approximation

Proposition. If an MDP with function class $\mathcal{F}$ has Bellman Rank M .

$$\Rightarrow \dim_{\text{BE}}(\mathcal{F}, \Pi_{\mathcal{F}}, \varepsilon) \leq O\left(1 + M \cdot \log\left(1 + \frac{\gamma}{\varepsilon}\right)\right)$$

Proof: Wlog assume $\max\{\|\mathcal{L}_h(\mathcal{F})\|_2, \|\mathcal{E}_h(\mathcal{F})\|_2\} \leq \sqrt{L}$

$$\exists \underline{h}: \dim_{\text{BE}}((I - \gamma \mathcal{T}_h)\mathcal{F}, \underline{\Pi_{\mathcal{F}, h}}, \varepsilon) \geq \underline{n}$$

Note: No completeness assumption is required!

$$\mu_1, \dots, \mu_n \in \Pi_{\mathcal{F}, h} \text{ s.t. } \exists f_1, \dots, f_n \quad \forall i \in [n],$$

$$\sqrt{\sum_{t=1}^{i-1} (\mathbb{E}_{\mu_t}[f_t - \gamma_h f_{t+1}])^2} \leq \varepsilon$$

$$|\mathbb{E}_{\mu_t}[f_t - \gamma_h f_{t+1}]| > \varepsilon \Rightarrow \mathcal{E}_h(f^i, \Pi_{g_t})$$

$t \in [i]$

$$\exists g_1, \dots, g_n \in \mathcal{F}: \mu_i = \underline{d}^{\Pi_{g_i}} \quad \forall i \in [n]$$

$$\textcircled{1} \sqrt{\sum_{t=1}^{i-1} \langle \mathcal{L}_h(g_t), \mathcal{E}_h(f_t) \rangle^2} \leq \varepsilon$$

$$\frac{\langle \mathcal{L}_h(g_i), \mathcal{E}_h(f_i) \rangle}{\Delta_i} > \varepsilon$$

$x_i \quad z_i$

$$\textcircled{2} |\mathcal{L}_h(g_i), \mathcal{E}_h(f_i)| > \varepsilon$$

Cumulative Covariance matrix: $V_i = \sum_{t=1}^{i-1} \underline{Z_t Z_t^\top} + \frac{\varepsilon^2}{L} \underline{I_M}$

$$\textcircled{2} \Leftrightarrow \langle x_i, z_i \rangle > \varepsilon \Leftrightarrow \|x_i\|_{V_i} \cdot \|z_i\|_{(V_i)^{-1}} > \varepsilon$$

$$\textcircled{1}: \|x_i\|_{V_i} = \sqrt{x_i^T V_{i-1} x_i} = \sqrt{x_i^T \underbrace{\sum_{t=1}^{i-1} Z_t Z_t^T}_{\frac{\xi^2}{\epsilon}} x_i + \underbrace{\frac{\xi^2}{\epsilon} x_i^T x_i}_{\frac{\xi^2}{\epsilon} \|x_i\|^2}}$$

$$\|Z_{i-1}^T x_i\|_2 \leq \epsilon \quad \Rightarrow \quad \sqrt{\underbrace{\|Z_{i-1}^T x_i\|_2^2}_{\|Z_{i-1} x_i\|_2^2} + \epsilon^2} \leq \sqrt{2} \epsilon.$$

$\underline{Z_{i-1}} = [Z_1, Z_2, \dots, Z_{i-1}]$

$$\Rightarrow \|Z_i\| (V_i)^{-1} \geq \frac{1}{\sqrt{2}}.$$

$$\det(V_n) = \det(V_{n-1} \cdot (1 + \|Z_n\|_{V_{n-1}}^2))$$

$$= \det(V_{n-1}) \cdot (1 + \|Z_n\|_{V_{n-1}}^2)$$

$$\geq \frac{3}{2} \det(V_{n-1}) \geq \dots \geq \det\left(\frac{\xi^2}{\epsilon} I_M\right) \cdot \left(\frac{3}{2}\right)^{n-1}$$

$$\det(V_n) \leq \left(\frac{\text{trace}(V_n)}{M}\right)^M$$

$$\leq \left(\frac{\frac{\xi^2(n-1)}{M} + \frac{\xi^2}{\epsilon}}{M}\right)^M$$

$$= \left(\frac{\xi^2}{\epsilon}\right)^M \cdot \left(\frac{3}{2}\right)^{n-1}$$

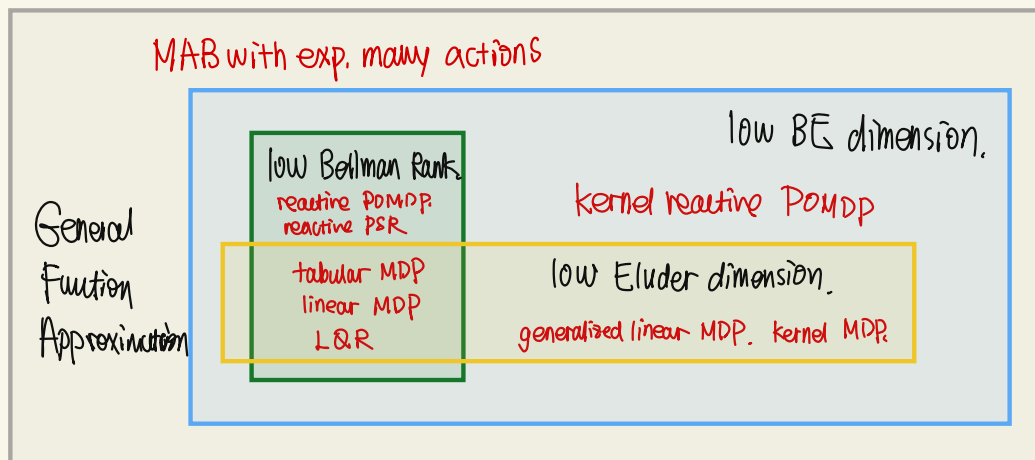
$$\Rightarrow \left(\frac{3}{2}\right)^{n-1} \leq \left(\frac{\xi^2(n-1)}{M\xi^2} + 1\right)^M$$

$$\Rightarrow n \leq 4 \cdot \left(1 + M \cdot \log\left(\frac{\xi^2(n-1)}{M\xi^2} + 1\right)\right)$$

$$= O\left(1 + M \cdot \log \frac{\xi}{\epsilon}\right) \quad \square$$

GOLF (Global Optimism based on Local Fitting)

Recall: RL Zoo of general function approximation



Another route: Bellman rank [Jiang et al. 2017]

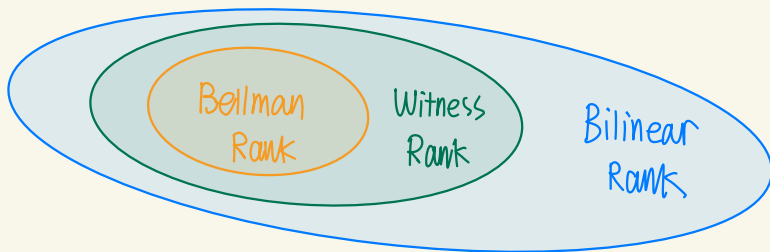
→ Witness rank [Sun et al. 2019]

→ Bilinear rank [Du et al. 2021]

e.g. factorized MDP
[Kearns and Koller, 1999]

FQI

(Fitted-Q Iteration)



GOLF: optimistic version of FQI

$F_1 = F$ // version space (confidence set)

$D_1, \dots, D_{H_1} = \phi$

$$F = F_1 \times F_2 \times \dots \times F_{H_1}$$

$$\underline{\pi} = \underline{\pi}_1 \times \underline{\pi}_2 \times \dots \times \underline{\pi}_{H_1}$$

for $k=1, 2, \dots, K$:

Optimism: $\underline{f_k} = \underset{f \in F_k}{\operatorname{argmax}} \underline{J_f}(\pi_k)$ $\pi_k = \pi_{f_k}$

Collect a trajectory $(s_1, a_1, r_1, \dots, s_H, a_H, r_H, s_{H+1})$ by following π_k .

Augment $D_h = D_h \cup \{(s_h, a_h, r_h, s_{h+1})\} \quad \forall h \in [H]$

Evaluate: $\hat{\Sigma}_h(f_h, \pi_k) = \underline{L}_h(f_h, f_{h+1}) - \inf_{g \in F_h} \underline{L}_h(g, f_{h+1}) \quad \forall f$

squared Bellman error (unbiased estimator) where $\underline{L}_h(\xi_h, \xi_{h+1}) = \frac{1}{|D_h|} \sum_{(s, a, s') \in D_h} (\xi_h(s, a) - r - \max_{a' \in A} \xi_{h+1}(s', a'))^2$

Eliminate: $F_{k+1} = \{f \in F_k : \hat{\Sigma}_h(f_h, \pi_k)^2 \leq \beta, \quad \forall h \in [H]\}$

① FQI: $\underline{f_h} \leftarrow \underset{\phi \in F_h}{\operatorname{argmin}} \underline{L}_h(\phi, f_{h+1})$

Average Bellman Error (ϕ -type)

$Q_h^*(s_h, a_h) = r_h + \max_{a' \in A} Q_h^*(s', a')$

$\underline{\Sigma}_h(f_h, \pi_k) = \mathbb{E}_{\substack{(s_h, a_h) \sim d\pi_k \\ r_h = r_h(s_h, a_h) \\ s_{h+1} \sim p_h(s_h, a_h)}} \left[\underline{f_h}(s_h, a_h) - r_h - \max_{a' \in A} \underline{f_{h+1}}(s_{h+1}, a') \right]$

$= \mathbb{E}_{(s_h, a_h) \sim d\pi_k} \left[\underline{f_h}(s_h, a_h) - \mathbb{E}_{r_h, s_{h+1} | s_h, a_h} \left[r_h + \max_{a' \in A} \underline{f_{h+1}}(s_{h+1}, a') \right] \right]$

Squared Bellman Error:

$$\Sigma_h(f_h, \pi_k)^2 = \mathbb{E}_{(S_h, a_h) \sim \pi_k} \left[\underbrace{\left(f_h(S_h, a_h) - \underbrace{\mathbb{E}[r_h + \max_{a \in A} f_{h+1}(S_{h+1}, a)]}_{\substack{\Gamma_h + \max_{a \in A} f_{h+1} \\ \mathbb{E}[Y|x]}} \right)^2}_{f(x)} \right]$$

$$\mathbb{E}[L_h(f_h, f_{h+1})] = \mathbb{E}_{S_h, a_h, \Gamma_h, S_{h+1}} \left[\underbrace{\left(f_h(S_h, a_h) - \underbrace{\Gamma_h - \max_{a \in A} f_{h+1}(S_{h+1}, a)}_{f(x)} \right)^2}_{\text{noise}} \right]$$

$S_h, a_h : X$

$$\Gamma_h + \max_{a \in A} f_{h+1}(S_{h+1}, a) : Y$$

$$(\Gamma_h f_{h+1})(S_h, a_h) : \mathbb{E}[Y|x]$$

$$\mathbb{E}[L_h(\Gamma_h f_{h+1}, f_{h+1})]$$

Double-Sampling Problem:

$$\begin{aligned} \mathbb{E}[(f(x) - Y)^2] &\rightarrow \mathbb{E}[L_h] \\ &= \mathbb{E}[(f(x) - \mathbb{E}[Y|x])^2] \rightarrow \Sigma_h^2 \\ &\quad + \mathbb{E}[(\mathbb{E}[Y|x] - Y)^2] \geq 0 \end{aligned}$$

$$\Rightarrow \Sigma_h(f_h, \pi_k)^2 = \mathbb{E} \left[L_h(f_h, f_{h+1}) - \underbrace{L_h(\Gamma_h f_{h+1}, f_{h+1})}_{\text{noise}} \right]$$

② Optimism: $J(\pi) = \mathbb{V}_1^{\pi}(S_1)$

$$\begin{aligned} \text{Reg}_k &= \sum_{k=1}^K \underbrace{J(\pi^*) - J(\pi_k)}_{\parallel} \\ &= \sum_{k=1}^K J_{Q^*}(\pi_{Q^*}) - J(\pi_k) \end{aligned}$$

$$\approx \sum_{k=1}^K \underline{J_{f_k}(\pi_k) - J(\pi_k)}$$

$$= \sum_{k=1}^K \sum_{h=1}^H \Sigma_h(\underline{f_{k,h}}, \pi_k)$$

(Bellman error-telescoping)

$$\inf_{g \in \mathcal{F}_h} \underline{L_h(g, f_{h+1})}$$

When Bellman completeness is satisfied,

$$\text{i.e. } \Gamma_h f_{h+1} \in \mathcal{F}_h$$

$Q^* \in \mathcal{F}_k$ i.e. Q^* is never being eliminated

$f_k = \arg\max_{A \in \mathcal{F}_k} J_f(\pi_k)$

$$= \sum_{h=1}^H \sum_{k=1}^K \sum_{\pi_k} (f_{k,h}, \pi_k)$$

It remains to bound Cumulative Bellman error by BE dimension

f is ρ -covering
($\|\cdot\|_\infty$)

③ Concentration: $\beta = c \cdot \log \frac{KH \mathcal{N}_F(\rho)}{\delta} + K \cdot \rho$

$$\sum_{i=1}^{K-1} \mathbb{E} [(f_{k,h}(S_h, a_h) - (\pi_h f_{k,h+1})(S_h, a_h))^2 \mid S_h, a_h \sim d_h^{\pi_h}] \leq O(\beta)$$

$$\sum_{i=1}^{K-1} (f_{k,h}(S_h^i, a_h^i) - (\pi_h f_{k,h+1})(S_h^i, a_h^i))^2 \leq O(\beta)$$

(Martingale concentration argument (Freedman's inequality))

filtration $\mathcal{F}_{k,h} = \sigma(\{S_1^i, a_1^i, \dots, a_i^i, \dots, S_{h+1}^i\}_{i=1}^{k-1}, \{S_1^k, a_1^k, \pi_1^k, \dots, S_h^k, a_h^k\})$

(Lemma) Function class $\underline{\Phi}$ over $\mathcal{X}$, $\forall \phi \in \underline{\Phi}, x \in \mathcal{X} : |\phi(x)| \leq C$.

For $\{\phi_k\}_{k=1}^K \subseteq \underline{\Phi}$, $\{\mu_k\}_{k=1}^K \subseteq \Pi \subseteq \Delta(\mathcal{X})$

s.t. $\sum_{i=1}^{k-1} (\mathbb{E}_{\mu_i} [\phi_k])^2 \leq \beta$. then $\forall k: \forall w$.

$$\sum_{i=1}^K |\mathbb{E}_{\mu_k} [\phi_k]| \leq O(\sqrt{d \cdot \beta \cdot K} + \min\{K, d\} \cdot C + K \cdot w)$$

where $d = \dim_{DE}(\underline{\Phi}, \Pi, w)$

Distributional Eluder Dimension.

$$\rho = \frac{1}{K}, \quad \omega = \frac{1}{\sqrt{K}}, \quad C=1, \quad \mathcal{X} = S \times A, \quad \Phi = (\mathbb{I} - \gamma_h) f.$$

$$\Pi \begin{cases} \Pi_{f,h} : \mu_k = d_h^{\pi_k}(s_h = \cdot, a_h = \cdot) \\ \Pi_{\bar{0},h} : \mu_k = \mathbb{1}_{\{(s_h^k, a_h^k) = \cdot\}} \end{cases}$$

$$\Phi_k = f_{k,h} - \gamma_h f_{k,h+1}$$

$$\Rightarrow \sum_{k=1}^K \sum_h (f_{k,h}, \pi_k) \leq O\left(\sqrt{K \cdot \dim_{BE}(f, \Pi, \sqrt{\frac{1}{K}}) \cdot \log \frac{KH \mathcal{N}_F(\frac{1}{K})}{\delta}}\right)$$

$$\Rightarrow \text{Reg}_K \lesssim H \cdot \sqrt{K \cdot \dim_{BE}(f, \Pi, \sqrt{\frac{1}{K}}) \cdot \log \frac{KH \mathcal{N}_F(\frac{1}{K})}{\delta}}$$

pure exploration

$$\Leftrightarrow \text{SC: } \hat{\pi} = \text{Unif}[\pi_1, \dots, \pi_K],$$

SC: output $\hat{\pi}$.

$$J(\pi^*) - J(\hat{\pi}) \leq \varepsilon.$$

To guarantee $J(\pi^*) - J(\hat{\pi}) \geq \varepsilon$ w.p. $\geq 1 - \delta$,

we need $\Omega\left(\frac{H^2 d}{\varepsilon^2} \cdot \log \sim\right)$ samples.

$$\text{Reg: } \text{Reg}_K = \sum_k J(\pi^*) - J(\pi_k) = o(K)$$

exploration & exploitation

$$\text{Proposition: } \sum_{i=1}^K \mathbb{1}_{\{|E_{\mu_i}[\Phi_i]| \geq \varepsilon_i\}} \leq \frac{\beta}{\varepsilon_i^2} \cdot \dim_{BE}(\Phi, \Pi, \omega)$$

$$\exists K: |E_{\mu_K}[\Phi_K]| \geq \varepsilon.$$

$$\{\mu_1^{(l)}, \dots, \mu_m^{(l)}\}_{l \in [L]}$$

ε -independence

$$\sum_{i=1}^{m^{(l)}} (E_{\mu_i^{(l)}}[\Phi_K])^2 \geq \varepsilon^2$$

$$\{ \mu_1, \dots, \mu_{K-1} \}$$

$$\Rightarrow L \varepsilon^2 \leq \beta \Rightarrow L \leq \frac{\beta}{\varepsilon^2}$$